

# Measurement of $\nu_\mu + \bar{\nu}_\mu$ Charged-Current Single Pion Production Differential Cross Sections on Argon with the MicroBooNE Detector in the NuMI Beam

Analysis Note, Full FHC, RHC, and Combined Exposure

Analysis note

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## Abstract

We measure  $\nu_\mu + \bar{\nu}_\mu$  charged-current single-charged-pion ( $\text{CC}\pi^\pm$ ) differential cross sections on argon using the full MicroBooNE NuMI exposure, in three configurations: forward horn current (FHC,  $8.857 \times 10^{20}$  POT), reverse horn current (RHC,  $11.082 \times 10^{20}$  POT) and their combination ( $19.939 \times 10^{20}$  POT).

Events are selected by requiring a muon candidate and exactly one contained charged-pion candidate from a fiducial vertex, an electromagnetic-shower veto, and a muon–pion opening angle below 2.6 rad, giving  $\approx 59\%$  purity at  $\approx 17\%$  efficiency. Cross sections are extracted by Wiener-SVD unfolding in six released distributions — muon and pion momentum, muon polar angle in both parameterisations, pion polar angle and the muon–pion opening angle — together with a flux-averaged total from a one-bin extraction of the same machinery. A proton-tagged subsample, in which a proton candidate is also required, is measured as secondary scope in the hadronic invariant mass, the transverse-kinematic-imbalance variables and two proton angles.

Each configuration carries its own integrated flux and exposure, and each result carries the complete configured covariance: flux, interaction model, hadron re-interaction, detector response, normalisation and statistical terms. Unfolded results must be compared with a prediction through the released additional-smearing matrix  $A_C$ , which is not normalisation-preserving;  $A_C$  and the covariance are released for every extraction.

**The analysis is blind.** Every “data” point in this note is Poisson-thrown from the central-value Monte Carlo, so no number here is a statement about beam-on data. The four background-control regions, which contain no signal-region events, have been opened on beam-on data; they show no evidence that the background model needs correcting.

The extraction is validated by same-sample self-closure, by an ensemble of independent Poisson realisations carried through the whole chain — which returns correctly sized statistical uncertainties — and by an independent bit-identical rebuild from the processed ntuples. [Sec. 1](#) explains how the note is organised; the studies behind the analysis choices are in the technical supplement, and the history of the analysis is in the change log.

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# 1 Introduction

Charged-current single charged-pion ( $\text{CC}\pi^\pm$ ) production,

$$\nu_\mu(\bar{\nu}_\mu) + \text{Ar} \longrightarrow \mu^\mp + \pi^\pm + X, \quad (1)$$

is among the most frequent neutrino interaction channels at energies of 1–5 GeV and constitutes a significant background for oscillation analyses at the same energy scale, most notably for  $\nu_e$  appearance measurements at experiments such as NOvA [3] and T2K [4], as well as the future DUNE [5]. Despite its importance, the cross section on nuclear targets remains poorly constrained: major neutrino interaction generators (GENIE v3, NuWro, NEUT) disagree by 20–30% in absolute rate and show significant shape differences for individual kinematic variables.

MicroBooNE is a single-phase liquid argon time-projection chamber (LArTPC) [33] at Fermilab that operated from 2015–2021, providing the world’s highest-statistics  $\nu_\mu$ -on-argon dataset below 5 GeV during its NuMI running period. This note presents a measurement of differential cross sections in this channel over the full NuMI exposure, in the forward-horn, reverse-horn and combined configurations, using Pandora pattern recognition [6, 12] and Wiener-SVD unfolding [7].

## What the measurement does

The measurement is built in five steps, and the note follows them in order.

1. **Exposure and flux** (§3). Each configuration is an independent measurement with its own exposure and its own integrated flux. The NuMI beam reaches MicroBooNE off-axis, and its direction is not the detector axis; every angle in this note is defined about the neutrino direction, which matters at the level of whole bins.
2. **Signal and selection** (§5–§7). The signal is defined by the particles leaving the nucleus, not by the interaction mechanism: one muon, one charged pion, no neutral pion, above stated momentum thresholds and inside the fiducial volume. The selection reconstructs that topology.
3. **Response and unfolding** (§9). The detector smears and loses events, so the measured spectrum is not the cross section. Wiener-SVD unfolding inverts the response with a regularisation that trades bias against variance, and reports what it did through the additional-smearing matrix  $A_C$ : a prediction must be multiplied by  $A_C$  before it is compared with these results.  $A_C$  is not normalisation-preserving, which is why integrated cross sections differ between observables of the same sample.
4. **Uncertainties** (§13). Flux, interaction model, hadron re-interaction and detector response are propagated as full covariance matrices through the same unfolding as the data, and are released with the results.
5. **Validation and blinding** (§14). The analysis is blind: results are extracted on fake data thrown from the central-value simulation. The background model is tested in four control regions that hold no signal-region event, which have been opened on beam-on data.

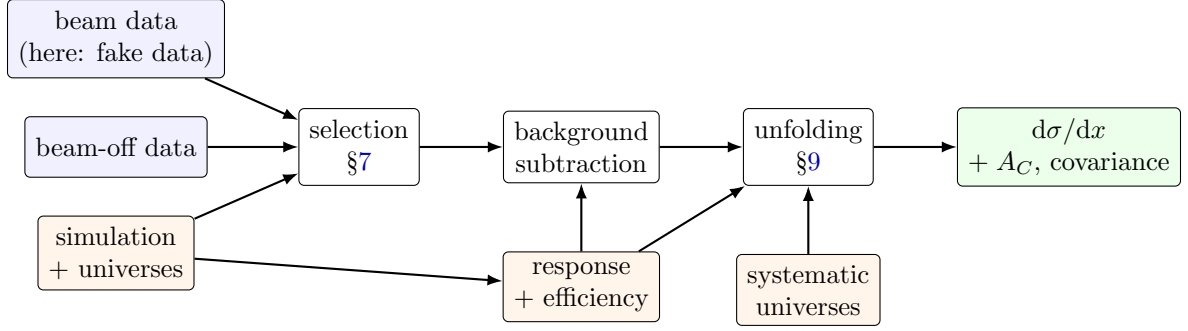


Figure 1: The extraction chain. The same selection is applied to data, beam-off data and simulation; the simulation additionally supplies the response and the systematic universes. What comes out is a cross section together with the additional-smearing matrix  $A_C$  and the covariance — all three are needed to compare it with a prediction.

## How to read this note

§10 is the results section and can be read on its own; §11 gives the proton-tagged subsample, which is secondary scope. Two companion documents carry material that would otherwise interrupt the argument: the *technical supplement* holds the studies behind the analysis choices — estimator surveys, binning and threshold optimisation, the closure programme, the control-region procedure — and the *change log* holds the history of the analysis, including every correction that changed a result and the values before and after. Neither is needed to follow this note. The additional-smearing and covariance matrices for every extraction are released as text in `report/data_release/`; a prediction cannot be compared with these results without the former, nor used quantitatively without the latter.

## 2 Physics Context

### 2.1 Production mechanisms and nuclear effects

At 0.5–5 GeV charged-current single charged-pion production proceeds through baryon resonance production, dominated by the  $\Delta(1232)$  and modelled by Rein–Sehgal-type form factors [35, 36, 37, 38, 39]; a non-resonant continuum and deep-inelastic scattering, together  $\sim 20\%$  of the rate here and added incoherently by every generator used; and coherent production off the whole nucleus ( $\lesssim 10\%$ ), which satisfies the topological signal definition of §5 and is therefore part of the signal, not a background. On argon, Fermi motion, Pauli blocking and binding modify the kinematics, and final-state interactions (absorption at  $\sim 25$ – $30\%$ , charge exchange, quasi-elastic scattering) mix the  $CC\pi^\pm$ ,  $CC\pi^0$  and  $CC0\pi$  topologies at the 30% level. That mixing is why the signal is defined by the final-state particle content and not by the production process, and why  $CC0\pi$  is the largest background: pion absorption and a proton taken for the pion feed it into the selection (Sec. C of the technical supplement).

### 2.2 Generators compared

Four generators are compared with the results; their relevant model choices are summarised in Table 1. The injected model and the source of the response is the MicroBooNE tune of GENIE v3 [16]. In the measured phase space the standalone generators span a factor of  $\sim 1.6$  in the total cross section, GENIE lowest and NEUT highest (Table 14 of the technical supplement), bracketing the tune; because the present results are extracted on central-value fake data, that spread is the context for the model comparisons, not a measured tension.

Table 1: Model content of the generators compared, restricted to what matters for interpreting this measurement.

| Generator        | Resonance model                                                            | Non-resonant / DIS                                  | Final-state interactions                                |
|------------------|----------------------------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------------|
| GENIE v3 [9, 20] | KLN-BS, 18 resonances to $W \lesssim 2$ GeV                                | Bodek-Yang, incoherent; AGKY transition             | INTRANUKE hA (one effective interaction)                |
| NuWro [21]       | $\Delta(1232)$ only (Paschos-Yu)                                           | separate amplitude; smooth transition to Bodek-Yang | Salcedo-Oset (in-medium $\Delta$ , two-body absorption) |
| NEUT [40, 23]    | modified Rein-Sehgal, 18 resonances, Graczyk-Sobczyk $\Delta$ form factors | GRV98 with Bodek-Yang above $W \approx 2$ GeV       | step-by-step intranuclear cascade                       |
| GiBUU [22]       | $\Delta$ and higher resonances with in-medium spectral functions           | PYTHIA fragmentation above $W \approx 1.6$ GeV      | transport equations with pion optical potential         |

### 2.3 Previous measurements

Table 2 lists the published  $CC\pi^+$  measurements on nuclear targets. The two most directly relevant are MicroBooNE’s own: the BNB measurement [52], on the same detector with an on-axis spectrum peaked at 0.7 GeV and negligible  $\bar{\nu}_\mu$  content, gives  $\sigma = (3.75 \pm 0.07_{\text{stat}} \pm 0.80_{\text{syst}}) \times 10^{-38} \text{ cm}^2/\text{Ar}$  and finds all tested generators above the data at very forward muon angles; and the NuMI  $\nu_e + \bar{\nu}_e$   $CC\pi^+$  measurement [51], on the same beam and reconstruction, which shares this analysis’s extraction framework and conventions (Sec. 7.1 of the technical supplement). The NuMI off-axis flux used here is softer and broader than the BNB one with a  $\sim 35\%$   $\bar{\nu}_\mu$  admixture in FHC, and the flux-averaged total is divided by the whole flux above 60 MeV while pion production opens near 0.3 GeV, so the  $\sim 1.0 \times 10^{-38} \text{ cm}^2/\text{Ar}$  found here is not to be compared directly with the BNB value (§3.4). The T2K off-axis measurement on carbon [49] overlaps this spectrum most closely among the other experiments. The extended survey is in the technical supplement (App. B of the technical supplement).

Table 2: Selected  $\text{CC}\pi^+$  cross-section measurements on nuclear targets.  $\langle E_\nu \rangle$  is the approximate mean or peak neutrino energy. “Type” indicates whether a total integrated (T) or differential (D) cross section was reported. “Target” gives the nucleus primarily used for cross-section extraction.

| Experiment                      | Target        | $\langle E_\nu \rangle$<br>(GeV) | Type | Key result / reference                                                                                                              |
|---------------------------------|---------------|----------------------------------|------|-------------------------------------------------------------------------------------------------------------------------------------|
| ANL<br>12-ft [41]               | D             | 1.0                              | T    | Bubble chamber; foundational<br>$\sigma$ vs $E_\nu$                                                                                 |
| BNL 7-ft [42]                   | D             | 1.5                              | T    | Higher statistics; 15%<br>normalisation tension with<br>ANL                                                                         |
| K2K [43]                        | C             | 1.3                              | T    | First accelerator $\text{CC}\pi^+$ on<br>carbon near detector                                                                       |
| MiniBooNE [44]                  | $\text{CH}_2$ | 0.7                              | D    | $Q^2$ , $W$ , $p_\mu$ , $\theta_\mu$ ; 20% excess over<br>NUANCE                                                                    |
| MINERvA [46]                    | CH            | 3.6                              | D    | $p_\pi$ , $\theta_\pi$ , Adler angles; NuMI<br>medium-energy                                                                        |
| T2K<br>ND280 [49]               | CH            | 0.85                             | D    | $p_\mu$ , $\cos\theta_\mu$ ; off-axis beam similar<br>to this work                                                                  |
| ArgoNEUT [50]                   | Ar            | 9.6                              | T    | First $\text{CC}\pi^+$ on Ar; 115 $\nu$ +<br>158 $\bar{\nu}$ events ( $1.25 \times 10^{20}$ POT)                                    |
| MicroBooNE<br>BNB [52]          | Ar            | 0.7                              | D    | Full BNB dataset<br>( $1.11 \times 10^{21}$ POT); $\sigma =$<br>( $3.75 \pm 0.80$ ) $\times 10^{-38}$ $\text{cm}^2/\text{Ar}$       |
| MicroBooNE<br>NuMI $\nu_e$ [51] | Ar            | 0.73                             | D    | First $\nu_e + \bar{\nu}_e$ $\text{CC}\pi^+$ on Ar;<br>$\sigma = (0.93 \pm 0.28) \times$<br>$10^{-39}$ $\text{cm}^2/\text{nucleon}$ |

### 3 Beam and Neutrino Flux

#### 3.1 NuMI beam and flux composition

The NuMI beam is produced by 120 GeV protons from the Fermilab Main Injector striking a graphite target. Charged secondary hadrons are focused by two magnetic horns whose polarity selects the sign of the focused mesons. In Forward Horn Current (FHC) mode the horns focus positive pions and kaons, giving a beam with predominantly  $\nu_\mu$  content; in Reverse Horn Current (RHC) mode the polarity is reversed to focus negatives, giving a predominantly  $\bar{\nu}_\mu$  beam. MicroBooNE is located 135 mrad ( $7.71^\circ$ ) off-axis from the NuMI beam axis<sup>1</sup>, reducing the peak neutrino energy to the 0.5–2.5 GeV range and narrowing the energy spectrum. Because MicroBooNE sits off-axis and at comparatively low energy, the horn sign-selection is imperfect and both modes carry a sizeable wrong-sign component, most notably RHC, whose wrong-sign  $\nu_\mu$  fraction is large.

The flux composition at the MicroBooNE site is taken from the new-G4 central-value flux histograms, integrated over  $E_\nu > 60$  MeV to exclude an unphysical sub-threshold pile-up near  $E_\nu = 0$  (removed by the flux group’s low-energy peak removal; below the reaction threshold and irrelevant to the measurement). The muon-flavour split of these central-value histograms matches the author’s authoritative integrated fluxes exactly (Sec. 3.4):

<sup>1</sup>From the detector position in beam coordinates, (5502, 7259, 67270) cm: 91.1 m transverse at 672.7 m downstream. Earlier versions of this note quoted 110 mrad. See §3.2.

Table 3: NuMI flux composition at the MicroBooNE site by neutrino type, for the FHC and RHC horn modes (new-G4 central-value flux, integrated over  $E_\nu > 60$  MeV, the same threshold as the authoritative integrals of Table 5; above 0.1 GeV the RHC split would read 42.0/55.2%). FHC is  $\nu_\mu$ -dominant; RHC is  $\bar{\nu}_\mu$ -dominant with a large wrong-sign  $\nu_\mu$  fraction. The combined row weights each mode by its fluence,  $\Phi \times \text{POT}$  (FHC 6.03, RHC  $7.14 \times 10^{11}$  cm<sup>-2</sup>; Tables 4 and 5), giving a near-balanced  $\nu_\mu/\bar{\nu}_\mu$  mix. The signal is charge-blind ( $|\nu_{\text{pdg}}| = 14$ ), so both  $\nu_\mu$  and  $\bar{\nu}_\mu$  contribute in every mode.

| Horn mode | $\nu_\mu$ | $\bar{\nu}_\mu$ | $\nu_e$ | $\bar{\nu}_e$ |
|-----------|-----------|-----------------|---------|---------------|
| FHC       | 63.4%     | 34.0%           | 1.7%    | 0.9%          |
| RHC       | 44.2%     | 53.3%           | 1.4%    | 1.2%          |
| Combined  | 53.0%     | 44.4%           | 1.5%    | 1.1%          |

Flux corrections from the Package to Predict the FluX (PPFX) [8] are applied per event via the central-value weight stored in the analysis ntuples. The absolute flux normalisation of Sec. 3.4 uses the same new-G4 histograms.

### 3.2 Beam direction in detector coordinates

Two different angles describe the MicroBooNE–NuMI geometry, and they must not be confused (Fig. 2). The *off-axis angle* is a property of the beamline: the detector sits 91.1 m off the NuMI axis at 672.7 m downstream of the target, a baseline of 678.8 m and an off-axis angle of  $7.71^\circ$  (134.6 mrad).<sup>2</sup> That angle says nothing about how the TPC is oriented.

The angle that enters the measurement is a different one: the direction in which the neutrinos travel *relative to the detector  $z$  axis*. Averaged over interactions in the active volume, the true neutrino direction in detector coordinates is (0.4624, 0.0489, 0.8853), which lies  $28.6^\circ$  away from  $z$ . The detector  $z$  axis is therefore a poor measurement coordinate for the beam: it is not the direction the neutrinos came from, nor the NuMI beamline axis (itself  $23.2^\circ$  from  $z$ ).

<sup>2</sup>Computed from the detector position in beam coordinates, (5502, 7259, 67270) cm, taken from the same rotation matrix used for the NuMI beamline geometry weights.

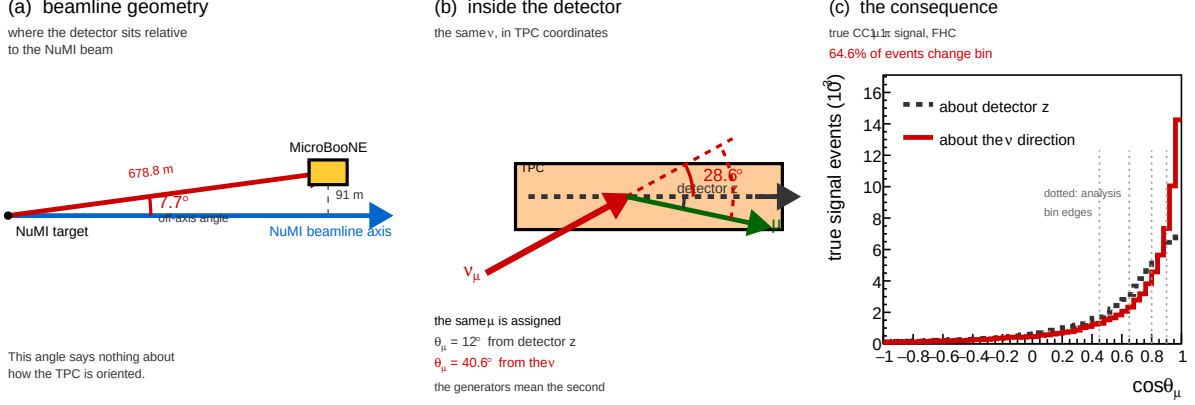


Figure 2: The NuMI geometry seen by MicroBooNE. (a) Beamline geometry, to scale: the detector sits 91.1 m off the NuMI axis at a baseline of 678.8 m, an off-axis angle of  $7.7^\circ$ . (b) The same neutrinos in TPC coordinates, to scale (the  $x$ - $z$  projection; the quoted  $28.6^\circ$  is the full three-dimensional angle): the beam arrives  $28.6^\circ$  away from the detector  $z$  axis, so one and the same muon is assigned  $\theta_\mu = 12^\circ$  about  $z$  but  $\theta_\mu = 40.6^\circ$  about the neutrino. The generators mean the second. (c) The resulting shift in the true  $\cos \theta_\mu$  distribution; dotted lines mark the analysis bin edges. Under the two conventions 64.6% of signal events fall in a different bin (44.6% for  $\cos \theta_\pi$ ).

This matters because every polar angle and every transverse-kinematic-imbalance variable is defined with respect to the neutrino direction. In the generator predictions the neutrino defines  $+z$  by construction — the GENIE event records used here have  $p_x^\nu = p_y^\nu = 0$  and  $p_z^\nu = E_\nu$  exactly — so a measurement that refers its angles to the detector  $z$  axis is not being compared with the prediction it appears to be compared with. The effect is not small: 64.6% of signal events change  $\cos \theta_\mu$  bin between the two conventions, and 44.6% change  $\cos \theta_\pi$  bin. For the TKI variables the axis error exceeds the observable itself, the mean  $\delta p_T$  being 0.742 GeV/ $c$  about  $z$  against 0.264 GeV/ $c$  about the beam.

Accordingly, all angular and TKI observables in this analysis are referred to the neutrino direction, following the  $\beta$  convention of the NuMI  $\nu_e$  CC1 $\pi$  internal note [1] and of the earlier NuMI  $\nu_e$  cross-section measurement it derives from [2]. At truth level the exact per-event neutrino direction is used, which reproduces the generator convention exactly. At reconstruction level the neutrino direction is approximated by the vector from the NuMI target — at  $(-31388, -3316, -60100)$  cm in detector coordinates, a 684 m baseline to the detector centre (678.8 m to the TPC origin quoted above) — to the reconstructed interaction vertex. Adopting the same  $\beta$  as the  $\nu_e$  analysis keeps the two NuMI measurements directly comparable.

The choice of reconstruction-level axis is in any case not a sensitive one. A single fixed axis and the per-event  $\beta$  differ by a median of 2.4 mrad (maximum 7.3 mrad), which moves only 0.42% of muons across a  $\cos \theta_\mu$  bin edge — to be compared with the 64.6% moved by the detector- $z$  convention. Against the true per-event direction the two perform almost identically, with a mean opening angle of  $1.95^\circ$  for  $\beta$  and  $2.00^\circ$  for a fixed axis. What neither can recover is the physical divergence of the beam: neutrinos reaching the detector come from decays distributed along the decay pipe, not from the target, and over a 684 m baseline a 2.5 m detector subtends only  $\sim 4$  mrad, so the vertex position carries almost no information about the direction. That residual is a resolution effect, absorbed by the response matrix like any other.

Quantities that are invariant under this choice of axis are unaffected:  $p_\mu$ ,  $p_\pi$ , the muon-pion opening angle  $\theta_{\mu\pi}$ , the invariant mass  $W_{\pi p}$ , and all integrated cross sections. The event selection is likewise unaffected, as no selection cut and no BDT input variable uses a polar angle.

### 3.3 Data exposure

The analysis is *blind*: the real beam-on data are not used. Each configuration is extracted on Poisson-thrown central-value fake data, generated and scaled *per run* to that run’s data POT (Table 4). The FHC measurement uses the full FHC exposure (Runs 1, 2, 4 and 5); RHC uses Runs 1, 2, 3, 4a, 4b and 4c; the combined measurement uses all fourteen overlay files. Every run is normalised independently to its own data exposure, so the prediction carries the data-POT mixture of run periods and the framework is ready for real per-run beam-on data (each fake-data sample is a drop-in replacement for its beam-on counterpart).

Table 4: Run-period exposure accounting (final, full FHC+RHC). “MC POT” is each period’s native simulated exposure; “Data POT” is the per-run exposure to which the (blind, per-run) fake data is scaled. *Each run is normalised independently* by its own factor,  $\text{Scale} = (\text{run Data POT})/(\text{run MC POT})$ , so the prediction is the data-POT-weighted mixture of run periods; the normalisation a real per-run beam-on dataset produces. This replaces an earlier single per-mode pooling factor: because the per-run MC/data POT ratios vary by up to a factor of three (Scale ranges 0.051–0.141 within FHC alone), a single factor would weight the runs by their MC statistics rather than by the data exposure. The RHC Run 4 row sums the 4a/4b/4c sub-files and Run 3 sums aa–ae; sub-files sharing a run’s data POT are scaled by the group MC total so they pool within that run. The combined measurement applies each run’s own factor. Run 5 uses the  $\nu_\mu$  overlay produced with reweighted PPFX flux weights (the only Run-5 overlay available). The fake data are thrown from the same per-run-scaled prediction, so the closure (Table 15) tests the bookkeeping of this scheme, not its correctness against data.

| Run period                | MC POT ( $\times 10^{20}$ ) | Data POT ( $\times 10^{20}$ ) | Scale          |
|---------------------------|-----------------------------|-------------------------------|----------------|
| Run 1 (FHC)               | 23.282                      | 3.283                         | 0.1410         |
| Run 2 (FHC)               | 24.934                      | 1.268                         | 0.0509         |
| Run 4 (FHC)               | 28.335                      | 2.075                         | 0.0732         |
| Run 5 (FHC)               | 19.300                      | 2.231                         | 0.1156         |
| <b>FHC total</b>          | <b>95.85</b>                | <b>8.857</b>                  | <i>per run</i> |
| Run 1 (RHC)               | 8.997                       | 0.605                         | 0.0673         |
| Run 2 (RHC)               | 57.865                      | 2.591                         | 0.0448         |
| Run 3 (RHC, aa–ae)        | 55.185                      | 5.003                         | 0.0907         |
| Run 4 (RHC, 4a–4c)        | 32.586                      | 2.883                         | 0.0885         |
| <b>RHC total</b>          | <b>154.63</b>               | <b>11.082</b>                 | <i>per run</i> |
| <b>Combined (FHC+RHC)</b> | <b>250.49</b>               | <b>19.939</b>                 | <i>per run</i> |

The full exposure substantially reduces the data-statistical term (Table 15 of the technical supplement–Table 17 of the technical supplement), leaving the measurement flux- and detector-systematic limited. When the analysis is unblinded, the real beam-on POT (and gate counts) replace the data-POT values above.

### 3.4 Integrated flux

The absolute flux normalisation uses the new-G4 GEANT4 NuMI flux integrated over  $E_\nu > 60$  MeV (the flux group’s low-energy peak removal, which removes an unphysical  $\sim 25$ – $30$  MeV artifact spike). The signal is charge-blind ( $|\nu_{\text{pdg}}| = 14$ , covering both  $\nu_\mu \rightarrow \mu^- \pi^+$  and  $\bar{\nu}_\mu \rightarrow \mu^+ \pi^-$ ), so the normalisation uses the combined  $\nu_\mu + \bar{\nu}_\mu$  flux  $\Phi = \Phi_{\nu_\mu} + \Phi_{\bar{\nu}_\mu}$ . Because RHC is  $\bar{\nu}_\mu$ -dominant with a different absolute flux than FHC, each configuration uses its *own* integrated flux:

Table 5: Integrated  $\nu_\mu$  and  $\bar{\nu}_\mu$  fluxes per POT used by the analysis (new-G4 flux,  $E_\nu > 60$  MeV), and the combined  $\nu_\mu + \bar{\nu}_\mu$  value used as the per-configuration normalisation. The combined row is the data-POT-weighted average of the two modes (weights 8.857 : 11.082; Table 4). Each configuration is normalised with its own flux: the RHC flux is 5.4% below the FHC one, so using a single value for both would bias the RHC cross section by that amount.

| Config   | $\Phi_{\nu_\mu}$<br>( $10^{-10} \text{ cm}^{-2} \text{ POT}^{-1}$ ) | $\Phi_{\bar{\nu}_\mu}$ | $\Phi_{\nu_\mu + \bar{\nu}_\mu}$ | $\bar{\nu}_\mu$ frac. |
|----------|---------------------------------------------------------------------|------------------------|----------------------------------|-----------------------|
| FHC      | 4.43515                                                             | 2.37644                | 6.81159                          | 34.9%                 |
| RHC      | 2.92348                                                             | 3.52298                | 6.44646                          | 54.6%                 |
| Combined | 3.59497                                                             | 3.01368                | 6.60865                          | 45.6%                 |

The mean energy of the  $\nu_\mu + \bar{\nu}_\mu$  flux above 60 MeV is 0.479 GeV (FHC) and 0.475 GeV (RHC) ( $\nu_\mu$  alone 0.468 and 0.544 GeV;  $\bar{\nu}_\mu$  alone 0.500 and 0.417 GeV, the RHC antineutrinos being the focused, softer component at this off-axis angle). Only 40% of that flux lies above 0.3 GeV, 26% above 0.5 GeV and 13% above 1 GeV. Because the flux-averaged cross section divides by the *whole* flux above 60 MeV while single pion production only opens near 0.3 GeV, the total is roughly a factor 2.5 smaller than it would be on the BNB spectrum peaked at 0.7 GeV; this, together with the measured phase space ( $p_\pi > 0.175$  GeV/ $c$ ,  $\theta_{\mu\pi} < 2.6$  rad), is why the  $\sim 1.0 \times 10^{-38} \text{ cm}^2/\text{Ar}$  found here is not to be compared directly with the  $3.75 \times 10^{-38} \text{ cm}^2/\text{Ar}$  of the BNB measurement of §2.3. These fluxes are set per configuration in the extraction. Because every simulated event carries the PPFX central-value flux correction (§4; mean 0.944 on selected signal), the integrated flux that divides the event rate must be the PPFX-corrected central value, not the raw GEANT4 flux; the central-value histograms of the new-G4 flux file are taken to be that central value, and this should be confirmed with the flux author before unblinding, since a raw-G4 integral would put every absolute cross section  $\sim 5\text{--}6\%$  low. **Open normalisation sign-off** (not required for the control-region opening, required before the signal region is opened): written confirmation by the NuMI flux contact that the integrated constants of Table 5 are PPFX-corrected central values consistent with the per-event PPFX central-value weighting. The integrated values were validated four independent ways: standalone NuWro and GENIE event generation on the same flux, the flux author’s numbers, and a GENIE closure.

The number of argon target nuclei in the fiducial volume is computed from first principles as

$$N_{\text{Ar}} = \frac{\rho_{\text{LAr}} \cdot V_{\text{FV}} \cdot N_A}{A_{\text{Ar}}} = 8.710 \times 10^{29}, \quad (2)$$

with  $\rho_{\text{LAr}} = 1.3836 \text{ g/cm}^3$ ,  $A_{\text{Ar}} = 39.948 \text{ g/mol}$  and  $N_A = 6.02214076 \times 10^{23} \text{ mol}^{-1}$ . The fiducial box is  $10 < x < 246$ ,  $-101 < y < 101$ ,  $10 < z < 986 \text{ cm}$  ( $4.6528 \times 10^7 \text{ cm}^3$ ), from which the  $675.1 < z < 775.1 \text{ cm}$  dead region is subtracted, leaving  $V_{\text{FV}} = 4.1761 \times 10^7 \text{ cm}^3$ . Without that subtraction the same expression would give  $9.705 \times 10^{29}$ . This is the value the code uses, computed by the extraction from the fiducial box with the dead slab removed, and it is the value entering the conversion factor of §9 and the one-bin total of §10.3. The per-configuration normalisation factor is  $N_{\text{norm}} = \Phi_{\text{mode}} \cdot N_{\text{POT,mode}} \cdot N_{\text{Ar}}$ , evaluated with each mode’s flux and data POT (Table 4).

## 4 Monte Carlo Samples

Monte Carlo events are generated with GENIE v3 [9, 20] (the MicroBooNE tune [16]) on argon-40, propagated through the detector with Geant4 [24], and overlaid with real cosmic-ray data (the “overlay” technique) so that simulated events experience in-situ detector noise and cosmogenic activity. Table 6 summarises the input samples.

Table 6: Input samples (final, full FHC+RHC). “ $n$ ” is the number of overlay files; “MC POT” is the summed native simulated exposure ( $\times 10^{20}$ ). The blind Poisson-thrown fake data stands in for the (unused) beam-on data; the detector variations use the native Run-4 FHC and RHC samples. The per-file lists and POT/trigger values are in the configuration tables of the release.

| Sample                  | Composition                                                          | $n$ | MC POT |
|-------------------------|----------------------------------------------------------------------|-----|--------|
| $\nu_\mu$ overlay (FHC) | Runs 1, 2, 4 (pandora) + Run 5 (reweighted-PPFX)                     | 4   | 95.85  |
| $\nu_\mu$ overlay (RHC) | Runs 1, 2, 3, 4a, 4b, 4c (pandora)                                   | 10  | 154.63 |
| Dirt MC                 | GENIE NuMI dirt overlays, per run (Run-1 file standing in for Run 2) | 7   | n/a    |
| EXT (beam-off)          | cosmic data, per run, horn-mode-independent                          | 7   | n/a    |
| Detector variations     | Run-4 FHC + Run-4 RHC, CV + 8 knobs each                             | 18  | n/a    |
| Fake data (blind)       | Poisson-thrown central value, per configuration                      | n/a | n/a    |
| Beam-on data            | blinded, not used until unblinding                                   | n/a | n/a    |

**Event weights.** Each MC event is weighted by

$$w = \text{Scale}[\text{run}] \times w_{\text{PPFX}} \times w_{\text{tune}}, \quad (3)$$

where  $\text{Scale}[\text{run}] = (\text{run Data POT})/(\text{run MC POT})$  is the *per-run* POT factor of Table 4: each run period is scaled independently to its own data exposure, so the prediction reproduces the data-POT mixture of runs rather than the MC-statistics mixture (see the caption of Table 4). Operationally this is exactly what the framework does natively each overlay file carries its run’s data POT via a matching per-run beam-on (here, per-run fake-data) sample, and is scaled by  $\text{run\_bnb\_pot}$  divided by that run’s MC exposure, taken as the sum of the *distinct* exposure values stored by the run’s files (the run total under both processed-file conventions; caption of Table 18).  $w_{\text{PPFX}}$  is the PPFX central-value flux correction and  $w_{\text{tune}}$  is the GENIE/GEANT4 reweighting correction. One weight rule is used everywhere:

$$w_{\text{safe}} = \begin{cases} 1, & w \text{ non-finite, } w < 0, \text{ or } w > 30, \\ w, & \text{otherwise (a weight of exactly zero is kept),} \end{cases} \quad (4)$$

applied to the product  $w = \text{Scale} \times w_{\text{PPFX}} \times w_{\text{tune}}$  (including the 0.65 normalisation weight the dirt overlay carries, which scales its generated exposure to the out-of-cryostat rate expected at the NuMI site). The universe construction, the covariance, the unfolding inputs, the cut-flow histograms, the count-level tables, the control-region yields, the transfer factors and the beam-on comparison at unblinding all use Eq. (4); one rule everywhere is what makes count-level and covariance-level numbers comparable to the per cent. At unblinding the per-run fake-data samples are replaced by the corresponding per-run beam-on files (same runs, same POT), so the normalisation is already correct.

**EXT scaling.** Beam-off data are cosmic-only, so a beam-off sample is matched to a beam-on run by *run period* alone, irrespective of the horn polarity it was recorded under. Each run period uses the beam-off sample(s) taken in that period, scaled by that run’s beam-on/beam-off gate ratio (times the 0.98 NuMI beam-occupancy factor); where a run has several beam-off files the framework sums their gate counts and pools them:

$$\text{EXT}_{\text{run}} = 0.98 \frac{\text{bnb\_gates}[\text{run}]}{\sum_{\text{files} \in \text{run}} \text{ext\_gates}} \sum_{\text{files} \in \text{run}} N_{\text{EXT}}^{\text{sel}}. \quad (5)$$

The gate counts are the analyser’s trigger table, released in `data.release/ext_gates.tsv`: Run 1 4 582 248.27, Run 3b 18 512 842.15 + 14 136 286.5 = 32 649 128.65, Run 4 4a+4b 9 338 778.225, 4c 8 060 024.70, 4d 17 432 345.70 (all four pooled under Run 4), Run 5 19 256 341.475; the beam-on gates per run are those of Table 4’s runs (9 846 635, 3 535 129, 4 131 149, 5 154 196 for FHC

Runs 1, 2, 4, 5 and 1 458 253, 5 422 907, 10 349 610, 6 304 167 for RHC Runs 1–4). No Run-2 beam-off sample exists and none will be produced, so the Run-1 sample, the nearest pre-CRT period, stands in for Run 2 permanently, scaled by its own gate count. The same beam-off files serve the FHC and RHC beam-on runs of a given period, and the combined configuration lists each (file, run) pair once. The selected beam-off rate per  $10^6$  gates is 1.75 (Run 1), 0.43 (Run 3b), 1.15 (Run 4) and 0.52 (Run 5): the pre-CRT Run-1 rate is 1.5–4 times the CRT-era ones (the rates carry 16–35% Poisson uncertainties, from 8, 14, 40 and 10 raw selected beam-off events), which is why period matching rather than a pooled sample is used. At the FHC final cut the beam-off background is 30 events (23 RHC, 53 combined), 1.9% of the selected sample. Figure 3 shows the per-period rates, scale factors and resulting yields; the machine-readable version is `data_release/ext_gates.tsv`, and the normalisation manifest (Sec. F of the technical supplement) asserts that every (beam-off file, run) pair enters each configuration exactly once, that every symlink resolves, and that every gate count is a gate count.

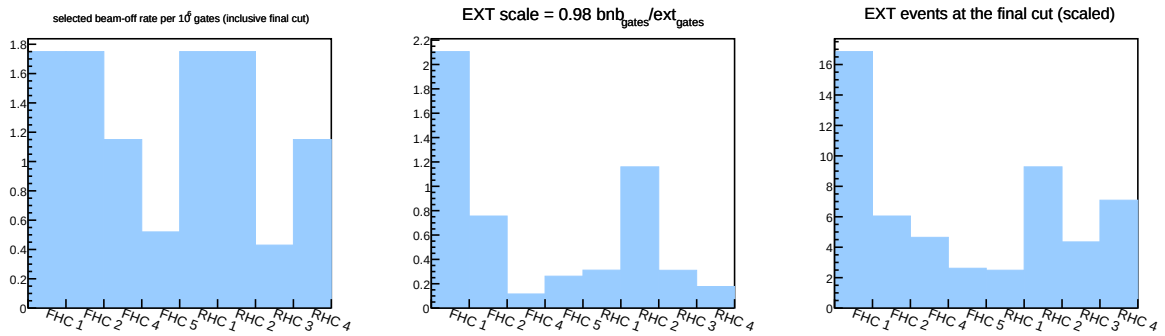


Figure 3: Beam-off validation by run period, from `data_release/ext_gates.tsv`: selected beam-off rate per  $10^6$  gates at the inclusive final cut (left), the per-run scale factor  $0.98 \text{ bnb\_gates}/\text{ext\_gates}$  (centre) and the resulting beam-off events at the final cut (right), for each beam-on run period of the FHC and RHC configurations. The Run-1 sample serves Run 2 of both modes as a stand-in (for RHC Run 1 it is the native period sample). The pre-CRT Run-1 rate is 1.5 to 4 times the CRT-era ones.

**No further cosmic-rejection cut.** Further cosmic-rejection cuts were not adopted because the beam-off contribution is 1.9% of the selected prediction and is removed by an unbiased data-driven subtraction, so a cut can only reduce the subtraction’s statistical uncertainty, about six events on a sample of 1612; every plausible cut loses more signal than that. The directional and calorimetric variants that were considered are in Sec. 8.2 of the technical supplement.

**Software trigger.** Data and EXT files are pre-filtered to require the software trigger; the CRT-era beam-off files (Runs 4–5) carry a vestigial trigger flag that is zero throughout, so for those files the pre-CRT trigger flag is used instead. MC events failing the software trigger are removed at the NeutrinoSlice cut stage. In practice this has no measurable effect at the current working point: all MC events passing the topological cut are verified to pass the software trigger (46 810/46 810 events confirmed numerically).

## 5 Signal Definition

The signal is defined at generator level and is used only to construct the selection efficiency and response matrix. It is *never* applied as a reconstruction cut. The definition is *charge-blind*: both  $\nu_\mu \rightarrow \mu^- \pi^+$  and  $\bar{\nu}_\mu \rightarrow \mu^+ \pi^-$  are treated as signal on the same footing.

### Neutrino flavour

$|\nu \text{ PDG}| = 14$  ( $\nu_\mu$  or  $\bar{\nu}_\mu$ ).

### Interaction type

Charged-current.

### True vertex

Inside the fiducial volume (Section 6).

### Primary multiplicity

Exactly one muon ( $|\text{PDG}| = 13$ ), exactly one charged pion ( $|\text{PDG}| = 211$ ) of *any momentum* among the primary final-state particles of the interaction (a second charged pion below threshold makes the event background, not signal; pions from the decay of primary hyperons are not primaries and are not counted), zero neutral pions ( $|\text{PDG}| = 111$ ), zero kaons ( $K^\pm$ ,  $K^0$ ,  $K_S$ ,  $K_L$ ) and no other mesons. Any number of protons (and neutrons) is allowed.

### Kinematic thresholds

$p_\mu > 0.15 \text{ GeV}/c$ ,  $p_\pi > 0.175 \text{ GeV}/c$ , and  $\theta_{\mu\pi} < 2.6 \text{ rad}$ . These define the *measured phase space*, chosen from the efficiency turn-on curves (Section 8). The momentum efficiency turns on sharply at  $\sim 0.15\text{--}0.20 \text{ GeV}/c$  (set by the 10 cm muon and 20 cm pion track-length cuts); below that it is only  $\sim 1\text{--}3\%$  with a strongly model-dependent correction. The opening-angle upper bound (2.6 rad) excludes the large-angle region, which is dominated by cosmic and dirt background ( $\sim 7\%$  purity in the  $\theta_{\mu\pi} > 2.513 \text{ rad}$  bin). No *lower* opening-angle cut is applied: the small-angle region is high-purity ( $\sim 58\%$ ), so cutting it would only remove signal (it is low-efficiency but not low-purity). The same cut is applied at reco level so the signal and selection acceptances match. The first  $p_\mu/p_\pi$  bins and the  $\theta_{\mu\pi}$  upper edge start/end at these thresholds.

(Earlier iterations used  $p_\mu, p_\pi > 0.1 \text{ GeV}/c$  with the full  $0\text{--}\pi$  angle; that left near-dead efficiency bins at low momentum and at the large-angle extreme. The phase space above raises the efficiency from 12.1% to 15.6% at  $\sim 55\%$  purity at the beam-off scaling then in use, see Section 8.)

**Pion threshold and the sibling  $\nu_e$  measurement.** The companion MicroBooNE NuMI  $\nu_e \text{ CC } 1\pi$  analysis defines its pion signal by kinetic energy,  $\text{KE}_\pi > 40 \text{ MeV}$ , equivalent to  $p_\pi > 0.113 \text{ GeV}/c$ . Our  $p_\pi > 0.175 \text{ GeV}/c$  ( $\text{KE}_\pi > 84 \text{ MeV}$ ) is therefore a higher threshold. Lowering it toward the  $\nu_e$  value is possible, the signal definition is a truth-level choice and the unfolding maps reco to truth but the pion momentum reconstruction (track range under the muon hypothesis, mass-corrected) degrades rapidly below  $0.175 \text{ GeV}/c$ . The table below is the study that led to *rejecting* a lower threshold; it is not part of the measurement. The measured phase space is  $p_\pi > 0.175 \text{ GeV}/c$  throughout this note, and no cross section is reported below it:

Table 7: Pion momentum reconstruction versus true  $p_\pi$  (correctly reconstructed signal pions). Below the  $0.175 \text{ GeV}/c$  threshold the bias and resolution blow up and the reconstructed fraction halves.

| True $p_\pi$ [GeV/c]           | Bias $\langle (p^{\text{reco}} - p^{\text{true}})/p^{\text{true}} \rangle$ | RMS resolution | Reco fraction |
|--------------------------------|----------------------------------------------------------------------------|----------------|---------------|
| 0.080–0.113                    | +131%                                                                      | 73%            | 3.4%          |
| 0.113–0.150                    | +66%                                                                       | 56%            | 3.4%          |
| 0.150–0.175                    | +30%                                                                       | 51%            | 4.3%          |
| <b>0.175–0.250</b> (threshold) | <b>−8%</b>                                                                 | <b>31%</b>     | <b>10.0%</b>  |
| 0.250–0.400                    | −30%                                                                       | 21%            | 10.4%         |

At the threshold the resolution is 31%; in the  $[0.113, 0.175]$  GeV/ $c$  region it is 51–56% with a +30–66% bias and only  $\sim 3$ –4% reconstructability. Extending the measurement to the  $\nu_e$  threshold would thus add bins carrying large, strongly model-dependent unfolding systematics. For consistency with the  $\nu_e$  measurement the threshold could be lowered to 0.113 GeV/ $c$  with the  $[0.113, 0.175]$  GeV/ $c$  region kept in a single coarse bin to contain those systematics, or both phase spaces reported; this is a systematics-versus-cross-analysis-consistency trade rather than a reconstruction limitation.

**Why the threshold was not lowered.** Extending to the sibling  $\nu_e$  threshold was investigated and rejected: the  $[0.113, 0.175]$  GeV/ $c$  region has 2.2% selection efficiency and a strongly model-dependent reconstruction correction. The study is recorded in the technical supplement. It is not part of this measurement, and no cross section is reported below  $p_\pi > 0.175$  GeV/ $c$ .

## 6 Fiducial Volume

The fiducial volume (FV) is:

$$x \in [10.0, 246.0] \text{ cm}, \quad y \in [-101.0, +101.0] \text{ cm}, \quad z \in [10.0, 986.0] \text{ cm}, \quad (6)$$

with the dead-wire region  $z \in [675.1, 775.1]$  cm excluded. The total FV volume is

$$\begin{aligned} V_{\text{FV}} &= (246 - 10) \times 202 \times (976 - 100) \text{ cm}^3 \\ &= 236 \times 202 \times 876 \text{ cm}^3 \approx 4.17 \times 10^7 \text{ cm}^3. \end{aligned} \quad (7)$$

A wider *containment* region (2 cm inset from the active-volume faces, except the top face, which is inset by 10 cm) is used to require that the pion candidate track endpoint is contained:

$$x \in [2.0, 254.35] \text{ cm}, \quad y \in [-113.53, +107.47] \text{ cm}, \quad z \in [2.1, 1034.9] \text{ cm}. \quad (8)$$

### 6.1 Which volume applies where

Three distinct volumes appear above and it is worth stating plainly which is used for what, since “contained” and “in the fiducial volume” are not the same requirement.

| volume        | extent                                                                                | used for                              |
|---------------|---------------------------------------------------------------------------------------|---------------------------------------|
| fiducial (FV) | $[10, 246] \times [-101, 101] \times [10, 986]$ cm<br>minus $z \in [675.1, 775.1]$ cm | neutrino vertex, true <i>and</i> reco |
| containment   | $[2, 254.35] \times [-113.53, 107.47] \times [2.1, 1034.9]$ cm                        | track endpoints                       |
| active TPC    | $[0, 256.35] \times [-115.5, 117.5] \times [0, 1036.9]$ cm                            | (reference only)                      |

**True and reco use identical dimensions.** The framework allows the true and reconstructed fiducial volumes to differ, and other selections use that freedom. Here they are defined as the same box. The signal definition tests the *true* neutrino vertex and the selection tests the *reconstructed* vertex, both through the same helper, so both inherit the dead-region veto as well. The truth and reconstructed boundaries are geometrically identical, so there is no *intentional* acceptance mismatch. Vertex-resolution and space-charge effects can nevertheless move a reconstructed vertex across these boundaries, and events that migrate across the FV or dead-region edge are represented in the simulated response (true-in, reco-out as inefficiency; true-out, reco-in as the out-of-FV background category of Table 10) rather than being absent.

**The dead-region veto applies to the vertex only.** An event whose vertex lies outside the unresponsive-wire slab is kept even if one of its tracks crosses that slab. This is the conventional choice, but it is an assumption rather than a necessity: tracks crossing dead wires have degraded calorimetry and range reconstruction, and the associated inefficiency is absorbed into the response matrix rather than removed by a cut.

**The dead slab is also removed from the target count.** The argon target number is computed from the FV with the slab subtracted, not from the full box. Retaining the slab would inflate the target count by  $100/876 \approx 11\%$  and bias every cross section low by the same factor. Note that this subtraction is applied on the NuMI code path specifically; the generic path computes the target number without it, so the two must not be mixed.

**Relation to the active volume.** The FV is inset from the active TPC by approximately 10 cm in  $x$ , 15 cm in  $y$ , 10 cm at the upstream  $z$  face and 51 cm at the downstream face. The containment box, by contrast, is inset by only 2 cm (10 cm at the top): track endpoints are required to be inside the detector, not inside the fiducial region, which is the correct requirement for range-based momentum and for the pion-containment cut.

## 7 Event Selection

The event selection is implemented as a selection class of the extraction framework, which produces the response matrices and spectra for all results in this note; a standalone re-implementation, kept as an independent cross-check, applies the identical cut sequence. The selection comprises ten sequential cuts (Cut 0 is the unselected reference sample).

### 7.1 BDT-based particle identification

Three gradient-boosted decision trees (TMVA [26]), trained on the MicroBooNE overlay MC, supply the calorimetric particle identification used in the muon- and pion-candidate cuts. All three share a common set of per-track inputs: the three Bragg-peak  $dE/dx$  likelihoods under the proton, muon and MIP hypotheses, the log-likelihood-ratio PID score, the Pandora track-vs-shower score, and the space-charge-corrected track end-point coordinates:

- **MIP BDT:** separates minimum-ionising tracks (muons, pions) from stopping protons. Both the muon and the pion candidate are required to have a MIP score above  $-0.1$  (Cuts 4–5).
- **Pion BDT:** a dedicated pion-versus-proton discriminant applied to the pion candidate, pion score above  $-0.1$  (Cut 5).
- **Muon BDT:** the same inputs plus the track length. It applies no threshold: the set of *qualifying* muon-candidate tracks is defined by the cuts of Table 8, and the muon BDT only *orders* that set; the qualifying track with the highest muon-BDT score is chosen as the muon (rather than the longest qualifying track, which the BDT ordering supersedes).

The two candidate-PID BDTs deliberately omit track length so the score does not bias the momentum spectra. A fourth score, a cosmic-rejection BDT, is available for cosmic rejection but is disabled by default (no benefit in an A/B test; the Pandora topological score already provides the cosmic rejection at Cut 3).

### 7.2 Cut definitions

**Cut 0. EventsInTrueFV.** All events. Reference denominator for truth-level studies.

- Cut 1. NeutrinoSlice.**  $\geq 1$  reconstructed track, the software trigger, and a cosmic-BDT requirement that is disabled (no benefit found in an A/B test).
- Cut 2. InFiducialVol.** The SCE-corrected reconstructed neutrino vertex lies inside the FV.
- Cut 3. Topological.** Topological score  $> 0.67$ . This Pandora-produced multivariate score distinguishes cosmic-ray topologies from neutrino-like topologies and provides the primary cosmic rejection: EXT events are reduced by a factor of  $\sim 28$ – $29$  between Cut 2 and Cut 3 (Table 10).
- Cut 4. MuonCandidate.** At least one primary PFP track satisfies the muon PID criteria of Table 8. Among the qualifying tracks, the one with the highest muon-BDT score (§7.1) is the muon candidate; track length enters only through the 10 cm qualifying cut and as a BDT input, not as the ordering.
- Cut 5. ContainedPion.** Exactly one primary track (other than the muon candidate) satisfies the pion PID criteria of Table 9 and has its endpoint inside the containment volume. The track with the highest LLR PID score is the pion candidate.
- Cut 6. MuonIn3Planes.** The muon candidate has  $\geq 1$  hit on each of the U, V, Y planes.
- Cut 7. PionIn3Planes.** The pion candidate has  $\geq 1$  hit on each of the U, V, Y planes.
- Cut 8. ShowerCut.** Zero primary-level EM showers, unless exactly one shower is present with fewer than 50 Y-plane hits and a vertex distance  $> 10$  cm (consistent with a delta-ray or low-energy nuclear photon).
- Cut 9. OpeningAngle.** The reconstructed  $\mu$ – $\pi$  opening angle satisfies  $\theta_{\mu\pi} < 2.6$  rad ( $\approx 149^\circ$ ), rejecting the large-angle cosmic/dirt background. No lower bound is applied (the small-angle region is high-purity). The same cut is imposed on the signal definition (Section 5).
- Cut 10.**  
**Nonprotons** (final cut). Fewer than 4 primary tracks have LLR PID  $> 0.1$  (non-proton-like), and the total number of primary reconstructed tracks is less than 5.

Table 8: Muon candidate PID thresholds (applied at Cut 4).

| Variable        | Threshold     | Description               |
|-----------------|---------------|---------------------------|
| Track score     | $> 0.80$      | Track-like PFP            |
| Vertex distance | $\leq 4.0$ cm | Attached to $\nu$ vertex  |
| Track length    | $> 10.0$ cm   | Minimum track length      |
| LLR PID         | $> 0.20$      | Muon-like                 |
| MIP BDT         | $\geq -0.10$  | Minimum-ionising-particle |

Table 9: Pion candidate PID thresholds (applied at Cut 5). Endpoint containment requires the track endpoint inside the containment volume. The Bragg-pion score is the pion-hypothesis Bragg-peak likelihood score (rejects stopping protons).

| Variable           | Threshold     | Description               |
|--------------------|---------------|---------------------------|
| Track score        | $\geq 0.50$   | Track-like PFP            |
| LLR PID            | $> 0.10$      | Pion / non-proton-like    |
| Vertex distance    | $\leq 4.0$ cm | Attached to $\nu$ vertex  |
| Track length       | $> 20.0$ cm   | Minimum track length      |
| MIP BDT            | $> -0.10$     | MIP-like                  |
| Pion BDT           | $> -0.10$     | Pion-like (vs. proton)    |
| Bragg-pion score   | $\geq 0.08$   | Proton rejection          |
| Endpoint contained | yes           | Inside containment volume |

### 7.3 Kinematic reconstruction

**Muon momentum.** A *hybrid* estimator, switched on containment of the muon-candidate track: the range-based momentum is used when the track ends inside the detector, and the multiple-Coulomb-scattering momentum when it exits (range is meaningless for an exiting track). The reported  $p_\mu$  is the hybrid, not either estimator alone. In the selected FHC sample the split is 37% contained (range) and 63% exiting (MCS [14]), i.e. the measurement is *MCS-dominated*: the NuMI muons are energetic and the MicroBooNE TPC is small, so most of them leave it. This matters for the  $p_\mu$  resolution and for the detector systematic, since the range and MCS estimators carry different uncertainties. The eight detector-variation knobs do not vary the multiple-Coulomb-scattering momentum scale or resolution directly, and no dedicated MCS systematic is assigned; the MCS estimator’s  $-10\%$  bias and  $30\%$  resolution (Fig. 20) enter through the response matrix only. A dedicated term is listed in §14.

**Pion momentum.** The starting estimator is the *muon-hypothesis* range momentum. Before a reconstructed  $p_\pi$  bin is assigned it is converted to the pion hypothesis by preserving the inferred kinetic energy,  $T = \sqrt{p_{\mu\text{-hyp}}^2 + m_\mu^2} - m_\mu$  and  $p_\pi = \sqrt{(T + m_\pi)^2 - m_\pi^2}$ ; the stored branch is not modified, the conversion is written into the reconstructed-bin definition of the bin configuration (Sec. 1.1 of the technical supplement, where it raises the lowest-bin reconstructed-to-true ratio from 0.888 to 0.976). The muon mass (105.7 MeV) is close to the pion mass (139.6 MeV), so the range-to-momentum conversion is a good proxy, and, crucially, the result is a *momentum*, directly comparable to the true  $p_\pi$  it is binned against. The proton-hypothesis range *kinetic energy* is not used: it is a different physical quantity from the momentum on the true axis, so binning it with the same edges would shift the migration matrix systematically rather than leave it near-diagonal (a true 0.2 GeV/ $c$  pion gives  $KE \approx 0.104$  GeV, two bins low, and a true 0.175 GeV/ $c$  pion falls below the first reco bin entirely). Range momentum is meaningful only for contained tracks, which the containment requirement in the selection guarantees.

**Angles.** Polar angles are defined with respect to the *neutrino direction*  $\hat{u}_\nu$ , not the detector  $z$ -axis:  $\cos\theta_\mu = \hat{u}_\mu \cdot \hat{u}_\nu$ ,  $\cos\theta_\pi = \hat{u}_\pi \cdot \hat{u}_\nu$ . The two are far from equivalent — the NuMI beam reaches MicroBooNE  $28.6^\circ$  away from  $z$ , and referring the angles to  $z$  would place 64.6% of signal events in a different  $\cos\theta_\mu$  bin than the generator predictions they are compared with (Sec. 3.2). For reconstructed quantities  $\hat{u}_\nu$  is the  $\beta$  direction of Refs. [1, 2], from the NuMI target to the reconstructed vertex; at truth level it is the exact per-event neutrino direction. The opening angle  $\theta_{\mu\pi} = \arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$  is computed from reconstructed track directions and, being an angle between two measured tracks, is independent of this choice.

## 7.4 Proton-tagged selection

A second selection, the *proton-tagged* one, is applied to the same events. It is used for the secondary-scope results of §11: with an identified proton the full visible hadronic system is reconstructed, which opens the hadronic invariant mass and the transverse kinematic imbalance between the muon and the hadronic system. It is defined here, next to the inclusive selection it extends, so that the two can be read together.

**Definition.** The proton-tagged selection applies the ten inclusive cuts of §7.2 unchanged and adds an eleventh, the leading-proton identification (“IdentProton”), which defines the subsample:

| Requirement     | Threshold                    | Description                                                         |
|-----------------|------------------------------|---------------------------------------------------------------------|
| Candidate set   | —                            | track-like primaries other than the muon and pion candidates        |
| Proton likeness | LLR PID < 0.05               | the most proton-like of the set is the candidate                    |
| Containment     | end point in FV              | range-based momentum is meaningful only if contained                |
| Momentum        | $p_p \geq 0.3 \text{ GeV}/c$ | tracking threshold; below it protons are not reconstructed reliably |

The proton momentum is taken from the proton-hypothesis calorimetric kinetic energy,  $p_p = \sqrt{T_p^2 + 2m_p T_p}$ ; truth-matched on the FHC overlay it carries a  $-9\%$  bias (underestimate) and a  $21\%$  resolution, both folded into the response matrix and corrected by the unfolding. The signal definition is correspondingly narrowed to the inclusive signal of §5 plus at least one true proton above  $0.3 \text{ GeV}/c$ ; the muon and pion phase space is unchanged.

**Performance.** At full exposure the proton tag yields a selection efficiency of  $11.4\%$  and a purity of  $52\%$  in FHC ( $50\%$  RHC,  $51\%$  combined), against  $17.5\%$  and  $\sim 59\%$  for the inclusive selection. Relative to the inclusive final cut evaluated under the proton-tagged signal definition, the tag rejects  $67\%$  of the neutrino background for a  $32\%$  signal loss, lifting the purity from  $34\%$  to  $52\%$  and improving  $S/\sqrt{B}$  by  $\sim 20\%$  ( $17.1 \rightarrow 20.5$ , FHC). Truth-matched,  $87\%$  of tagged events contain a true proton above threshold; the  $\sim 13\%$  mis-tag is dominated by a charged pion or other track taken for the proton, which is why the proton-tagged results need a control region of their own before they are promoted (Sec. D.6 of the technical supplement). Its clearest benefit is cosmic rejection: requiring a contained proton above  $0.3 \text{ GeV}/c$  removes beam-off background by a factor  $3.3\text{--}3.6$  at the final cut and by up to a factor of six in the CC0 $\pi$  control region (Sec. D.7 of the technical supplement). The full-exposure cut-flow, stage by stage and per category, is Table 6 of the technical supplement (stacks in Fig. 5 of the technical supplement); it uses the same per-run exposure scaling as the inclusive cut-flow of Table 10, and its final row is the proton-tagged cut-and-count entry of Table 19.

## 8 Cut-Flow and Selection Performance

Figure 4 shows the full-exposure cut-flow yields for all three configurations as stacked plots, and Table 10 gives the detailed per-category, per-cut breakdown for each.

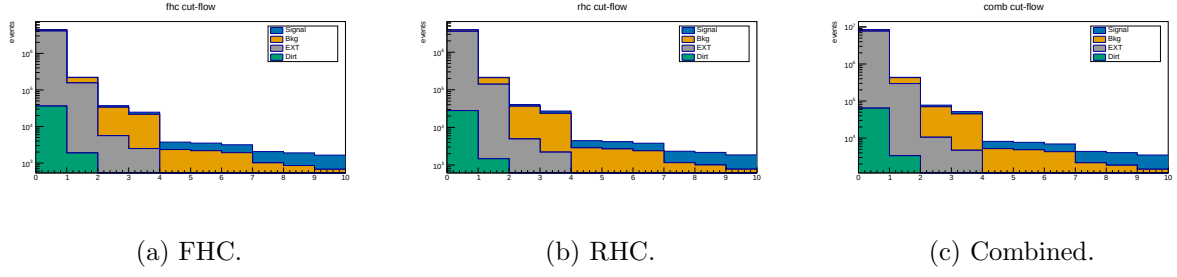


Figure 4: Cut-flow yields at *full exposure* for FHC, RHC and combined (log scale; MC-prediction stack: signal blue, background orange, EXT grey, dirt green; colourblind-safe Okabe-Ito), signal and background POT-scaled per mode. Data is blind, so this shows the MC-prediction stack; the detailed per-category breakdown is in Table 10.

Table 10: Cut-flow yields at *full exposure* for FHC, RHC and combined (MC prediction; data is blind: the fake-data *expectation* equals the central-value prediction by construction, while the realised pseudo-data fluctuate according to the Poisson throw). The histograms behind this table carry the framework’s full central-value weight,  $\text{tune} \times \text{PPFX} \times \text{normalisation}$  (§4, with the safe-weight rule of Eq. (4)), the same weight as the fake-data throw and the unfolding, so the final-cut yields agree with the cut-and-count Table 19 to  $< 1\%$ . Signal and “Bkg” (all non-signal MC: OOFV, NC,  $\text{CC}1\pi^0$ ,  $\text{CC}0\pi$ ,  $\text{CCN}\pi$ , CCK, CCothers) are from the  $\nu_\mu$  overlays, POT-scaled *per run* (each run by its own  $D_{\text{run}}/\text{MC}_{\text{run}}$ , Table 4); EXT (beam-off cosmic) is scaled by the (per-run summed) per-run beam-on/beam-off *gate* ratio times the 0.98 NuMI beam-occupancy factor, each run period using the beam-off sample taken in that period (§4); dirt from the GENIE NuMI dirt overlay. Pred. = Signal + Bkg + EXT + Dirt. Eff. is the cumulative signal efficiency relative to the generated signal (the None stage, where all true signal is counted); Pur. = Signal/Pred.

| Cut                                                                     | Signal       | Bkg MC       | EXT       | Dirt       | Pred.        | Eff. [%]    | Pur. [%]    |
|-------------------------------------------------------------------------|--------------|--------------|-----------|------------|--------------|-------------|-------------|
| <i>FHC (Runs 1,2,4,5; <math>8.857 \times 10^{20}</math> POT)</i>        |              |              |           |            |              |             |             |
| None                                                                    | 5 417        | 268 316      | 4 049 292 | 33 638     | 4 356 662    | 100.0       | 0.1         |
| InFiducialVol                                                           | 4 504        | 60 235       | 153 533   | 1 763      | 220 036      | 83.1        | 2.0         |
| Topological                                                             | 3 407        | 27 640       | 5 409     | 192        | 36 647       | 62.9        | 9.3         |
| MuonCandidate                                                           | 3 056        | 18 786       | 2 363     | 129        | 24 333       | 56.4        | 12.6        |
| ContainedPion                                                           | 1 358        | 2 080        | 223       | 24         | 3 685        | 25.1        | 36.8        |
| MuonIn3Planes                                                           | 1 299        | 1 964        | 185       | 16         | 3 464        | 24.0        | 37.5        |
| PionIn3Planes                                                           | 1 204        | 1 759        | 144       | 6.8        | 3 114        | 22.2        | 38.7        |
| ShowerCut                                                               | 1 016        | 891          | 108       | 5.8        | 2 021        | 18.8        | 50.3        |
| OpeningAngle                                                            | 1 009        | 807          | 41        | 1.9        | 1 859        | 18.6        | 54.3        |
| <b>Nonprotons</b>                                                       | <b>943</b>   | <b>637</b>   | <b>30</b> | <b>1.3</b> | <b>1 612</b> | <b>17.4</b> | <b>58.5</b> |
| <i>RHC (Runs 1,2,3,4a,4b,4c; <math>1.108 \times 10^{21}</math> POT)</i> |              |              |           |            |              |             |             |
| None                                                                    | 5 778        | 286 540      | 3 560 975 | 26 089     | 3 879 381    | 100.0       | 0.1         |
| InFiducialVol                                                           | 4 828        | 65 474       | 139 122   | 1 367      | 210 791      | 83.6        | 2.3         |
| Topological                                                             | 3 743        | 30 446       | 4 767     | 149        | 39 104       | 64.8        | 9.6         |
| MuonCandidate                                                           | 3 363        | 20 709       | 2 090     | 100        | 26 262       | 58.2        | 12.8        |
| ContainedPion                                                           | 1 488        | 2 576        | 191       | 19         | 4 274        | 25.7        | 34.8        |
| MuonIn3Planes                                                           | 1 426        | 2 430        | 158       | 13         | 4 028        | 24.7        | 35.4        |
| PionIn3Planes                                                           | 1 329        | 2 194        | 121       | 5.3        | 3 650        | 23.0        | 36.4        |
| ShowerCut                                                               | 1 101        | 1 026        | 86        | 4.5        | 2 218        | 19.1        | 49.7        |
| OpeningAngle                                                            | 1 093        | 933          | 33        | 1.5        | 2 060        | 18.9        | 53.1        |
| <b>Nonprotons</b>                                                       | <b>1 015</b> | <b>720</b>   | <b>23</b> | <b>1.0</b> | <b>1 759</b> | <b>17.6</b> | <b>57.7</b> |
| <i>Combined FHC+RHC (<math>1.994 \times 10^{21}</math> POT)</i>         |              |              |           |            |              |             |             |
| None                                                                    | 11 195       | 554 856      | 7 610 266 | 59 727     | 8 236 044    | 100.0       | 0.1         |
| InFiducialVol                                                           | 9 332        | 125 709      | 292 655   | 3 130      | 430 826      | 83.4        | 2.2         |
| Topological                                                             | 7 149        | 58 086       | 10 176    | 340        | 75 752       | 63.9        | 9.4         |
| MuonCandidate                                                           | 6 419        | 39 495       | 4 453     | 228        | 50 595       | 57.3        | 12.7        |
| ContainedPion                                                           | 2 845        | 4 656        | 414       | 43         | 7 958        | 25.4        | 35.8        |
| MuonIn3Planes                                                           | 2 725        | 4 394        | 343       | 29         | 7 492        | 24.3        | 36.4        |
| PionIn3Planes                                                           | 2 533        | 3 954        | 266       | 12         | 6 764        | 22.6        | 37.4        |
| ShowerCut                                                               | 2 117        | 1 917        | 194       | 10         | 4 238        | 18.9        | 50.0        |
| OpeningAngle                                                            | 2 102        | 1 740        | 74        | 3.3        | 3 919        | 18.8        | 53.6        |
| <b>Nonprotons</b>                                                       | <b>1 959</b> | <b>1 357</b> | <b>53</b> | <b>2.3</b> | <b>3 372</b> | <b>17.5</b> | <b>58.1</b> |

**Why the phase space and the opening-angle cut are what they are.** The thresholds were chosen from the efficiency turn-ons and purity studies on the Run-1 FHC sample (Sec. 8.1 of the technical supplement); the measured phase space removes the near-zero-efficiency momentum regions and the low-purity large-opening-angle region. On the full exposure the events the  $\theta_{\mu\pi} < 2.6$  rad cut removes are exactly those of the cosmic control region (Table 21): in FHC

6.0 signal, 69 neutrino background, 53 beam-off and 2.2 dirt, a signal purity of 4.6%. Dropping the cut would lower the signal-region purity from 58.5% to 54.5%, raise its beam-off content from 30 to 83 events and recover only 2.7% of the generated signal. No lower opening-angle cut is used because the small-angle region is high-purity ( $\sim 58\%$ ), so cutting it lowers the selected signal without improving purity. The per-run exposure scaling of the yields differs from a single per-mode factor by about 2% in the final-cut signal, a nearly uniform re-weighting that moves efficiency and purity by  $< 0.5\%$ .

## 8.1 Key N–1 distributions

Figure 5 shows the N–1 distributions (all cuts applied except the one under study) for the two cuts instrumented per event; the topological score (Cut 3) and the  $\mu$ – $\pi$  opening angle (Cut 9), for all three configurations. The shapes are consistent across horn modes, as expected since the selection is identical; RHC and combined simply carry the larger exposure. Signal (blue) and background (orange) are stacked and the dashed line marks the applied threshold. The remaining cut variables are shown at the final cut in Sec. 8.3.

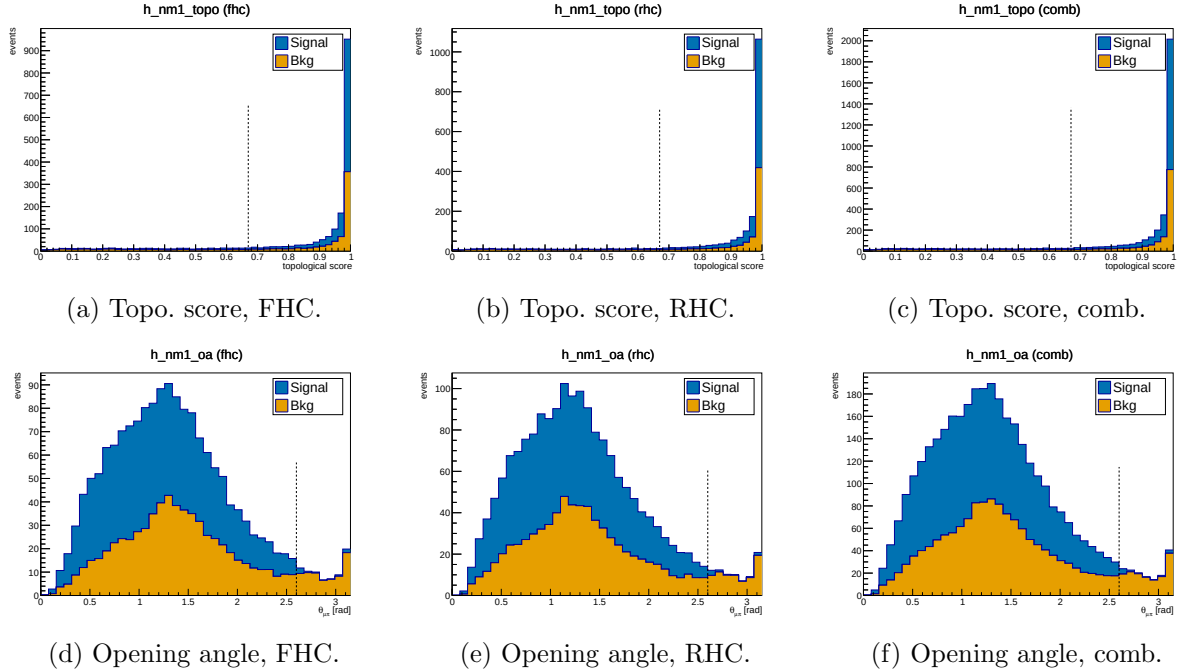
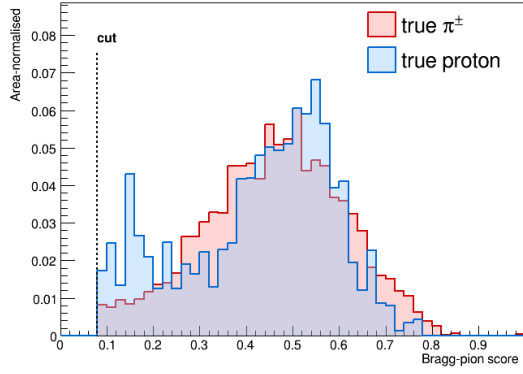


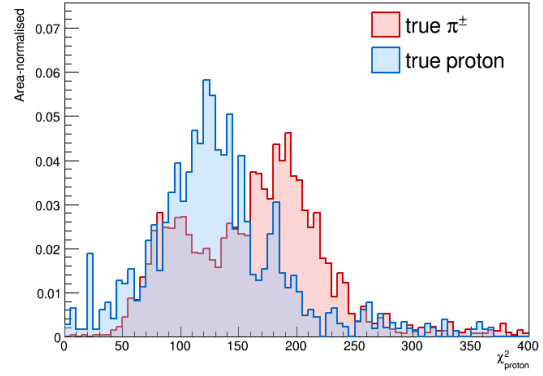
Figure 5: N–1 distributions for the topological score (Cut 3, top) and  $\mu$ – $\pi$  opening angle (Cut 9, bottom), for FHC, RHC and combined. Signal (blue) and background (orange) stacked; dashed line = applied threshold. Colourblind-safe Okabe-Ito palette.

## 8.2 Pion proton-rejection variables

Figure 6 shows the two main proton-rejection variables on the pion candidate, evaluated on MC truth-labelled pions ( $\pi^\pm$ ) and protons at the final cut stage, used to tune the working-point thresholds.



(a) Bragg-pion score. Working-point threshold:  $\geq 0.08$ .

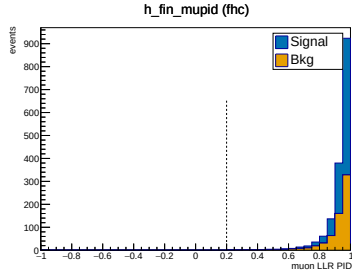


(b)  $\chi^2$  proton PID.

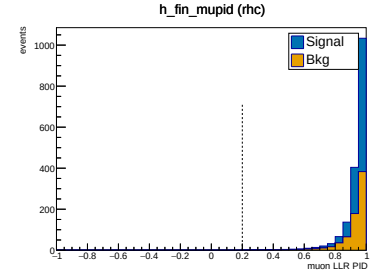
Figure 6: Pion-candidate PID distributions for true  $\pi^\pm$  (red) and true protons (blue) at the final cut stage. The Bragg-pion score provides the primary proton rejection;  $\chi_p^2$  is studied as cross-check.

### 8.3 Final-cut data/MC distributions

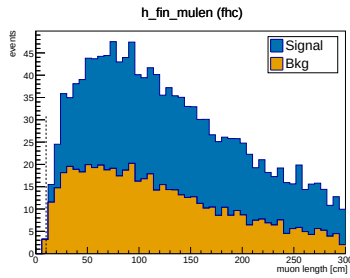
Figures 7 and 8 show the candidate-level distributions at the final selection cut for the four instrumented PID and track-length variables, for FHC and RHC. Signal (blue) and background (orange) are stacked (colourblind-safe Okabe-Ito palette).



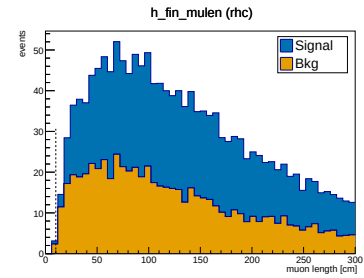
(a) Muon LLR PID, FHC.



(b) Muon LLR PID, RHC.

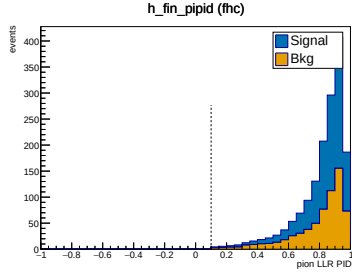


(c) Muon length, FHC.

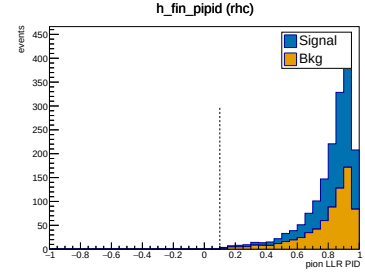


(d) Muon length, RHC.

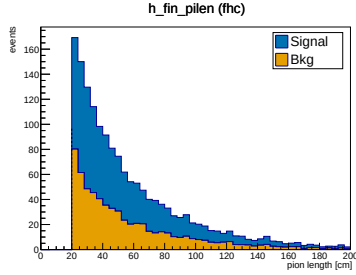
Figure 7: Muon-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC and RHC (the combined configuration is consistent with both and is not shown). Signal (blue) / background (orange) stacked.



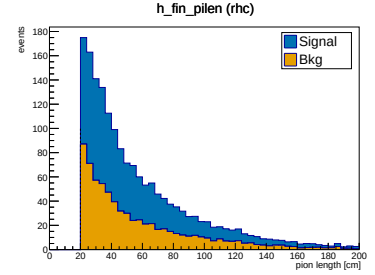
(a) Pion LLR PID, FHC.



(b) Pion LLR PID, RHC.

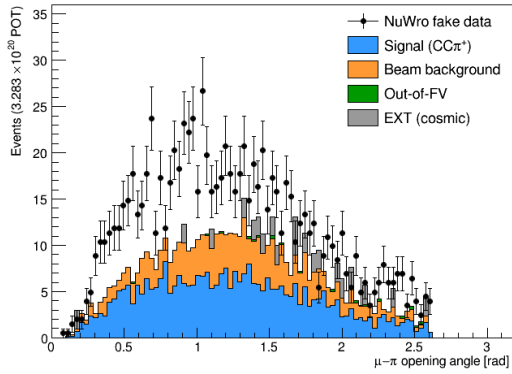


(c) Pion length, FHC.

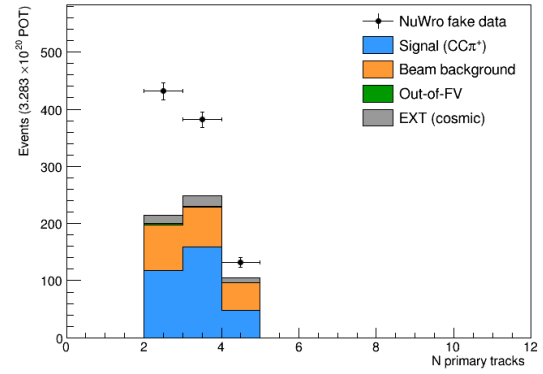


(d) Pion length, RHC.

Figure 8: Pion-candidate LLR PID (top) and track length (bottom) at the final cut, for FHC and RHC (the combined configuration is consistent with both and is not shown). Signal (blue) / background (orange) stacked.



(a)  $\mu$ - $\pi$  opening angle.



(b) Number of primary tracks.

Figure 9: Fake-data/MC comparison for topological variables at the final cut.

## 8.4 Background decomposition

Table 11 shows the truth-level background breakdown at the final cut.

Table 11: Background composition at the final cut, FHC full exposure (truth classes of the selection’s counters, CV-weighted and per-run POT-scaled; the same classes and yields as the signal-region column of Table 23 of the technical supplement). Fractions are of the 637 neutrino-background events; beam-off and dirt are listed for scale.

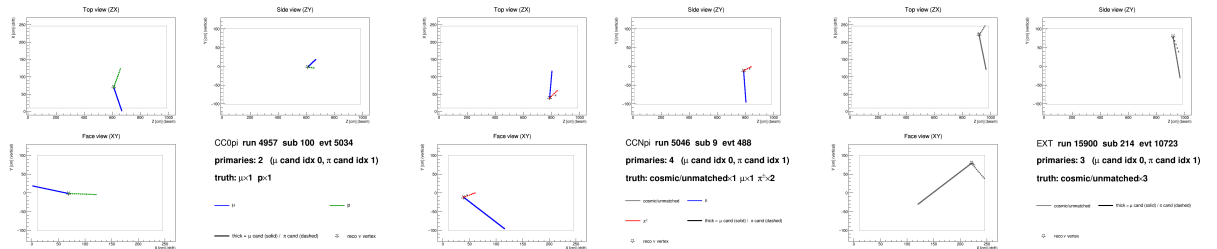
| Category                                                 | Events | Fraction |
|----------------------------------------------------------|--------|----------|
| $\nu_\mu$ CC $0\pi$ (proton or other track as pion)      | 244    | 38%      |
| $\nu_\mu$ CC $\geq 2\pi^\pm$ , no $\pi^0$                | 145    | 23%      |
| NC and $\nu_e$ CC                                        | 89.5   | 14%      |
| $\nu_\mu$ CC with a $\pi^0$                              | 58.0   | 9%       |
| $\nu_\mu$ CC $1\pi^\pm$ outside the measured phase space | 52.7   | 8%       |
| Out of fiducial volume                                   | 47.4   | 7%       |
| EXT (cosmic)                                             | 30     | n/a      |
| Dirt                                                     | 1.3    | n/a      |

The dominant background (38%) is CC $0\pi$  interactions in which a proton is reconstructed as the pion candidate. This mis-identification was verified directly by 2D event displays (truth-labelled by backtracked PDG): the pion candidate track shows a green (proton) truth colour with a stopping Bragg peak, while the genuine muon track is correctly identified.

At the current working point (Bragg-pion score  $\geq 0.08$ ), the pion candidate is truth-matched to a proton in  $\approx 18\%$  of cases. Tighter Bragg-score thresholds (0.20–0.30) were studied but shown not to improve the figure of merit: the pion-to-proton ratio in CC $\pi^\pm$  signal is  $\approx 4 : 1$ , so rejecting protons removes genuine pions at nearly the same rate.

## 8.5 Event displays

Figure 10 shows representative 2D truth-labelled event displays for the two dominant backgrounds and the EXT cosmic sample. Tracks are coloured by backtracked truth PDG:  $\mu$  (blue),  $\pi^\pm$  (red),  $p$  (green), cosmic/unmatched (grey). The muon candidate is drawn solid-thick; the pion candidate dashed-thick; the reconstructed  $\nu$  vertex is shown as a star.



(a) CC $0\pi$  background: a proton (green, dashed) is reconstructed as the pion candidate. (b) CC $N\pi$  background: two genuine charged pions present; one is selected as the pion candidate. (c) EXT cosmic event: all tracks are grey (no truth match), faking a neutrino vertex topology.

Figure 10: Truth-labelled 2D event displays (ZX projection) for representative background events at the final cut. Three views (ZX top, ZY side, XY face) are generated per event.

## 9 Unfolding

### 9.1 The extraction chain, stage by stage

Before the formalism, it is useful to see the measurement assembled in the order it is actually performed. Each stage is shown for all three beam configurations in the flat reco-bin space in which the unfolding operates (every analysis slice concatenated).

**Stage 1, selected events, POT-scaled (Fig. 11).** The starting point is the selected reconstructed spectrum, scaled to the full exposure of each configuration with the per-run POT factors of §4. The Monte Carlo is split into signal and background. At the final cut the purity is  $\simeq 59\%$  (Table 10), so some 40% of the selected events have to be removed before anything can be interpreted as a signal rate. Because the analysis is blind, the points are the Poisson-thrown fake data rather than beam-on data.

**Stage 2, background subtraction (Fig. 12).** The estimated background is subtracted bin by bin, leaving the background-subtracted data alongside the central-value MC signal. The neutrino background is taken from the simulation, the cosmic component from beam-off (EXT) data, and the dirt component from the dedicated overlay; the four control regions of §10.5 are what validate this step. This subtracted spectrum, not the raw selection, is the input to the unfolding.

**Stage 3, unfolding.** The subtracted spectrum still folds in the detector response and the selection efficiency, encoded in the response matrix of §9.3. With 50–99% bin migration a direct inversion is unusable (§9.4), so the measurement is regularised with Wiener–SVD. The price is the additional-smearing matrix  $A_C$ : what is recovered is  $A_C \times (\text{true signal})$ , and *every* model must therefore be multiplied by  $A_C$  before being compared with the data.

**Stage 4, comparison with models.** The unfolded result is then compared with the Micro-BooNE tune and the four external generators, all  $A_C$ -smeared into the same space (§10). It is worth separating two effects that are easily conflated.  $A_C$  suppresses the integral of each prediction, but it does so by a *common* factor: for FHC  $p_\mu$  it is 0.78, 0.78, 0.76 and 0.79 for GENIE, GiBUU, NEUT and NuWro respectively (smeared integral over truth-level total, from the released curves). The truth-level totals of those same generators span  $0.82$  to  $1.32 \times 10^{-38} \text{ cm}^2/\text{Ar}$  (Table 14 of the technical supplement), a factor 1.6. A common  $\simeq 0.78$  factor cannot produce a  $1.6\times$  spread, so the differences between the generators are genuine model differences and not an artefact of the regularisation. The one-bin total of §10.3 provides the independent check, since with a single bin there is no regularisation and  $A_C \equiv 1$ .

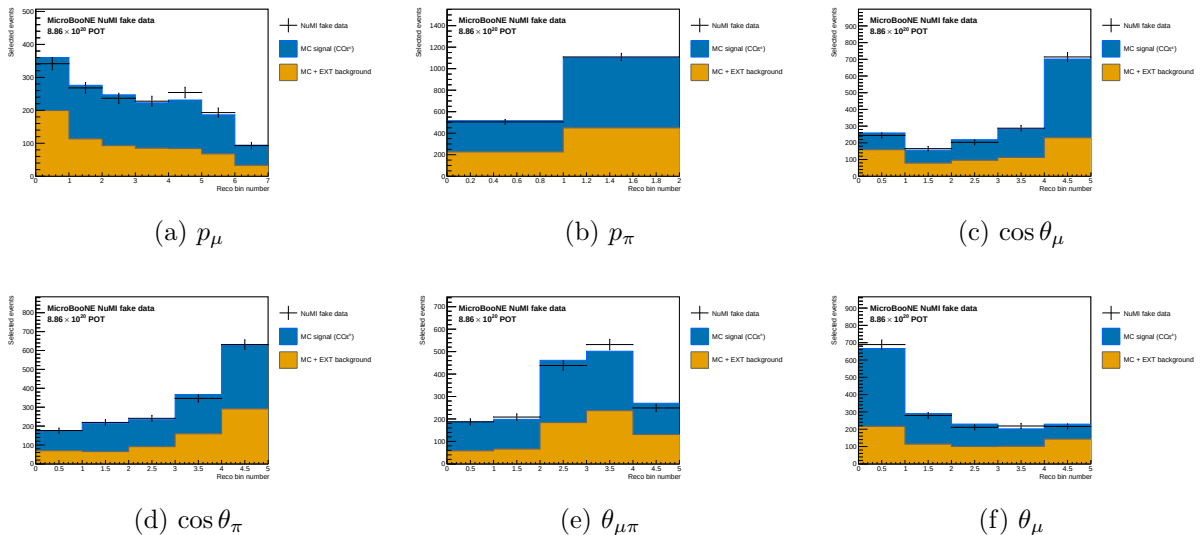


Figure 11: Stage 1, FHC: selected reconstructed spectra scaled to the full exposure, with the MC split into signal and background. Points are the Poisson-thrown fake data. The RHC and combined equivalents are Figs. 13 (the combined configuration is in Fig. 6 of the technical supplement).

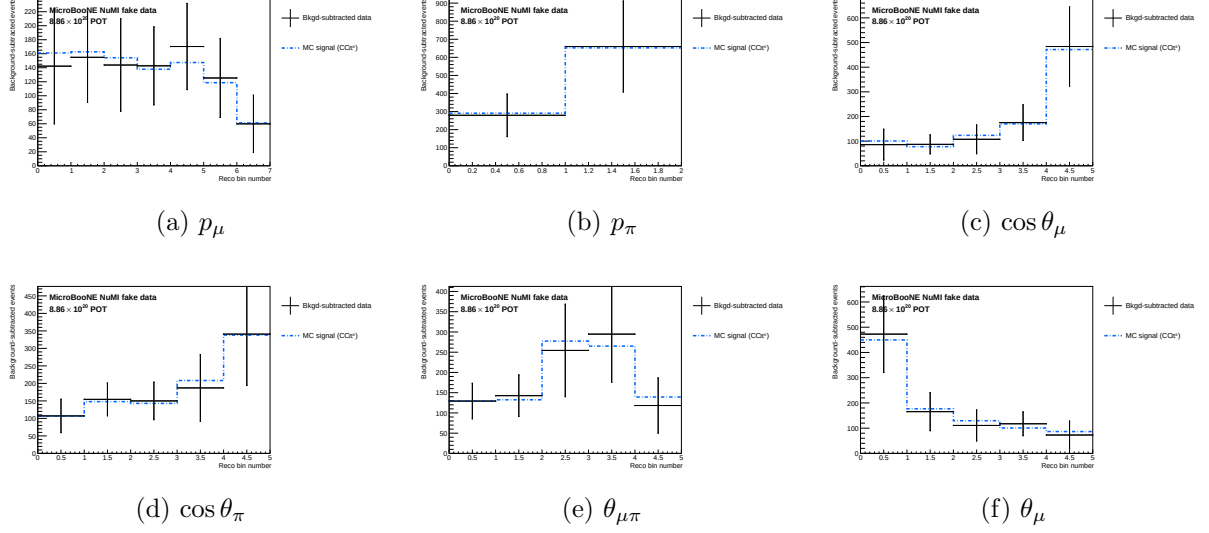


Figure 12: Stage 2, FHC: background-subtracted data compared with the central-value MC signal; the input to the unfolding. RHC and combined: Figs. 14 (combined: Fig. 7 of the technical supplement).

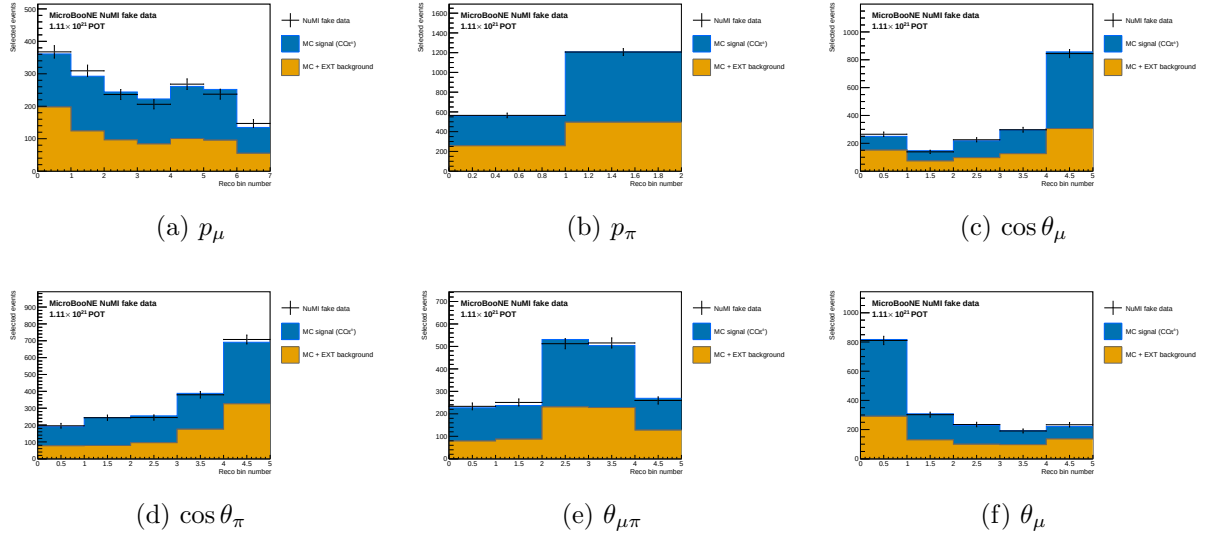


Figure 13: Stage 1, RHC: selected reconstructed spectra scaled to the full RHC exposure, MC split into signal and background.

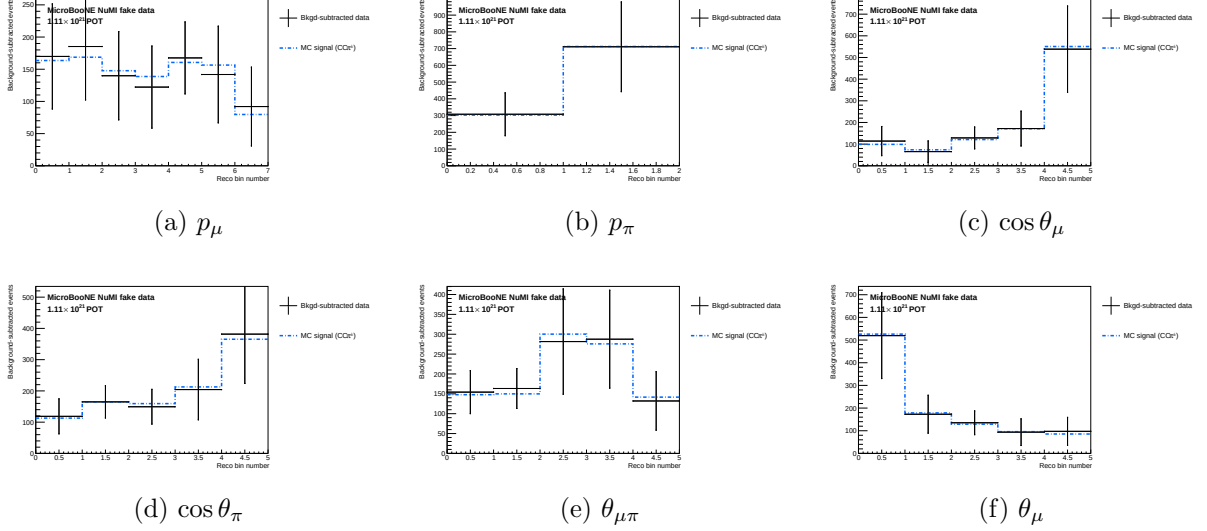


Figure 14: Stage 2, RHC: background-subtracted data compared with the central-value MC signal.

## 9.2 Signal extraction

After the event selection, the background-subtracted reco-space signal spectrum for observable  $x$  is:

$$d_i^{\text{sig}} = d_i^{\text{data}} - b_i^{\nu\text{-bkg}} - b_i^{\text{EXT}} - b_i^{\text{dirt}}, \quad (9)$$

where  $i$  runs over reconstructed bins.

## 9.3 Response matrix

Two distinct matrices are in play and it is worth separating them, since they differ by exactly the efficiency. The *migration* matrix

$$M_{ij} = P(\text{reco in bin } i \mid \text{true in bin } j, \text{ selected}), \quad \sum_i M_{ij} = 1, \quad (10)$$

describes only where selected events move between bins. The *smearceptance* matrix additionally carries the probability of being selected at all,

$$S_{ij} = \varepsilon_j M_{ij}, \quad \sum_i S_{ij} = \varepsilon_j, \quad (11)$$

where  $\varepsilon_j$  is the ratio of selected to generated signal events in true bin  $j$ .

It is  $S$ , not  $M$ , that enters the unfolding: the unfolders receives the smearceptance matrix, and the efficiency of Fig. 19 is recovered as its column sums.  $S$  is built from selected signal MC using the same binning on both axes. The folding relation the unfolders inverts is therefore

$$n_i^{\text{reco}} = \sum_j S_{ij} N_j^{\text{true}} + b_i, \quad S_{ij} = \varepsilon_j M_{ij}, \quad (12)$$

and the unfolded  $\hat{N}_j^{\text{true}}$  is already efficiency-corrected: no separate division by  $\varepsilon_j$  is applied afterwards (§13). Figure 18 plots  $S$  itself, exactly the matrix passed to the unfolders (colour scale  $P(\text{reco} \mid \text{true}) \times \varepsilon$ , so each column sums to the efficiency of that true bin rather than to unity). The migration diagonals quoted in this note, for example 91% and 76% for the two  $p_\pi$  bins, are elements of the column-normalised  $M$ , obtained from  $S$  by dividing each column by its sum; they are not read off the colour scale of Fig. 18.

## 9.4 Regularised unfolding: the bias–variance trade-off

Unfolding inverts the folding relation  $\vec{m} = A \vec{s} + \vec{b}$  to estimate the true signal spectrum  $\vec{s}$  from the measured  $\vec{m}$ . The naive inverse  $A^{-1}(\vec{m} - \vec{b})$  is unbiased but useless: bin migration makes  $A$  nearly singular, so its inverse amplifies statistical fluctuations into large, anticorrelated bin-to-bin oscillations. Every practical method therefore *regularises*, trading a small, controlled bias for a large reduction in variance. This analysis uses two complementary methods.

**Wiener-SVD (primary).** Diagonalising the response with an SVD, each mode  $k$  is weighted by the Wiener filter  $W_k = \sigma_k^2 / (\sigma_k^2 + N_k)$ ; the minimum-mean-squared-error weighting,  $\simeq 1$  for well-measured modes and  $\rightarrow 0$  for noise-dominated ones, so no regularisation strength is tuned by hand. The price is a *known* linear bias: the unfolded data estimates not  $\vec{s}$  but the smeared spectrum  $A_C \vec{s}$ , where the additional-smearing matrix  $A_C$  is published so that every theory prediction is smeared by the same  $A_C$  before comparison.

**D’Agostini iterative Bayesian [27] (cross-check).** Bayes’ theorem inverts the migration,  $P(C_j|E_i) = R_{ij}\pi_j / \sum_l R_{il}\pi_l$ , and the unfolded estimate is  $\hat{s}_j = \varepsilon_j^{-1} \sum_i P(C_j|E_i)(m_i - b_i)$ . The MC prior  $\pi_j$  is updated with the result and iterated; the *number of iterations* is the regularisation knob (few  $\Rightarrow$  prior-biased, many  $\Rightarrow$  noisy). It carries no  $A_C$ , so its integral is essentially unbiased and theory is compared raw.

Table 12: Trade-offs of the two unfolding methods. Neither is “more correct”; they make complementary choices about how to trade bias for variance and how to expose the residual bias.

|                   | Wiener-SVD                          | D’Agostini (iter. Bayes)             |
|-------------------|-------------------------------------|--------------------------------------|
| Regularisation    | Wiener filter (automatic, from S/N) | Iteration count $n$ (chosen)         |
| Residual bias     | Explicit $A_C$ , published          | Implicit; set by early stopping      |
| Result integral   | Suppressed, observable-dependent    | Essentially unbiased (physical)      |
| Theory comparison | Smear theory by $A_C$               | Raw theory                           |
| Positivity        | Not guaranteed                      | Guaranteed for positive input        |
| Uncertainties     | Closed-form, linear                 | Correlated across iters.; needs care |

Wiener-SVD is the primary result, for its objective regularisation and its explicit, reportable bias  $A_C$ ; D’Agostini serves as the quantitative cross-check (Sec. 4.4 of the technical supplement), where its flat, iteration-stable integral, 23–31% above Wiener-SVD (config-dependent), is consistent with the observable-to-observable spread of the Wiener-SVD integrated cross sections (Table 18) being the  $A_C$  smearing rather than a physical or acceptance effect; the row sums of §10.2 establish that directly. That is all it can establish here: with central-value pseudo-data and the same central value as its prior, D’Agostini returns its prior, so its flatness and iteration stability are guaranteed and say nothing about its bias on a different truth. The alternative-generator fake-data pass of §14 would make it a real test.

## 9.5 Wiener-SVD unfolding

The primary method is Wiener-SVD [7]. Every released result is produced by the Wiener-SVD unfolders of the XSecAnalyzer framework [18, 17]: Cholesky whitening of the data covariance, regularisation inside the SVD of  $RC^{-1}$ , and the Wiener filter of Eq. 3.24 of Ref. [7], whose strength is set by the data covariance rather than by hand. The regularisation matrix  $C$  is the second-derivative operator on the *density*, corrected for the non-uniform bin widths, with one-sided first-derivative rows at the ends. The width correction matters: on these binnings, whose widths vary by up to a factor of 14 ( $\cos\theta_\mu$ ; 10 for  $\theta_\mu$ , 6 for  $p_\mu$ ), a uniform-grid stencil penalises the density’s curvature by the wrong weights and drives  $A_C$  towards rank one, so that every

model is smeared onto one shape and the measurement carries no shape information (Sec. 5.1 of the technical supplement). A standalone replica of the unfold exists for cross-pipeline checks; it imports the framework’s covariance because it throws no universes, and with a statistics-only covariance it under-regularises and over-corrects the integral by  $\sim 26\%$ , which is why it is not used for any released number.

The differential cross section is:

$$\left. \frac{d\sigma}{dx} \right|_j = \frac{\hat{N}_j^{\text{true}}}{\Delta x_j \cdot \Phi_{\text{total}} \cdot N_{\text{Ar}}}, \quad (13)$$

where  $\hat{N}_j^{\text{true}}$  is the Wiener-SVD estimate of the true signal count in true bin  $j$  obtained by inverting Eq. (12),  $\Phi_{\text{total}} = \Phi_{\text{per POT}} \times T_{\text{POT}}$  is the total integrated flux, and  $N_{\text{Ar}}$  is the number of argon target nuclei in the FV. Because the response supplied to the unfold is the smear acceptance matrix  $S = \varepsilon M$ , the efficiency correction is part of the inversion and *no explicit*  $1/\varepsilon_j$  appears here (one would apply the correction twice). The implementation forms each bin as  $\hat{N}_j^{\text{true}}/(\text{conv} \cdot \Delta x_j)$  with the events-to-cross-section conversion  $\text{conv} = N_{\text{Ar}} \Phi_{\text{total}}/10^{-38}$  and applies no further efficiency factor. The per-configuration inputs at the final exposures are:

| Quantity                                                                                       | FHC     | RHC                    | Combined |
|------------------------------------------------------------------------------------------------|---------|------------------------|----------|
| $N_{\text{Ar}}$ (FV, dead-region excluded)                                                     |         | $8.710 \times 10^{29}$ |          |
| Flux constant $\phi$ ( $E_\nu > 60$ MeV; $10^{-10}$ cm $^{-2}$ /POT)                           | 6.81159 | 6.44646                | 6.60865  |
| Data POT ( $10^{20}$ )                                                                         | 8.857   | 11.082                 | 19.939   |
| Integrated flux $\Phi_{\text{total}} = \phi \times \text{POT}$ ( $10^{11}$ cm $^{-2}$ )        | 6.033   | 7.144                  | 13.177   |
| $\text{conv} = N_{\text{Ar}} \Phi_{\text{total}}/10^{-38}$ (events per $10^{-38}$ cm $^2$ /Ar) | 5255    | 6223                   | 11478    |

An integrated cross section is computed as the width-weighted sum,

$$\sigma_{\text{int}} = \sum_j \left. \frac{d\sigma}{dx} \right|_j \Delta x_j. \quad (14)$$

It is *not* required to agree between observables: the Wiener-SVD result estimates  $A_C x_{\text{true}}$  and  $A_C$  is not norm preserving, so  $\sigma_{\text{int}}$  is observable-dependent (§10.2). The observable-independent total is the one-bin total of §10.3.

## 9.6 Reconstructed spectra and response matrices

Figure 15 shows the background-subtracted reco-space signal spectra for all six observables, overlaid with the MC signal prediction and the individual background components.

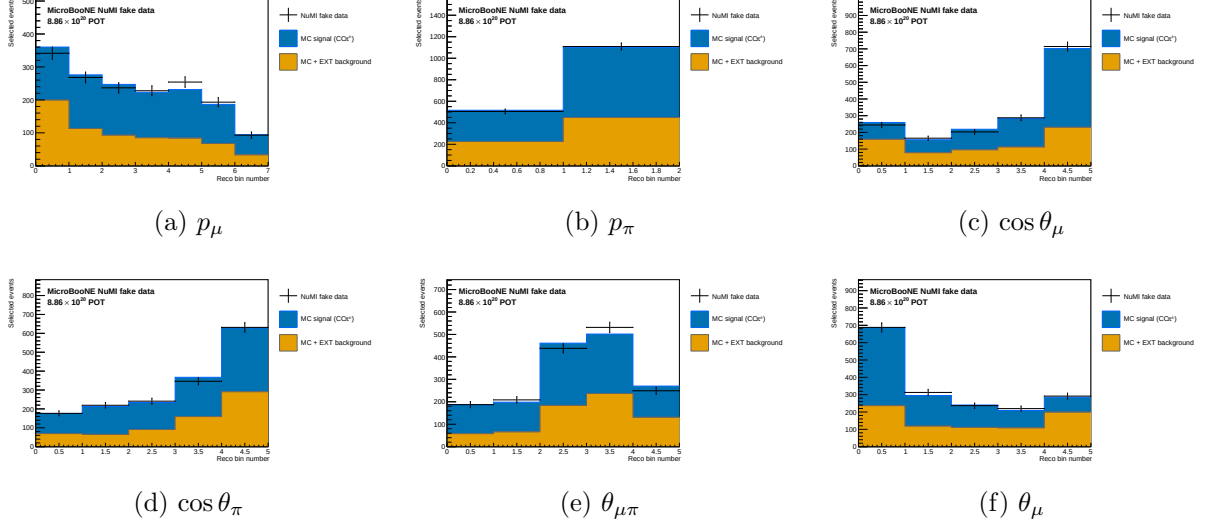


Figure 15: Reconstructed-space spectra at the final cut for all six observables, FHC full exposure (Runs 1+2+4+5): Poisson-thrown central-value fake data (black points), stacked signal (blue), beam background (orange), dirt (green), and EXT (grey). The fake data is drawn from the central-value Monte Carlo and so agrees with the stacked prediction by construction (a reco-level self-closure); the differential results and their closure against the  $A_C$ -smeared truth are shown in Fig. 21.

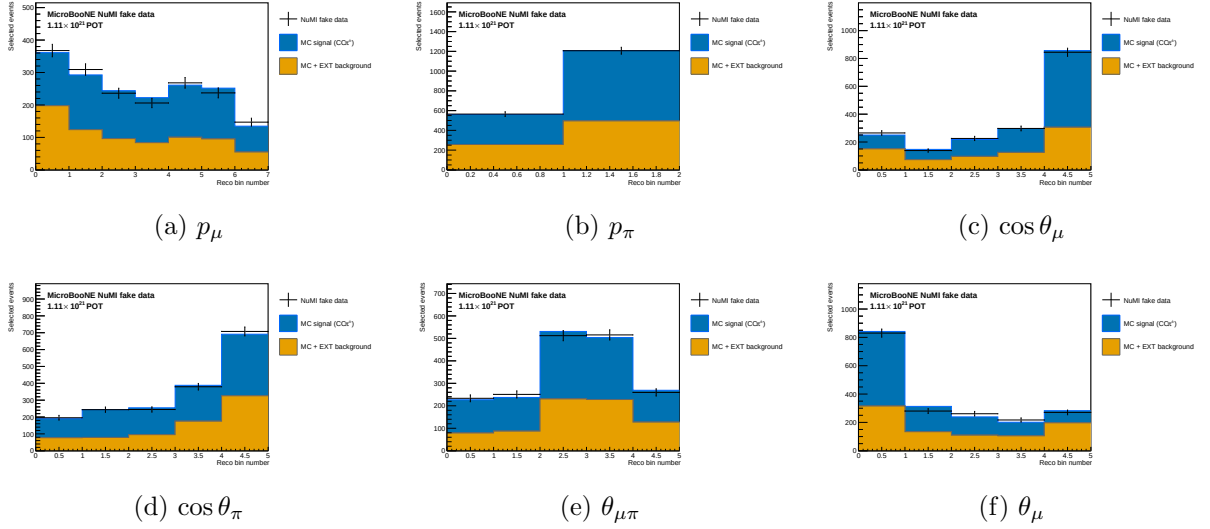


Figure 16: Reconstructed-space spectra, RHC full exposure (Runs 1+2+3+4a+4b+4c); same components as Fig. 15.

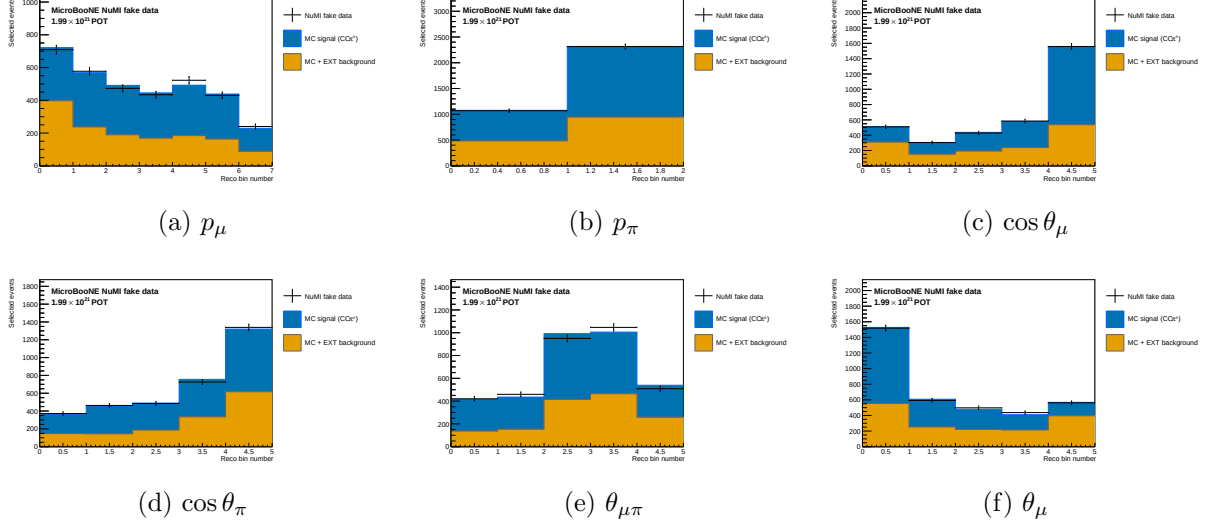


Figure 17: Reconstructed-space spectra, combined FHC+RHC (per-mode normalised); same components as Fig. 15.

Figure 18 shows the smearceptance matrices  $S$  (§9.3; colour scale includes the efficiency, so the columns sum to  $\varepsilon_j$ , not to unity) for the six observables at the released binning. The column-normalised diagonals of the released FHC responses are 0.86–0.95 for the best-resolved bin of each angular observable (the forward bin of  $\cos \theta_\mu$ , the first bin of the others), 0.67–0.88 elsewhere in  $\cos \theta_\mu$ ,  $\theta_\mu$ ,  $\cos \theta_\pi$  and  $\theta_{\mu\pi}$ , and 0.81, 0.65, 0.57, 0.50, 0.45, 0.38, 0.33 for the seven  $p_\mu$  bins:  $p_\mu$  is *not* near-diagonal, because its bins (0.2–0.5 GeV/ $c$  wide) are narrower than the 30% resolution of the MCS estimator, which reconstructs 63% of the selected sample and is intrinsically less precise than range. The 0.68 diagonal criterion quoted for  $p_\pi$  below is therefore not a criterion this analysis applies uniformly: for  $p_\mu$  the Wiener-SVD regularisation absorbs the migration (the released  $A_C$  row sums stay at 0.66–0.91, §10.2) and the closure and ensemble tests pass, whereas for  $p_\pi$  the migration is accompanied by a physical saturation of the estimator, which no regularisation recovers. A finer  $p_\mu$  binning is excluded by the same resolution (Sec. 6 of the technical supplement).

$p_\pi$ . At any fine binning the  $p_\pi$  response is strongly non-diagonal, because the range estimator saturates near 0.34 GeV/ $c$  once pions interact hadronically before stopping; that is a physical limit rather than an algorithmic one, and the fine-bin study is in the technical supplement (Sec. 1.1 of the technical supplement). The adopted two-bin response shown here is sufficiently diagonal, with column-normalised diagonals of 91% and 76%, but it carries only threshold-versus-above-threshold information: the upper bin is an inclusive region, not a resolved spectral point (§10.1).

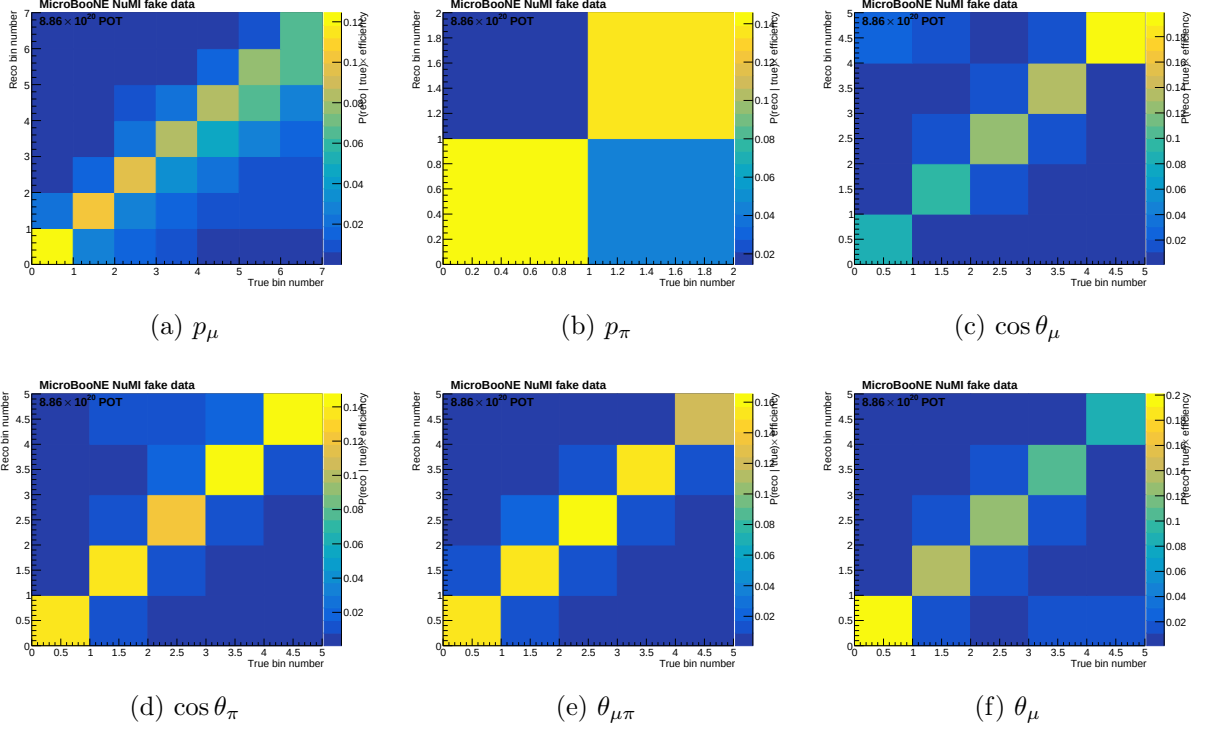


Figure 18: Smearing matrices  $S_{ij} = \varepsilon_j M_{ij}$ , FHC, exactly as passed to the unfoldr. Five panels are at the released binning, including the released two-bin  $p_\pi$  response (91% and 76% column-normalised diagonals); the rejected five-bin  $p_\pi$  response, which shows the reconstructed events piling into the lowest bin from every true bin, is Fig. 8 of the technical supplement.

Figure 19 shows the selection efficiency per true bin for all six observables, computed as the ratio of selected to generated signal MC.

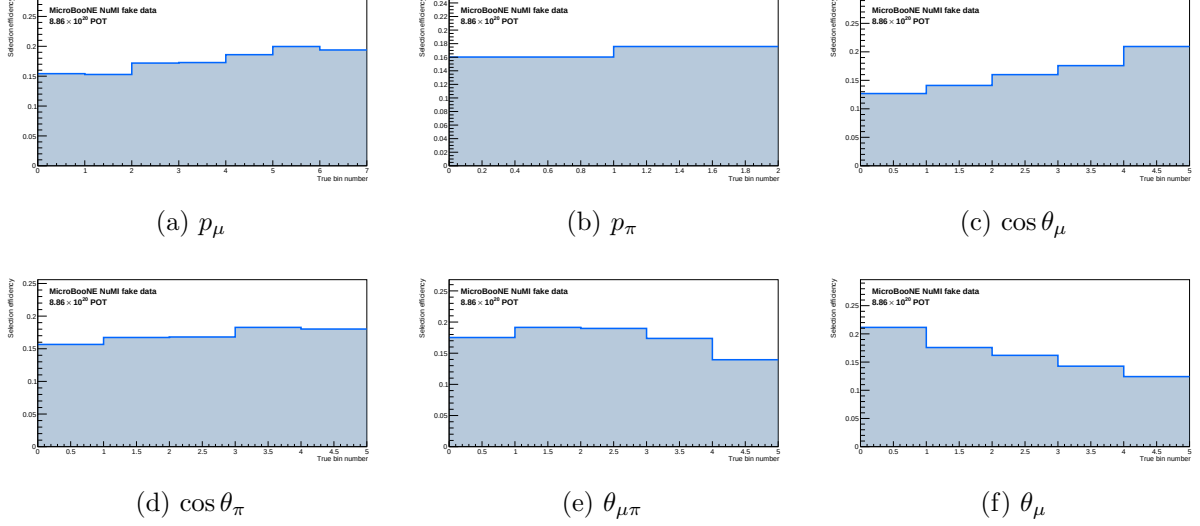


Figure 19: Selection efficiency as a function of true observable value, in the measured phase space ( $p_\mu > 0.15$ ,  $p_\pi > 0.175$  GeV/c,  $\theta_{\mu\pi} < 2.6$  rad). Efficiency is the ratio of selected to generated signal MC ( $\varepsilon_i = \sum_j A_{ij}$ , integrated over reco bins). With the momentum thresholds and the 2.6 rad opening-angle cut the efficiency is flat at  $\sim 13$ – $20\%$  across all observables; the former  $\sim 5\%$  low-momentum turn-on bins and the cosmic-dominated large-angle region are removed. The lowest point is the smallest opening-angle bin ( $\sim 9\%$  efficiency); it is retained because it is high-purity ( $\sim 58\%$ ), not low-purity.

## 9.7 Reconstruction resolution

Figure 20 shows the reconstruction resolution of each measured observable, evaluated on selected signal events (GENIE  $\nu_\mu$  overlay, measured phase space). For the momenta the relative resolution  $(\text{reco} - \text{true})/\text{true}$  is shown; for the angular observables the absolute residual  $(\text{reco} - \text{true})$  is used, since the relative metric diverges as  $\cos \theta \rightarrow 0$  /  $\theta_{\mu\pi} \rightarrow 0$ . The red points are the mean residual per true bin; the quoted numbers are the overall mean (bias) and RMS (resolution).

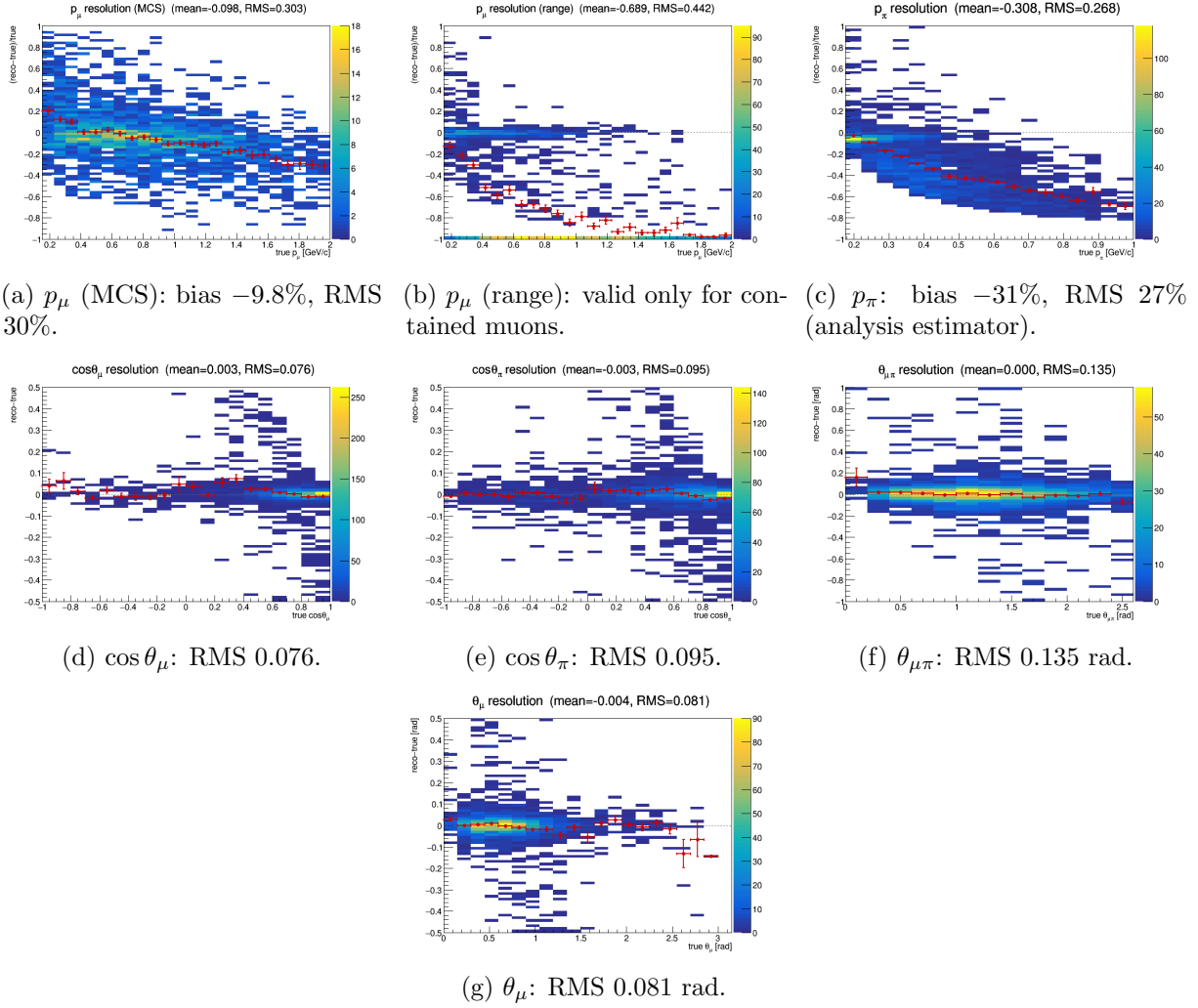


Figure 20: Reconstruction resolution vs. true value, selected signal. Momenta: relative ( $\text{reco} - \text{true}$ )/ $\text{true}$ ; angles: absolute ( $\text{reco} - \text{true}$ ). Red points: mean residual per true bin.  $\theta_\mu$  is derived as  $\arccos(\cos \theta_\mu)$ , exactly as the bin config does it, so the panel shows the quantity the analysis bins on; it is unbiased ( $-0.004$  rad) with RMS  $0.081$  rad, the drift below zero above  $\sim 2.5$  rad being a handful of backward-going muons. The muon MCS momentum slightly over-estimates at low  $p_\mu$  and increasingly under-estimates above  $\sim 1$  GeV/c; the range estimator is badly biased for *exiting* muons (range = distance to wall), which is why the analysis uses range only for contained muons (range-or-MCS). The pion panel shows the estimator the analysis actually bins on, not the raw muon-hypothesis range momentum (Sec. 1.1 of the technical supplement): mean bias  $-31\%$ , RMS  $27\%$ . That average is dominated by high momenta; the bias is only  $-2\%$  in the first true bin and reaches  $-65\%$  above  $0.8$  GeV/c; because the reconstructed value saturates near  $0.34$  GeV/c once pions interact hadronically before stopping. The angular variables are essentially unbiased with narrow resolution.

## 9.8 Model comparison: $\chi^2$ against truth, tune and generators

Table 13 collects the bin-by-bin  $\chi^2$  of the unfolded fake data against each prediction, for every observable and every beam configuration. All predictions are  $A_C$ -smeared, so the comparison is made in the measurement space, and the complete configured covariance is used. The *truth* column is the closure test: it is small throughout (with the caveat on its  $p$ -values in §10.1).

**One application of  $A_C$ .** Every prediction in these tables is smeared by  $A_C$  exactly once, by the extractor; the plotting path reads the smeared curves and does not smear again. That is verified externally: the released prescription  $\frac{1}{w_i} \sum_j (A_C)_{ij} p_j$  applied to the released truth-level predictions reproduces the released curves to a worst per-bin difference of  $1.2 \times 10^{-10}$  (Sec. F of the technical supplement). The point matters because a second application would over-smooth the sharply peaked generators and manufacture an order-of-magnitude  $\chi^2$  separation in the muon-angle observables that is not there: NEUT and NuWro in FHC  $\theta_\mu$  give 8.46/5 and 9.10/5, and in  $\cos \theta_\mu$  3.88/5 and 4.04/5 (Table 13), in line with the other observables.

Table 13:  $\chi^2/\text{ndf}$  of the unfolded fake data against each  $A_C$ -smeared prediction, for the inclusive selection. “truth” is the fake-data truth (the closure test).

| Beam | Observable        | truth  | uB tune | GENIE  | GiBUU  | NEUT   | NuWro  |
|------|-------------------|--------|---------|--------|--------|--------|--------|
| FHC  | $p_\mu$           | 1.45/7 | 1.54/7  | 2.51/7 | 1.69/7 | 3.08/7 | 1.42/7 |
|      | $p_\pi$           | 0.11/2 | 0.10/2  | 0.37/2 | 1.25/2 | 0.74/2 | 0.57/2 |
|      | $\cos \theta_\mu$ | 0.67/5 | 1.09/5  | 1.67/5 | 1.27/5 | 3.88/5 | 4.04/5 |
|      | $\cos \theta_\pi$ | 0.68/5 | 0.69/5  | 1.68/5 | 1.91/5 | 1.26/5 | 1.42/5 |
|      | $\theta_{\mu\pi}$ | 0.30/5 | 0.86/5  | 1.20/5 | 1.44/5 | 3.87/5 | 3.82/5 |
|      | $\theta_\mu$      | 1.52/5 | 2.14/5  | 5.55/5 | 6.07/5 | 8.46/5 | 9.10/5 |
| RHC  | $p_\mu$           | 0.64/7 | 2.51/7  | 3.44/7 | 2.81/7 | 3.19/7 | 3.31/7 |
|      | $p_\pi$           | 0.02/2 | 0.01/2  | 0.56/2 | 0.30/2 | 0.25/2 | 0.11/2 |
|      | $\cos \theta_\mu$ | 0.35/5 | 0.37/5  | 1.11/5 | 0.24/5 | 2.71/5 | 2.67/5 |
|      | $\cos \theta_\pi$ | 0.49/5 | 0.43/5  | 1.27/5 | 2.86/5 | 1.79/5 | 1.08/5 |
|      | $\theta_{\mu\pi}$ | 0.19/5 | 0.34/5  | 2.52/5 | 3.54/5 | 0.46/5 | 1.08/5 |
|      | $\theta_\mu$      | 0.12/5 | 0.21/5  | 0.78/5 | 0.34/5 | 5.60/5 | 5.76/5 |
| comb | $p_\mu$           | 0.41/7 | 0.74/7  | 2.08/7 | 0.86/7 | 1.46/7 | 0.98/7 |
|      | $p_\pi$           | 0.01/2 | 0.01/2  | 0.45/2 | 0.92/2 | 0.47/2 | 0.30/2 |
|      | $\cos \theta_\mu$ | 0.05/5 | 0.09/5  | 0.84/5 | 0.47/5 | 3.27/5 | 2.92/5 |
|      | $\cos \theta_\pi$ | 0.73/5 | 0.54/5  | 1.84/5 | 2.72/5 | 1.43/5 | 1.23/5 |
|      | $\theta_{\mu\pi}$ | 0.06/5 | 0.54/5  | 1.76/5 | 2.64/5 | 2.41/5 | 2.89/5 |
|      | $\theta_\mu$      | 0.56/5 | 0.83/5  | 3.16/5 | 4.71/5 | 8.20/5 | 7.87/5 |

Table 14: The same comparison for the muon angle in both parameterisations. The two binnings span the same full range but are different partitions of it: the  $\theta_\mu$  edges (0.43, 0.62, 0.84, 1.17 rad) are not the arccos images of the  $\cos \theta_\mu$  edges (0.45, 0.64, 0.86, 1.10 rad). The  $\theta_\mu$   $\chi^2$  are systematically the larger for every column including the truth, a consequence of the different partition and of  $A_C$ . Model statements in this note are made on  $\cos \theta_\mu$ ;  $\theta_\mu$  is released as the second parameterisation.

| Beam | Variable          | truth  | uB tune | GENIE  | GiBUU  | NEUT   | NuWro  |
|------|-------------------|--------|---------|--------|--------|--------|--------|
| FHC  | $\cos \theta_\mu$ | 0.67/5 | 1.09/5  | 1.67/5 | 1.27/5 | 3.88/5 | 4.04/5 |
| FHC  | $\theta_\mu$      | 1.52/5 | 2.14/5  | 5.55/5 | 6.07/5 | 8.46/5 | 9.10/5 |
| RHC  | $\cos \theta_\mu$ | 0.35/5 | 0.37/5  | 1.11/5 | 0.24/5 | 2.71/5 | 2.67/5 |
| RHC  | $\theta_\mu$      | 0.12/5 | 0.21/5  | 0.78/5 | 0.34/5 | 5.60/5 | 5.76/5 |
| Comb | $\cos \theta_\mu$ | 0.05/5 | 0.09/5  | 0.84/5 | 0.47/5 | 3.27/5 | 2.92/5 |
| Comb | $\theta_\mu$      | 0.56/5 | 0.83/5  | 3.16/5 | 4.71/5 | 8.20/5 | 7.87/5 |

## 10 Full-exposure results: FHC, RHC, and combined

## 10.1 Scope and status of each result

All fifty-one extractions carry the complete configured systematic covariance, detector variations included (“complete” relative to the configured uncertainty model; the identified terms that are not in it are listed in §14). The proton-tagged detector terms run from 10.5% to 108.5% with a median of 33.0%, against 15.6%–35.0% for the inclusive results; the wide range is the sparse six-bin samples, where the unfolding amplifies them.

The proton-tagged results are secondary scope because the exposure does not support the differential  $W_{\pi p}$  shape at any binning, the proton-mistag background lacks a dedicated inverted-proton-PID control region (Sec. D.6 of the technical supplement), and ensemble coverage has not been demonstrated for this sparser family (§14).

| Result                                                                           | Status                                                        | Basis                                                                                                                                                                                                                                                         |
|----------------------------------------------------------------------------------|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Primary scope</i>                                                             |                                                               |                                                                                                                                                                                                                                                               |
| Total flux-averaged cross section (one-bin extraction), all three configurations | Analysis definition final; result blinded                     | Not suppressed by $A_C$ ; the most directly interpretable total.                                                                                                                                                                                              |
| $d\sigma/dp_\mu$                                                                 | Analysis definition final; result blinded                     | Complete configured covariance including detector variations.                                                                                                                                                                                                 |
| $d\sigma/d\cos\theta_\mu$ and $d\sigma/d\theta_\mu$                              | Analysis definition final; result blinded                     | Complete configured covariance; discriminates between models.                                                                                                                                                                                                 |
| $d\sigma/d\cos\theta_\pi$                                                        | Analysis definition final; result blinded                     | Complete configured covariance. (No $\theta_\pi$ parameterisation is extracted; only the muon angle is released in both forms.)                                                                                                                               |
| $d\sigma/d\theta_{\mu\pi}$                                                       | Analysis definition final; result blinded                     | Complete configured covariance; the best-conditioned shape observable.                                                                                                                                                                                        |
| $d\sigma/dp_\pi$ , two-bin                                                       | Analysis definition final, coarse; result blinded             | Complete configured covariance, but a two-region partition rather than a spectrum: one near-threshold interval and one broad inclusive interval above 0.205 GeV/c. See below.                                                                                 |
| <i>Secondary scope — complete configured covariance, secondary results</i>       |                                                               |                                                                                                                                                                                                                                                               |
| Proton-tagged integrated cross section                                           | Complete (configured covariance)                              | Detector systematics propagated, closure $\chi^2$ recomputed, $A_C$ and covariance released.                                                                                                                                                                  |
| Proton-tagged differential results, including transverse kinematic imbalance     | Validation products, not proposed for approval in this review | Detector systematics propagated; angles and transverse quantities in the beam frame; $A_C$ and covariance released; the $W_{\text{had}}$ edge scan has not been run (§14) and the differential $W_{\pi p}$ shape is withdrawn.                                |
| Two-dimensional cross sections                                                   | Proposed secondary scope, combined configuration only         | $\theta_p \times \delta p_T$ and $\cos\theta_\pi \times \cos\theta_\mu$ ; eight pairs screened and extracted, per-mode results are too poorly conditioned to invert (§11.4).                                                                                  |
| Proton angular observables $\theta_p$ , $\theta_{\pi p}$                         | Proposed secondary scope                                      | Binning, generator predictions and all three extractions complete on fake data (§11.3); $\theta_{\pi p}$ needed a gated reprocess of the proton-tagged tree; $\theta_p$ is rank-deficient in RHC and is proposed for FHC and the combined configuration only. |
| <i>Withdrawn — do not quote</i>                                                  |                                                               |                                                                                                                                                                                                                                                               |
| Differential $W_{\pi p}$ shape                                                   | Withdrawn                                                     | Appendix C.                                                                                                                                                                                                                                                   |
| Five-bin $d\sigma/dp_\pi$                                                        | Withdrawn                                                     | Appendix C.                                                                                                                                                                                                                                                   |

**What the reader should know before the tables.** Three points condition how the numbers below are read, and are established in Sec. E of the technical supplement. *First*,  $d\sigma/dp_\pi$  is released in two bins only: charged pions interact before ranging out, so the reconstructed momentum saturates near 0.34 GeV/c and no finer binning survives the migration criterion; the upper bin is open-ended and its released density represents a bin integral. *Second*, the closure  $\chi^2$  quoted with each extraction uses the complete systematic covariance, so its  $p$ -values are not a sensitive test of the extraction — they confirm consistency and nothing more. *Third*, an ensemble of independent Poisson realisations carried through the whole chain gives per-bin pull widths of 0.97–1.08, so the quoted statistical uncertainties are correctly sized; the study covers three extractions in one configuration, and RHC and the proton-tagged family have not been tested.

**On the blinding status.** Every “data” point in this note is Poisson-thrown central-value fake data. No beam-on signal region has been unblinded. Statements comparing configurations, or comparing the result to a generator, are therefore properties of the injected sample and the extraction machinery, not measurements.

This is the primary result of the analysis: the  $\text{CC1}\mu 1\pi^\pm$  cross section extracted over the full available exposure in forward horn current (FHC), reverse horn current (RHC), and their combination, all on Poisson-thrown central-value fake data. The FHC measurement uses Runs 1, 2, 4 and 5 (data POT  $8.857 \times 10^{20}$ ); RHC uses Runs 1, 2, 3, 4a, 4b, 4c ( $1.108 \times 10^{21}$ ); the combined measurement uses all fourteen overlay files at the joint POT  $1.994 \times 10^{21}$ . All three close cleanly against the  $A_C$ -smeared truth for every observable (Tables 15–17); the differential results are shown in Figs. 21–23.

Table 15: FHC (Runs 1, 2, 4, 5; data POT  $8.857 \times 10^{20}$ ): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against  $A_C$ -smeared truth. All values are read from the released covariance and results table; the anatomy of the flux term is in §13.2.

| Observable        | $\sigma_{\text{int}}$ | Flux  | Detector | Total | Closure $\chi^2/\text{ndf}$ |
|-------------------|-----------------------|-------|----------|-------|-----------------------------|
| $p_\mu$           | 0.838                 | 28.0% | 24.7%    | 40.7% | 1.45/7 ( $p = 0.98$ )       |
| $p_\pi$           | 0.822                 | 32.5% | 30.3%    | 50.2% | 0.11/2 ( $p = 0.95$ )       |
| $\cos \theta_\mu$ | 0.754                 | 28.0% | 28.3%    | 42.0% | 0.67/5 ( $p = 0.98$ )       |
| $\cos \theta_\pi$ | 0.674                 | 27.0% | 22.9%    | 38.8% | 0.68/5 ( $p = 0.98$ )       |
| $\theta_{\mu\pi}$ | 0.819                 | 28.3% | 19.6%    | 36.5% | 0.30/5 ( $p = 1.00$ )       |
| $\theta_\mu$      | 0.960                 | 24.7% | 15.6%    | 31.0% | 1.52/5 ( $p = 0.91$ )       |

We extend the measurement to the reverse-horn-current beam as both a standalone measurement and a combined  $\nu_\mu + \bar{\nu}_\mu$  result. The RHC *flux* is  $\bar{\nu}_\mu$ -dominant (54.6%  $\bar{\nu}_\mu$ , versus FHC’s 35%), with  $> 60$  MeV integrals  $\Phi_{\nu_\mu}^{\text{RHC}} = 2.923 \times 10^{-10}$ ,  $\Phi_{\bar{\nu}_\mu}^{\text{RHC}} = 3.523 \times 10^{-10}$ , total  $6.446 \times 10^{-10} \text{ cm}^{-2}\text{POT}^{-1}$  (the FHC-convention computation reproduces the authoritative FHC values exactly). The *measured event rate*, however, remains  $\nu_\mu$ -dominant (76%  $\nu_\mu$ , 20%  $\bar{\nu}_\mu$ ) because the  $\nu_\mu$  cross section is  $\sim 2\text{--}3\times$  larger than the  $\bar{\nu}_\mu$  one. The RHC overlays (Runs 1, 2, 3, 4a, 4b, 4c) are all verified as genuine  $\nu_\mu$  overlays (flavour composition 100%  $\nu_\mu + \bar{\nu}_\mu$  at truth level, events per POT uniform across files) and processed,  $4.05 \times 10^6$  events, combined MC POT  $1.546 \times 10^{22}$ , mutually consistent ( $2.62 \times 10^{-16}$  events/POT) with the PPF weights, covering the full RHC exposure (Run 3 is the largest period,  $5.52 \times 10^{21}$  MC POT and  $1.44 \times 10^6$  events). Native Run-4 RHC detector-variation samples (the standard eight-knob set: light-yield down/Rayleigh, recombination, space charge, and the four wire-modification variations) provide the RHC detector systematic, verified as genuine  $\nu_\mu$  overlays with a uniform  $3.80 \times 10^{-16}$  events/POT across all knobs. The RHC dirt prediction uses the FHC dirt overlay, scaled to the RHC exposure, as no RHC dirt sample exists. The standalone result below uses Poisson-thrown central-value fake data at the RHC data POT  $1.108 \times 10^{21}$ . The generator predictions are re-averaged to the RHC flux composition; the combined FHC+RHC predictions are the fluence-weighted combination of the two horn modes (0.458 FHC / 0.542 RHC, each mode weighted by  $\Phi_{\text{tot}} \times \text{POT}$ ).

The combined FHC+RHC measurement uses the summed 14-file response over the full exposure ( $5.6 \times 10^6$  events, combined data POT  $D_{\text{comb}} = 1.994 \times 10^{21} = D_{\text{FHC}} + D_{\text{RHC}}$ ), with each horn mode normalised to its *own* data exposure. This is a genuine subtlety: the detector-variation covariance is absolute (POT-scaled reco counts), so a single POT scale applied to all files would weight the modes by their Monte-Carlo POT (0.62:1 FHC:RHC) rather than by data exposure (0.80:1). Each mode’s numuMC and detVar samples therefore carry a per-mode stored exposure (FHC  $2.158 \times 10^{22}$ , RHC  $2.782 \times 10^{22}$ ,  $= S_{\text{mode}} \times D_{\text{comb}}/D_{\text{mode}}$ ) so the framework’s data-over-stored-exposure scaling reduces to  $D_{\text{FHC}}/S_{\text{FHC}}$  and  $D_{\text{RHC}}/S_{\text{RHC}}$ ; the fake data is Poisson-thrown with the matching per-mode rate.

The joint *flux systematic* needs no hand-built cross-horn correlation. All fourteen numuMC files carry the *same* 600 PPFX multisim universes, which the universe maker accumulates index-matched across both horn modes in a single chain, so universe  $u$  is the same hadron-production draw in FHC and RHC. Per-universe summing therefore propagates one parameter draw through both beam configurations and lets each mode’s flux response set the result, capturing the physical (not unit) cross-horn correlation automatically; universe indices are shared across FHC and RHC and summed index by index, which preserves the implemented cross-horn correlations without a hand-built block covariance. The detector variations are carried per mode, the Run-4 FHC and Run-4 RHC samples of each knob accumulated under one label, so the combined detector term is exposure-weighted like the central value. The combined closure passes for every observable (Table 17), with an unfolded flux systematic of 24–30% of the same size as the standalone FHC (25–33%) and RHC (25–28%) values (Tables 15–17), as expected for a common flux model propagated through three similarly conditioned responses.

The standalone RHC extraction (native Run-4 RHC detector variations) closes cleanly for every observable (Table 16): the realised  $A_C$ -smeared truth is recovered with  $\chi^2 \leq 0.7$  on two to seven degrees of freedom and  $p \geq 0.98$  throughout. The reconstructed-level flux uncertainty is  $\sim 15\%$ , below the FHC  $\sim 17\%$ , because the  $\bar{\nu}_\mu$ -dominant flux carries a smaller PPFX hadron-production uncertainty; the unfolded flux term (25–28%, Table 16) is nevertheless comparable to the FHC one, since the background subtraction multiplies both by the same  $1 + B/S$  (§13.2).

Table 16: RHC (Runs 1, 2, 3, 4; data POT  $1.1082 \times 10^{21}$ ): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against  $A_C$ -smeared truth. All values are read from the released covariance and results table; the anatomy of the flux term is in §13.2.

| Observable        | $\sigma_{\text{int}}$ | Flux  | Detector | Total | Closure $\chi^2/\text{ndf}$ |
|-------------------|-----------------------|-------|----------|-------|-----------------------------|
| $p_\mu$           | 0.690                 | 27.6% | 21.5%    | 37.7% | 0.64/7 ( $p = 1.00$ )       |
| $p_\pi$           | 0.748                 | 28.4% | 34.7%    | 48.6% | 0.01/2 ( $p = 0.99$ )       |
| $\cos \theta_\mu$ | 0.733                 | 25.8% | 35.0%    | 46.3% | 0.35/5 ( $p = 1.00$ )       |
| $\cos \theta_\pi$ | 0.615                 | 25.2% | 25.5%    | 38.5% | 0.49/5 ( $p = 0.99$ )       |
| $\theta_{\mu\pi}$ | 0.831                 | 25.2% | 16.6%    | 32.6% | 0.18/5 ( $p = 1.00$ )       |
| $\theta_\mu$      | 0.805                 | 25.6% | 22.4%    | 36.1% | 0.12/5 ( $p = 1.00$ )       |

Table 17: Combined (data POT  $1.9939 \times 10^{21}$ ): integrated cross section, the two largest systematic terms, the total, and the fake-data closure against  $A_C$ -smeared truth. All values are read from the released covariance and results table; the anatomy of the flux term is in §13.2.

| Observable        | $\sigma_{\text{int}}$ | Flux  | Detector | Total | Closure $\chi^2/\text{ndf}$ |
|-------------------|-----------------------|-------|----------|-------|-----------------------------|
| $p_\mu$           | 0.827                 | 24.2% | 21.4%    | 34.9% | 0.41/7 ( $p = 1.00$ )       |
| $p_\pi$           | 0.775                 | 30.2% | 30.8%    | 47.9% | 0.01/2 ( $p = 1.00$ )       |
| $\cos \theta_\mu$ | 0.812                 | 25.7% | 23.8%    | 37.3% | 0.04/5 ( $p = 1.00$ )       |
| $\cos \theta_\pi$ | 0.658                 | 25.6% | 22.5%    | 37.1% | 0.73/5 ( $p = 0.98$ )       |
| $\theta_{\mu\pi}$ | 0.879                 | 25.7% | 17.7%    | 33.2% | 0.06/5 ( $p = 1.00$ )       |
| $\theta_\mu$      | 0.851                 | 25.2% | 20.6%    | 34.4% | 0.56/5 ( $p = 0.99$ )       |

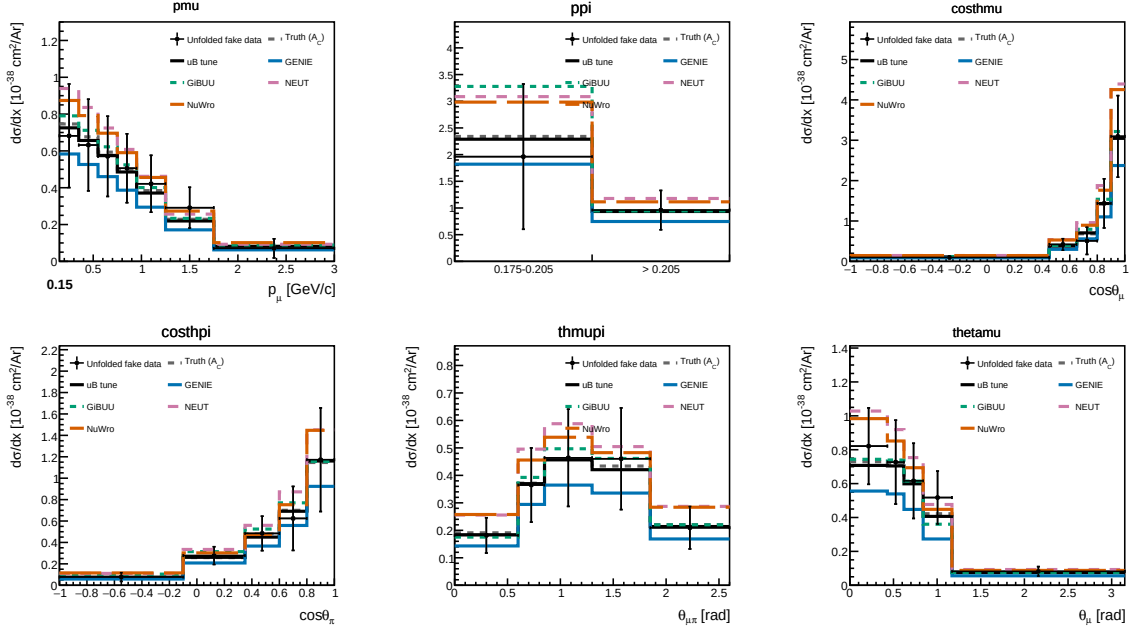


Figure 21: Differential cross sections  $d\sigma/dx$  for the six released distributions on the current Poisson-thrown central-value fake data, FHC full exposure: unfolded fake data (black points),  $A_C$ -smeared fake-data truth (grey dashed), the MicroBooNE tune (black solid), and the four generator predictions (GENIE, GiBUU, NEUT, NuWro). The unfolded points recover the injected truth for every observable, the same closure summarised in Table 15.

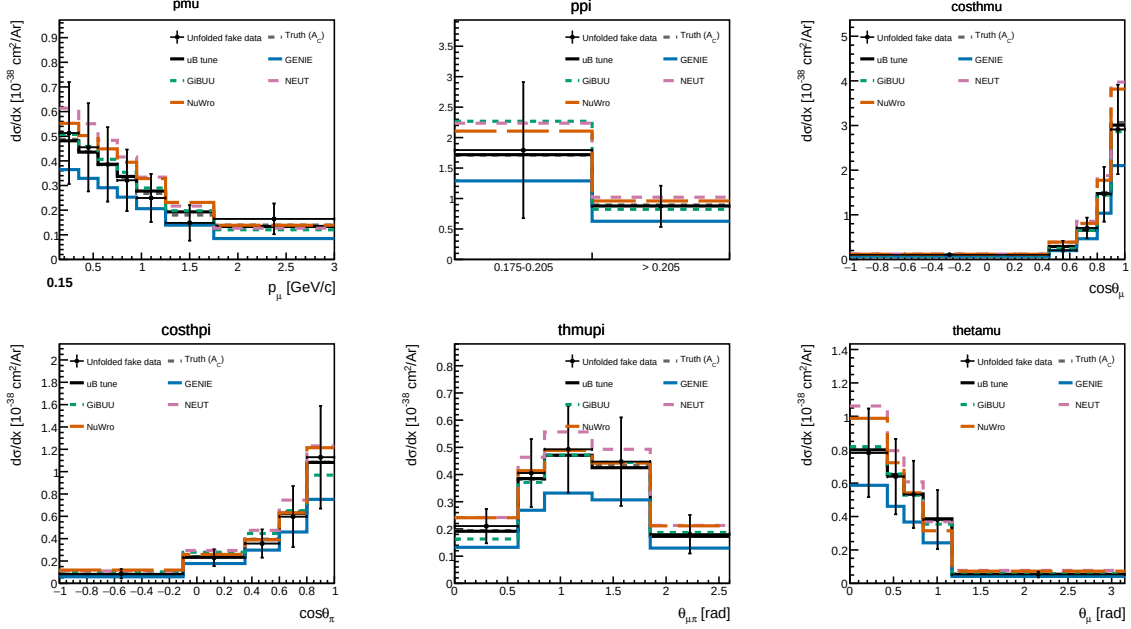


Figure 22: Standalone RHC ( $\bar{\nu}_\mu$ -dominant) differential cross sections on the current Poisson-thrown fake data, with the RHC-flux-averaged generator predictions. Same layout as Fig. 21.

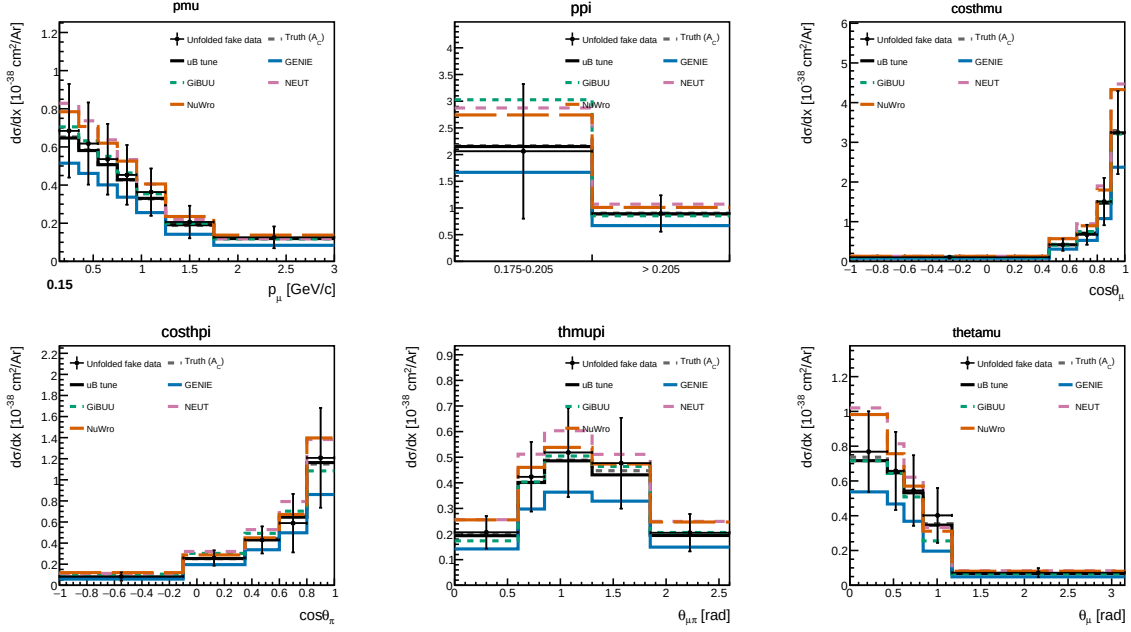


Figure 23: Combined FHC+RHC ( $\nu_\mu + \bar{\nu}_\mu$ ) differential cross sections on the current per-mode-normalised fake data, with the fluence-weighted combined generator predictions.

Table 18: Integrated (unfolded) cross section  $\sigma_{\text{int}}$  over the measured phase space, for each observable and configuration (current fake data, five-bin forward-split  $\cos\theta_\pi$  and coarsened  $\theta_{\mu\pi}$  binning, per-configuration flux;  $10^{-38} \text{ cm}^2/\text{Ar}$ ). The value differs between observables because the Wiener-SVD additional-smearing  $A_C$  is not norm-preserving; the D’Agostini cross-check (Sec. 4.4 of the technical supplement) recovers a flat, 23–31% higher total, consistent with the spread being a smearing effect (the  $A_C$  row sums of §10.2 establish it directly). All entries come from one pass of the current code with the cached universes rebuilt. The MC normalisation is evaluated run by run using the sum of the *distinct* stored exposure values of that run’s files: this gives the run total when that total is repeated in each split file (the processed-file convention of the inclusive family; Run 3 is split across 5 files and Run 4 across 3) and the sum of the true file exposures when each file stores its own (the proton-tagged family), so it handles both conventions without weighting a run by its split multiplicity; FHC (one MC file per run) is unaffected by the convention. The final row gives the alternative muon-angle parameterisation  $\theta_\mu$  (Sec. 5.1 of the technical supplement), complete for all three configurations; no  $\theta_\pi$  parameterisation is extracted. (Values are fake-data closures; the signal is  $\nu_\mu + \bar{\nu}_\mu$ , so the RHC flux-average is not expected to lie far below FHC.)

| Observable        | FHC   | RHC   | Combined |
|-------------------|-------|-------|----------|
| $p_\mu$           | 0.838 | 0.690 | 0.827    |
| $p_\pi$           | 0.822 | 0.748 | 0.775    |
| $\cos\theta_\mu$  | 0.754 | 0.733 | 0.812    |
| $\cos\theta_\pi$  | 0.674 | 0.615 | 0.658    |
| $\theta_{\mu\pi}$ | 0.819 | 0.831 | 0.879    |
| $\theta_\mu$      | 0.960 | 0.805 | 0.851    |

**Uncertainties shown.** Every differential cross section and every integrated value quoted in this note carries *both* statistical and systematic uncertainties. The error bars on the unfolded points are the square root of the diagonal of the *total* covariance matrix, propagated through the unfolding: flux (PPFX multisims), cross-section (GENIE multisims and unisims), detector variations [15], hadron re-interaction [25], POT counting and target number, together with the

data, MC and EXT statistical terms. For  $p_\mu$  in FHC the bin-averaged total is 40.7% (Table 27), dominated by flux (28.0%) and detector (24.7%), with data statistics 14.3% and cross-section 10.7%. The one-bin total cross section of §10.3 carries its own breakdown from the same universes, written there as systematic  $\oplus$  data-statistical (FHC: 38.5%  $\oplus$  4.3%); without regularisation its flux and detector terms are not amplified.

## 10.2 How far $A_C$ is from preserving normalisation

Because the Wiener-SVD result estimates  $A_C x_{\text{true}}$  rather than  $x_{\text{true}}$ , and  $A_C$  is not norm preserving, summing the differential result over its bins does not give the physical total cross section. This subsection quantifies the effect.

The row sums of  $A_C$  are the diagnostic used here: they give the response to a flat bin-integrated truth, for which a row summing to unity passes the normalisation of that bin through unchanged and a row summing to 0.2 passes one fifth of it. For a non-flat prediction the output of bin  $i$  is  $\sum_j (A_C)_{ij} t_j$ , which depends on the individual elements and on the shape of  $t$ ; the row sums describe the dominant behaviour, and the model-specific integral factors of §9.1 (0.76–0.79 for the four generators in FHC  $p_\mu$ ) show how far it carries over.

| Observable        | FHC       | RHC       | combined  |
|-------------------|-----------|-----------|-----------|
| $p_\mu$           | 0.66–0.91 | 0.58–0.99 | 0.68–1.01 |
| $p_\pi$ (2-bin)   | 0.35–0.54 | 0.40–0.51 | 0.29–0.54 |
| $\cos \theta_\mu$ | 0.17–0.39 | 0.05–0.45 | 0.09–0.43 |
| $\cos \theta_\pi$ | 0.68–0.77 | 0.70–0.84 | 0.71–0.83 |
| $\theta_{\mu\pi}$ | 0.52–1.03 | 0.58–1.15 | 0.63–1.21 |
| $\theta_\mu$      | 0.58–0.78 | 0.22–1.16 | 0.53–0.67 |

The suppression is large (inclusive  $\cos \theta_\mu$  has a row sum as low as 0.05 in RHC) and differs strongly between observables, which is why the integrated cross section obtained by summing a differential result varies from observable to observable while the physical total cannot; the D’Agostini cross-check, whose integral is flat across observables, confirms that the spread is the smearing (Sec. 4.4 of the technical supplement). The proton-tagged row sums, and the two matrices whose least-determined row sums to a small negative value (RHC  $W_{\pi p}$  and the first  $\cos \theta_\mu$  bin of the proton-tagged RHC sample), are tabulated and discussed in Sec. 8.5 of the technical supplement; a negative row is not forbidden for a regularisation operator, but those two rows should be read with corresponding caution.

**Numerical data release.** All fifty-one differential extractions are released in `report/data_release/`, indexed by `index_extractions.tsv`: the additional-smearing matrix, the full per-component covariance in cross-section units, and `curves*.tsv` holding the unfolded data together with every model curve shown here. The six one-bin totals, with their uncertainty breakdown and realised truth, are in `total_xsec.tsv`.

A prediction supplied as a per-bin cross section is compared with these results by

$$\left(\frac{d\sigma}{dx}\right)_i^{\text{smeared}} = \frac{1}{w_i} \sum_j (A_C)_{ij} p_j,$$

with  $p_j$  the predicted cross section in true bin  $j$  and  $w_i$  the width of bin  $i$ . This reproduces the published generator curves exactly — the worst per-bin deviation over the released matrices is  $1.2 \times 10^{-10}$  — so an external user can verify their implementation against any `curves*.tsv` column before applying it to a new model.

The check is part of the release procedure because it is sensitive to the one error an internal closure cannot see, a prediction smeared twice, which would miss it by up to 55%.

The practical consequence is stated in the results section and repeated here because it is easy to get wrong: quote the physical total from the one-bin extraction, which is not subject to  $A_C$ , and treat the sum of a Wiener-SVD differential result as a quantity in smeared space rather than as a cross section.

### 10.3 Total flux-averaged cross section (one-bin extraction)

The total flux-averaged cross section in the measured phase space is extracted with the same machinery as the differential results, as a *one-bin distribution*: a single true bin containing every signal event, defined on  $\cos \theta_\mu$  over its full range  $[-1, 1]$  so that no overflow bin exists, and a single reconstructed bin over the same range (the proton-tagged version uses the same construction on the proton-tagged selection). With one bin the response  $S = \epsilon M$  reduces to the efficiency, there is nothing to regularise, and the Wiener filter is switched off, so the “unfolding” is the plain inversion

$$\sigma = \frac{N_{\text{sel}} - N_{\text{bkg}}}{\epsilon \Phi N_{\text{Ar}}}, \quad (15)$$

with  $A_C \equiv 1$  exactly (released with the result in `data_release/total_xsec.tsv`). Here  $N_{\text{sel}}$  is the selected count,  $N_{\text{bkg}}$  the predicted background ( $\nu$ -background + EXT + dirt),  $\epsilon = N_{\text{sel}}^{\text{sig}}/N_{\text{gen}}^{\text{sig}}$  the selection efficiency in the measured phase space ( $\simeq 17.5\%$  for all configurations),  $\Phi$  the integrated flux ( $\Phi = \text{POT} \times \Phi_{\text{POT}}$ , Table 5) and  $N_{\text{Ar}} = 8.71 \times 10^{29}$  the number of argon nuclei in the fiducial volume (the dead  $z$  region removed), the same normalisation  $N_{\text{Ar}} \Phi / 10^{-38}$  used for the differential result. Every term of Eq. (15) is evaluated universe by universe, so the covariance contains the flux, cross-section, re-interaction and detector variations, the POT and target-count terms and the MC, EXT and data statistical terms, propagated exactly as for the differential results. Its flux term, 28–30% inclusive and 34–35% proton-tagged, is the flux normalisation times  $(1 + B/S)$  with  $B$  the neutrino background (§13.2); the detector term is 20–22% (12–27% proton-tagged) and the data-statistical term 3–4% (5–7%). The differential results carry the same terms bin by bin (flux 24–33%, detector 16–35%, statistics 6–18% bin-averaged); the regularisation redistributes them between bins rather than amplifying their sum. Because no additional smearing is applied the total is the physical one: it lies above the  $A_C$ -suppressed  $\sigma_{\text{int}}$  of Table 18, most visibly for RHC, and agrees with the D’Agostini cross-check (Sec. 4.4 of the technical supplement).

The closed-form cut-and-count evaluation of Eq. (15) on the central-value prediction is retained as the check of the normalisation chain: it reproduces the one-bin extraction of the central-value prediction to four digits (1.0309, 0.9285, 0.9753 inclusive; 0.6270, 0.5506, 0.5856 proton-tagged) and supplies the  $N_{\text{sel}}$ ,  $N_{\text{bkg}}$  and  $\epsilon$  columns of Table 19. The MC counts carry the framework’s full central-value weight,  $\text{tune} \times \text{PPFX} \times \text{normalisation}$  (§4); the PPFX central-value weight averages 0.944 on selected signal, so a count that omitted it would sit 3–6% above the prediction the unfolding actually uses.

Table 19 is the headline total-cross-section table, for the inclusive selection and for the proton-tagged subsample. The  $\sigma$  column is the one-bin extraction on the Poisson-thrown fake data, so it fluctuates about the injected tune by the throw; the realised truth of the throw is listed alongside, and the closure ratio is 0.975–0.998 with  $\chi^2/1 \ll 1$  in all six cases. The uncertainty column is what the measurement will carry at unblinding; it is the one-bin extraction’s own, not a proxy borrowed from a differential extraction, which would carry that extraction’s regularisation and depend on the throw through its unfolding operator. The RHC/FHC ratio of the injected tune is 0.901 (proton-tagged 0.878), the number to be compared with data once unblinded; on this throw the extracted ratio is 0.908 (0.860). Because the extraction is observable-independent, this is also the only proton-tagged total that should be quoted: the integrals of Tables 22–24 are  $A_C$ -suppressed and observable-dependent.

Table 19: Headline total flux-averaged cross sections ( $10^{-38} \text{ cm}^2/\text{Ar}$ ) in the measured phase space, from the one-bin extraction (§10.3) on the Poisson-thrown central-value fake data.  $N_{\text{sel}}$  and  $N_{\text{bkg}}$  are the predicted selected count and background ( $\nu$ -background + EXT + dirt) of the central-value prediction and  $\varepsilon$  the selection efficiency, from the cut-and-count check;  $\sigma$  is the one-bin extraction with its full universe-propagated uncertainty, “stat” its data-statistical term, “syst” the prediction total (flux, cross section, re-interaction, detector, POT, targets, MC and EXT statistics) and “total” their quadrature sum; “truth” is the realised truth of the throw and  $\chi^2/1$  the closure against it. MC counts carry the framework’s central-value weight (tune  $\times$  PPFX  $\times$  normalisation), the same as Table 10; the two differ by  $\lesssim 1\%$  because the yield comparison uses the Run-1 dirt overlay with per-mode scale factors rather than the per-run dirt files.

| Config                                                                                      | $N_{\text{sel}}$ | $N_{\text{bkg}}$ | $\varepsilon$ [%] | $\sigma$          | stat [%] | syst [%] | total [%] | truth | $\sigma/\text{truth}$ | $\chi^2/1$ |
|---------------------------------------------------------------------------------------------|------------------|------------------|-------------------|-------------------|----------|----------|-----------|-------|-----------------------|------------|
| <i>Inclusive CC <math>1\mu 1\pi X p</math></i>                                              |                  |                  |                   |                   |          |          |           |       |                       |            |
| FHC                                                                                         | 1612             | 668              | 17.4              | $1.026 \pm 0.397$ | 4.3      | 38.5     | 38.7      | 1.048 | 0.979                 | 0.003      |
| RHC                                                                                         | 1759             | 744              | 17.6              | $0.932 \pm 0.350$ | 4.1      | 37.4     | 37.6      | 0.941 | 0.990                 | 0.001      |
| Combined                                                                                    | 3371             | 1412             | 17.5              | $0.975 \pm 0.368$ | 3.0      | 37.7     | 37.8      | 0.990 | 0.985                 | 0.002      |
| <i>Proton-tagged CC <math>1\mu 1\pi 1p</math> (<math>p_p &gt; 0.3 \text{ GeV}/c</math>)</i> |                  |                  |                   |                   |          |          |           |       |                       |            |
| FHC                                                                                         | 723              | 347              | 11.4              | $0.618 \pm 0.255$ | 7.2      | 40.7     | 41.3      | 0.633 | 0.975                 | 0.004      |
| RHC                                                                                         | 786              | 395              | 11.4              | $0.531 \pm 0.251$ | 7.4      | 46.6     | 47.2      | 0.532 | 0.998                 | 0.000      |
| Combined                                                                                    | 1509             | 742              | 11.4              | $0.571 \pm 0.240$ | 5.2      | 41.8     | 42.1      | 0.579 | 0.986                 | 0.001      |

Against the standalone flux-averaged generator totals in the inclusive phase space (GENIE 0.82, GiBUU 1.13, NEUT 1.32, NuWro 1.25 for FHC; Table 14 of the technical supplement for all three configurations) the one-bin total sits  $0.5\text{--}0.8\sigma$  above GENIE and within  $0.8\sigma$  of the other three; the generator spread itself, a factor 1.6, is larger than any of these differences. For the proton-tagged phase space the FHC generator totals are GENIE 0.50, NuWro 0.65, GiBUU 0.74 and NEUT 0.78, bracketing the injected 0.63.

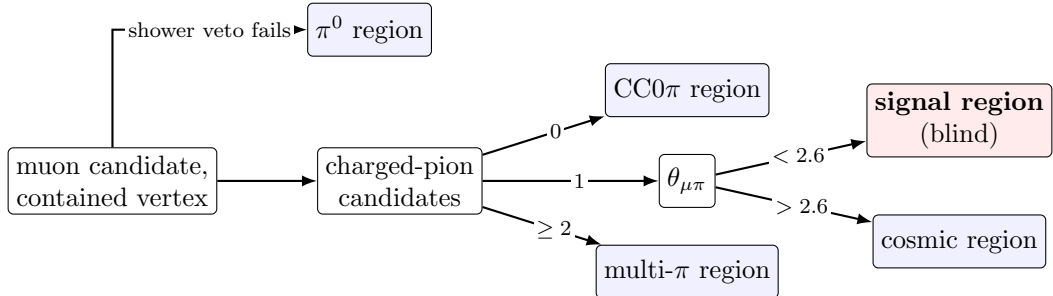


Figure 24: How the four control regions relate to the signal region. Each is one inverted requirement of the signal selection, so no event can be in both: the regions are disjoint from the signal region by construction, which is what allows them to be inspected on beam-on data while the signal region stays blind.

## 10.4 What the control regions show

The four regions have been opened on beam-on data. They test the background model at reconstruction level, with the signal region untouched, and the procedure they follow — the frozen definitions, the pre-agreed test statistic and the conditions under which a correction may be applied — is given in Sec. C of the technical supplement. Three things come out of them.

- **The background model needs no correction.** A joint FHC+RHC fit of the background class normalisations returns the dominant CC0 $\pi$  class at  $0.984 \pm 0.191$ , unity within a fifth of its own uncertainty (Sec. 6 of the control-sample note). The measurement keeps its nominal background prediction.

- **One run’s exposure is discrepant.** The global test passes in every period except FHC Run 1, where a free exposure factor fits  $0.520 \pm 0.020$  with  $\chi^2 = 16/17$ . The size and the sharpness of that number point to a bookkeeping error in the run’s exposure rather than to a mismodelled background; it is an open item (§14), and no result here rests on FHC Run 1 alone.
- **The remaining deficit is not signal contamination.** The regions sit 8–16% below prediction. Tightening each region so that it holds less signal — the CC0 $\pi$  region goes from 8.7% to 2.1% signal while keeping 58% of its events — leaves the data-to-prediction ratio where it was,  $0.84 \rightarrow 0.85$  (Sec. 7 of the control-sample note). Whatever causes the deficit does not scale with the signal content of the region.

## 10.5 Background-control sidebands

**The pseudo-data studies in this section are not a closure test of the complete beam-on sideband prediction.** The pseudo-data contain only the central-value neutrino-MC component, so the ratios reported here validate the selection logic, the run scaling and the central-value weighting only; the full-stack variant of Sec. D.2 of the technical supplement additionally exercises the bookkeeping path. At sideband unblinding, beam-on data will be compared with the full neutrino-MC + EXT + dirt prediction under the frozen procedure of Sec. D.5 of the technical supplement.

Because the analysis is blind, the background model cannot yet be confronted with beam-on data in the signal region. To exercise it before unblinding, four signal-depleted control regions (“sidebands”) are defined, each anchored by the inversion of one nominal signal requirement, with the requirements that the inversion leaves undefined dropped explicitly (branch-level definitions in Table 22 of the technical supplement):

- **CC0 $\pi$ :** the muon selection with no reconstructed charged pion (charged-current quasi-elastic /  $2p2h$ -enriched);
- **multi- $\pi$ :** two or more charged pions (deep-inelastic / resonant multi-pion);
- **$\pi^0$ :** a reconstructed shower present but the  $\pi^0$  shower veto failed (CC  $\pi^0$  / NC  $\pi^0$ );
- **cosmic:** the full signal selection with the muon–pion opening angle inverted ( $\theta_{\mu\pi} > 2.6$ ), enriched in out-of-time cosmic activity.

Each region is evaluated with the same per-run POT scaling and CV event weighting (§4) used for the measurement. Because the current “data” is a Poisson throw of the neutrino Monte Carlo central value only (it contains no EXT or dirt component), the meaningful closure quantity here is the ratio of the pseudo-data to the CV-weighted *neutrino* MC,  $N_{\text{data}}/N_{\nu\text{MC}}$ ; the EXT and dirt yields are shown for scale but enter the background-model test only once real beam-on data are available. Table 20 summarises the four regions for the three beam configurations.

Table 20: Background-control sidebands. “Signal fraction” is the signal contamination of the (CV-weighted neutrino) MC in the region; all four are strongly signal-depleted.  $N_{\text{data}}/N_{\nu\text{MC}}$  is the pseudo-data to neutrino-MC ratio: the high-statistics regions ( $\text{CC}0\pi$ ,  $\pi^0$ ) close to unity across FHC/RHC/combined, validating the per-run weighting and POT-scaling machinery; the low-statistics regions (multi- $\pi$ , cosmic) scatter within Poisson expectation ( $\sqrt{N} \approx 6\text{--}10\%$ ). The background-model test against the full stack (with EXT and dirt) is performed at control-region unblinding with beam-on data.

| Sideband        | Signal frac.<br>(of $\nu\text{MC}$ ) | $N_{\text{data}}/N_{\nu\text{MC}}$ |      |          |
|-----------------|--------------------------------------|------------------------------------|------|----------|
|                 |                                      | FHC                                | RHC  | Combined |
| $\text{CC}0\pi$ | $\sim 10\%$                          | 1.00                               | 0.99 | 1.00     |
| multi- $\pi$    | $\sim 23\%$                          | 1.01                               | 1.10 | 1.06     |
| $\pi^0$         | $\sim 10\%$                          | 1.00                               | 1.01 | 1.00     |
| cosmic          | $\sim 8\%$                           | 1.09                               | 0.99 | 1.04     |

**Yields and composition.** Table 21 gives the full breakdown for the three configurations. The regions differ enormously in size and in what they are sensitive to, which is what makes them complementary rather than redundant:

- **CC0 $\pi$**  is by far the largest ( $\sim 12\text{k}$  pseudo-data events per beam,  $\sim 24\text{k}$  combined). Its neutrino component is dominated by the quasi-elastic and  $2p2h$  modes that feed the signal region only through pion *absorption* in the final state. It is therefore the natural handle on the FSI model that converts  $\text{CC}1\pi^\pm$  into  $\text{CC}0\pi$  and vice versa; the largest single migration path into and out of the signal definition (§5).
- $\pi^0$  is the second largest (6.6–14.8k). It is enriched in  $\text{CC}/\text{NC} \pi^0$  events, whose showers are precisely what the shower veto (Cut 8) is designed to remove; it therefore tests the veto efficiency and the  $\pi^0$  production normalisation together. These are the events that leak into the signal region when a photon is missed.
- **multi- $\pi$**  is small (125–300 events) but has the highest signal contamination of the four ( $\sim 23\%$ ), because a true multi-pion event with one pion lost is exactly the wrong-topology background the selection fights. Its size is what limits its constraining power at present.
- **cosmic** is the region with by far the largest beam-off (EXT) share of the prediction: 53 EXT against 75 neutrino-MC events for FHC, a ratio of 0.7, against 0.1 in  $\text{CC}0\pi$  (with the per-run beam-off treatment of §4). It is the region that tests the beam-off normalisation and shape, and the one whose interpretation changes most at unblinding, since the current pseudo-data contain no EXT component at all. Its beam-off purity is only 41% (FHC; 34% RHC), so under the target-purity requirement of Sec. D.5 of the technical supplement it is diagnostic: it cannot by itself produce a fitted EXT normalisation correction, only an additional uncertainty.

**What the present numbers do and do not show.** With Poisson-thrown neutrino-MC pseudo-data, the ratios of Table 20 test the *machinery*, per-run POT scaling, CV weighting, the selection logic in each inverted region, and not the background model. That test passes: the two high-statistics regions sit at  $N_{\text{data}}/N_{\nu\text{MC}} = 0.99\text{--}1.01$  in all three configurations, which checks that the per-run scaling is applied identically on the throw and the prediction paths, not the normalisation itself. The apparent deviations in the small regions are statistical: with 82 cosmic-sideband events for FHC, a  $\sqrt{N}$  fluctuation is 11%, so the observed 1.09, and the RHC multi- $\pi$  1.10 (on 175 events, 8%), are within about one standard deviation.

**Use at unblinding.** Once the control regions are opened with beam-on data the same table becomes a genuine background test, and the statistical precision available is set by the yields above: the combined  $\text{CC0}\pi$  and  $\pi^0$  samples have 0.6% and 0.8% *raw data statistical* precision. Their effective constraint on the signal-region background is weaker and depends on the correlated transfer uncertainty (Sec. D.3 of the technical supplement): the transfer factors are small (0.03 and 0.02) but the transfer from a  $\text{CC0}\pi$  event to a selected  $\text{CC1}\pi$  topology is itself model dependent, so the achievable constraint is set by the cross-section and detector terms of that transfer rather than by the sample size. The cosmic region would constrain the EXT normalisation only to  $\sim 50\%$  (94 EXT events among  $\sim 250$  combined:  $\sqrt{257} = 16$  events of data statistics against the 32% prediction systematic on the 145 neutrino-MC events in the region,  $\pm 46$ ;  $\sim 17\%$  from data statistics alone); with the beam-off at 1.9% of the signal-region sample that is adequate. No uncertainty is currently assigned to the gate ratios themselves or to the Run-1 sample standing in for Run 2 (14% of the FHC and 23% of the RHC exposure); both are listed in §14. The multi- $\pi$  region, at  $\sim 300$  combined events, gives  $\sim 6\%$ , useful as a cross-check but not competitive with the others. The intended procedure is to unblind the control regions first, use them to test, and where item 4 of Sec. D.5 of the technical supplement permits to correct the normalisation of, the dominant background classes, and only then open the signal region.

Table 21: Sideband yields at full exposure. “Signal” is the signal contamination of the neutrino MC (events satisfying the inclusive signal definition of §5; all yields and the pseudo-data throw apply the safe-weight rule of Eq. (4)), “ $\nu$  bkg” the remaining neutrino MC, EXT the per-run beam-off (cosmic) samples scaled by the run gate ratios (§4), and dirt the out-of-cryostat overlay. The pseudo-data column is a Poisson throw of the neutrino MC only, so it should be compared with Signal +  $\nu$  bkg; EXT and dirt are listed for scale and enter the comparison only with real beam-on data.

| Beam | Sideband        | Pseudo-data | Signal | $\nu$ bkg | EXT   | Dirt | Sig. frac. |
|------|-----------------|-------------|--------|-----------|-------|------|------------|
| FHC  | $\text{CC0}\pi$ | 12 181      | 1 124  | 11 068    | 1 358 | 56   | 9.2%       |
| FHC  | multi- $\pi$    | 125         | 30     | 94        | 7.2   | 0.1  | 24.0%      |
| FHC  | $\pi^0$         | 6 628       | 662    | 5 985     | 778   | 12   | 10.0%      |
| FHC  | cosmic          | 82          | 6.0    | 69        | 53    | 2.2  | 7.9%       |
| RHC  | $\text{CC0}\pi$ | 12 590      | 1 206  | 11 452    | 1 214 | 43   | 9.5%       |
| RHC  | multi- $\pi$    | 175         | 35     | 124       | 5.9   | 0.1  | 22.0%      |
| RHC  | $\pi^0$         | 8 176       | 780    | 7 300     | 678   | 9.1  | 9.6%       |
| RHC  | cosmic          | 81          | 6.4    | 75        | 42    | 1.7  | 7.9%       |
| Comb | $\text{CC0}\pi$ | 24 771      | 2 329  | 22 520    | 2 572 | 99   | 9.4%       |
| Comb | multi- $\pi$    | 300         | 65     | 218       | 13    | 0.2  | 22.9%      |
| Comb | $\pi^0$         | 14 804      | 1 441  | 13 285    | 1 455 | 21   | 9.8%       |
| Comb | cosmic          | 163         | 12     | 145       | 94    | 3.9  | 7.9%       |

## 11 Secondary scope: proton-tagged subsample (hadronic invariant mass and transverse kinematic imbalance)

**Scope note.** This section is secondary scope. Its extractions carry the complete configured covariance, detector variations included, with detector terms of 10.5% to 108.5% (median 33.0%) against 15.6%–35.0% for the inclusive results: the same order, as expected for the same detector, except for the six-bin observables on the sparsest samples, where the unfolding amplifies them. The additional-smearing and covariance matrices are released with the rest (Sec. 10.2). What makes the scope secondary is a binning limitation that no systematic treatment addresses: the differential  $W_{\pi p}$  shape is withdrawn, because at this exposure no binning satisfies both the migration and

the statistics requirement, so the proton-tagged sample supports an integrated cross section, not a resonance-shape measurement; and the proton-mistag background has no dedicated control region yet (Sec. D.6 of the technical supplement). See Sec. 10.1.

**On the six-bin  $W_{\pi p}$  material below.** Any six-bin  $W_{\pi p}$  distribution in this section is *diagnostic*: it shows why the shape is not measurable and is not a result. The differential  $W_{\pi p}$  shape is withdrawn (Appendix C); the proton-tagged deliverable is the integrated cross section.

Requiring an identified proton in the final state, in addition to the muon and charged pion, reconstructs the full visible hadronic system and opens two classes of observable inaccessible to the inclusive  $\text{CC } 1\mu 1\pi Xp$  measurement: the *hadronic invariant mass*, which probes the  $\Delta(1232)$  and higher resonances directly, and the *transverse kinematic imbalance* (TKI) between the muon and the hadronic system, which is sensitive to Fermi motion and final-state interactions. The proton-tagged selection, its signal definition and its performance are given in §7.4; in brief, it adds a leading-proton identification to the inclusive selection, at an efficiency of 11.4% and a purity of 50–52%, and trades signal statistics for a fully reconstructed hadronic final state.

### 11.1 Extracted cross sections

Eleven observables are extracted in the proton-tagged subsample; ten are carried as *secondary validation products*, with their covariances but not proposed for approval in this review (the binning validation of  $W_{\text{had}}$  and the dedicated control region are incomplete, §14); the eleventh, the differential  $W_{\pi p}$  shape, is withdrawn (its files are in the release for completeness, flagged as such in `index_extractions.tsv`), and the robust proton-tagged deliverable proposed as a headline is the one-bin total cross section (Table 19). The ten are: the five hadronic-mass and transverse-kinematic-imbalance (W/TKI) variables specific to this selection ( $W_{\text{had}}$  and the four TKI observables), plus the five muon/pion kinematic observables of the inclusive analysis ( $p_\mu$ ,  $p_\pi$ ,  $\cos\theta_\mu$ ,  $\cos\theta_\pi$ ,  $\theta_{\mu\pi}$ ) re-measured on the proton-tagged phase space so that the two selections share a common set of kinematic distributions. The two hadronic-mass variables are the *directly measured*  $\pi p$  invariant mass,  $W_{\pi p} = [(E_\pi + E_p)^2 - (\vec{p}_\pi + \vec{p}_p)^2]^{1/2}$ , which makes no assumption on the neutrino direction, and the *calorimetric* hadronic mass  $W_{\text{had}} = [m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2]^{1/2}$  built from the reconstructed calorimetric energy. The four TKI observables; the transverse-momentum imbalance  $\delta p_T$ , its two angles  $\delta\alpha_T$  and  $\delta\phi_T$  [28], and the STV-inferred nucleon-momentum proxy  $p_n$  [29] (an approximation, see below), are computed from the muon versus the summed ( $\pi + p$ ) hadronic system using the framework STV tool. Proton momenta are obtained from the proton-hypothesis range-based kinetic energy.  $W_{\text{had}}$  is measured in six equal-population bins ( $W_{\pi p}$  was binned the same way for the diagnostic study in the supplement); the four TKI variables use the coarsened two-bin scheme adopted after the finer binning was found to sit inside the resolution (Sec. 4.1 of the technical supplement); the reverse-horn-current sample uses the same binning as FHC. The five inherited kinematic observables use the inclusive-analysis binning (two to seven bins; the proton-tagged  $\cos\theta_\pi$  keeps the former four-bin inclusive edges). All bin edges are set from the validated true-signal distributions (Table 7 of the technical supplement).

### Reconstruction from input quantities

All six observables are built from the three candidate tracks (muon, charged pion, proton) using only reconstructed quantities stored in the input ntuples: the track direction cosines  $\hat{n}$ , the muon-hypothesis range and multiple-scattering momenta and the proton-hypothesis calorimetric kinetic energy.

*Candidate three-momenta.* The muon momentum magnitude is range-based when the muon track ends inside the containment volume (§6.1), and multiple-Coulomb-scattering based otherwise.

The charged pion, being minimum-ionising, starts from the muon-hypothesis range momentum, converted to the pion hypothesis at fixed kinetic energy in the reconstructed-bin definition exactly as in the inclusive analysis (§7); the same two-bin configuration formula is used. The proton momentum follows from its calorimetric kinetic energy under the proton hypothesis,  $p_p = \sqrt{T_p^2 + 2m_p T_p}$  with  $T_p$  the calorimetric kinetic energy. Each three-momentum is  $\vec{p}_i = p_i \hat{n}_i$ , and the energies are  $E_\mu = \sqrt{p_\mu^2 + m_\mu^2}$ ,  $E_\pi = \sqrt{p_\pi^2 + m_\pi^2}$ ,  $E_p = T_p + m_p$  (masses  $m_\mu = 105.7$ ,  $m_\pi = 139.6$ ,  $m_p = 938.3$  MeV).

*Directly measured mass  $W_{\pi p}$ .* The invariant mass of the pion-proton four-vector sum, using no neutrino-direction assumption:

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}.$$

*Transverse kinematic imbalance.* The TKI observables treat the muon against the visible hadronic system  $\vec{p}_{\text{had}} = \vec{p}_\pi + \vec{p}_p$ ,  $E_{\text{had}} = E_\pi + E_p$ . With the components transverse to the beam ( $\hat{z}$ ) written  $\vec{p}_T^\mu$  and  $\vec{p}_T^{\text{had}}$ ,

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|, \quad \delta \alpha_T = \arccos \frac{-\vec{p}_T^\mu \cdot (\vec{p}_T^\mu + \vec{p}_T^{\text{had}})}{|\vec{p}_T^\mu| \delta p_T}, \quad \delta \phi_T = \arccos \frac{-\vec{p}_T^\mu \cdot \vec{p}_T^{\text{had}}}{|\vec{p}_T^\mu| |\vec{p}_T^{\text{had}}|},$$

both angles in degrees folded onto  $[0, 180]$ . For a stationary nucleon with no final-state interactions  $\delta p_T \rightarrow 0$ ; Fermi motion broadens it and FSI drive  $\delta \alpha_T \rightarrow 180^\circ$ .

*Calorimetric mass  $W_{\text{had}}$  and struck-nucleon momentum  $p_n$ .* The calorimetric available energy is  $E_{\text{cal}} = E_\mu + (E_{\text{had}} - m_p) + E_b$  with the argon removal energy  $E_b = 24.78$  MeV, giving the momentum transfer  $Q^2 = -[(0, 0, E_{\text{cal}}, E_{\text{cal}}) - p^\mu]^2$  and

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2}$$

(returned as 0 where reconstruction fluctuations make the argument negative). The nucleon-momentum proxy  $p_n$  combines the transverse imbalance with a longitudinal component  $p_L$  from energy-momentum balance,  $p_n = \sqrt{\delta p_T^2 + p_L^2}$ , using the MINERvA prescription [45] as implemented in the framework's transverse-kinematics tool (option 1):

$$p_L = \frac{R}{2} - \frac{M_{A'}^2 + \delta p_T^2}{2R}, \quad R = M_A + p_z^\mu + p_z^{\text{had}} - E_\mu - E_{\text{had}},$$

with the target mass  $M_A = 22 m_n + 18 m_p - 0.34381$  GeV (argon-40), the residual mass  $M_{A'} = M_A - m_n + E_b$  and  $E_b = 24.78$  MeV;  $z$  is the beam axis (§3.2). The tool was written for a single-proton hadronic system: it receives the summed three-momentum  $\vec{p}_\pi + \vec{p}_p$  and energy  $E_{\text{had}} = E_\pi + E_p$  in the proton slot and forms the hadronic kinetic energy as  $E_{\text{had}} - m_p$ , which is why  $m_p$  appears in  $E_{\text{cal}}$  and  $W_{\text{had}}$  above.  $\delta p_T$ ,  $\delta \alpha_T$  and  $\delta \phi_T$  depend only on the transverse momenta and are exact for the  $\mu + (\pi p)$  system;  $p_L$ ,  $p_n$  and  $W_{\text{had}}$  inherit the single-nucleon residual-mass and removal-energy assumptions of a quasi-elastic-oriented prescription and are therefore *proxies*, not the initial-state nucleon momentum. The generator predictions apply the identical formulas to the generator final states (Sec. 3.1 of the technical supplement), so the comparison is like for like; a truth-level closure of this prescription against a pion-production-specific definition, and a quantification of the bias from the residual-mass assumption, have not been performed and are listed in Sec. 14.

The pipeline is validated end-to-end on FHC with a fake-data closure: the Poisson-thrown central-value pseudo-data are unfolded with the same Wiener-SVD procedure (§9) and compared to the  $A_C$ -smeared truth. Table 22 shows the result. The bin-by-bin  $\chi^2/\text{ndf}$  is well below unity for every observable, confirming the unfolded pseudo-data are statistically consistent with

truth and that the response, background subtraction and normalisation are self-consistent. The integrated-cross-section ratios (0.96–1.08) are ratios to the  $A_C$ -smeared truth, so they measure the statistical scatter of this throw (and any bias), not the  $A_C$  suppression, which is common to numerator and denominator; the largest departures are in  $\cos\theta_\mu$  and  $p_\mu$ . The systematic treatment here includes the flux, cross-section, reinteraction and detector-variation terms.

The reverse-horn-current measurement is complete at the same binning (Table 23). All ten released RHC observables close positive, with  $\sigma_{\text{int}}$  in  $0.31\text{--}0.44 \times 10^{-38} \text{ cm}^2/\text{Ar}$  against 0.39–0.50 for FHC; these are  $A_C$ -suppressed integrals with different  $A_C$  per configuration, so their ratio is not the RHC/FHC ratio (that is 0.860 from the one-bin totals, Table 19). The closure gives  $\chi^2/\text{ndf} \lesssim 1$  for almost every observable.

One normalisation point is specific to this subsample and worth stating. The processed ntuples use two POT conventions: the files behind the inclusive selection store the *total* run MC POT duplicated in every file of a multi-file run, whereas the proton-tagged files store each file’s *own* POT. Normalising by the individual file POT would be correct for the first set but would scale every split of a multi-file run to the full run exposure in the second, inflating the simulated background by the split multiplicity ( $\sim 3\times$  for RHC) and driving the background-subtracted spectrum negative; normalising by the summed per-file POT has the mirror problem. The analysis therefore normalises each run by the sum of the *distinct* per-file POT values within that run, which reduces to the correct run total under either convention; FHC, with one MC file per run, is unaffected by the choice.

The combined-sample measurement is complete on the same binning, is included in the results tables and the data release, and covers all ten released combined proton-tagged observables.

Table 22: FHC fake-data closure for the twelve proton-tagged observables.  $\sigma_{\text{int}}$  is the integrated (unfolded) cross section over the measured phase space ( $10^{-38} \text{ cm}^2/\text{Ar}$ ); “unf./truth” is the ratio of integrals;  $\chi^2$  is the bin-by-bin comparison of unfolded pseudo-data to  $A_C$ -smeared truth (ndf = number of bins). The upper block is the five released W/TKI observables;  $W_{\text{had}}$  is at six bins, while the four TKI variables use the coarsened two-bin scheme adopted after the finer binning was found to sit inside the resolution; the six-bin  $W_{\pi p}$  closure rows are diagnostic only and are given in the technical supplement (Sec. 4.1 of the technical supplement); the lower block is the five kinematic observables inherited from the inclusive analysis. The small  $\chi^2/\text{ndf}$  validates the chain. The integral ratios compare the unfolded result with the  $A_C$ -smeared truth, so departures from unity reflect the statistical realisation of the throw and any residual extraction bias; they do not measure the suppression introduced by  $A_C$ . The third block ( $^\dagger$ ) is the two angular observables proposed as secondary scope (§11.3).

| Observable               | $\sigma_{\text{int}}$ | unf./truth | $\chi^2/\text{ndf}$ |
|--------------------------|-----------------------|------------|---------------------|
| $W_{\text{had}}$         | 0.389                 | 1.00       | 0.88/6              |
| $\delta p_T$             | 0.461                 | 0.98       | 0.01/2              |
| $\delta\alpha_T$         | 0.485                 | 0.96       | 0.02/2              |
| $\delta\phi_T$           | 0.474                 | 0.99       | 0.11/2              |
| $p_n$                    | 0.490                 | 0.98       | 0.06/2              |
| $p_\mu$                  | 0.496                 | 1.04       | 1.89/7              |
| $p_\pi$                  | 0.498                 | 0.98       | 0.01/2              |
| $\cos\theta_\mu$         | 0.445                 | 1.08       | 0.98/5              |
| $\cos\theta_\pi$         | 0.394                 | 0.99       | 0.30/4              |
| $\theta_{\mu\pi}$        | 0.469                 | 1.02       | 0.50/5              |
| $\theta_p^\dagger$       | 0.532                 | 0.98       | 1.92/4              |
| $\theta_{\pi p}^\dagger$ | 0.456                 | 0.99       | 0.16/3              |

Figure 25 shows the five released FHC W/TKI differential cross sections ( $W_{\text{had}}$  and the four TKI observables), overlaid with all four generator predictions; the withdrawn six-bin  $W_{\pi p}$  distribution is deliberately not shown here, and its diagnostic closure and generator curves are in the technical supplement (Sec. 4.1 of the technical supplement). The generator W/TKI predictions were regenerated at this binning (GENIE and GiBUU natively, NuWro and NEUT from their event vectors inside the SL7 container). All curves, data, tune and generators, are  $A_C$ -smeared, so the comparison is made consistently in the measurement space (Sec. 3.1 of the technical supplement). The predictions are built at truth level from each generator’s event-level final state, GENIE gst events, GiBUU final-event records, and the NuWro and NEUT event vectors reprocessed in their respective SL7-container environments, by applying the inclusive signal plus a leading true proton above 0.3 GeV/ $c$  and evaluating the six observables with the same formulas as the selection (§11.1), then combining the  $\nu_\mu$  and  $\bar{\nu}_\mu$  samples over the FHC flux to a per-argon cross section. The four generators span a factor of  $\sim 1.6$  in the integrated proton-tagged cross section, GENIE lowest (0.50), then NuWro (0.65), GiBUU (0.74) and NEUT highest ( $0.78 \times 10^{-38}$  cm<sup>2</sup>/Ar), bracketing the injected tune (proton-tagged one-bin total, Table 19). In the released two-bin TKI scheme the differences between generators are almost entirely normalisation: their shapes, and that of the tune, agree with the unfolded points within the uncertainties in every panel, so the present exposure does not discriminate between the generators’ treatments of Fermi motion and final-state interactions. Data and predictions are both computed about the beam axis (§3.2); a comparison in which one side used the detector  $z$  axis would show apparent shape disagreements of a factor of several, which is why the frame is stated with every transverse quantity.

Table 23: RHC fake-data closure for the twelve proton-tagged observables, binning and columns as in Table 22. All observables close positive with the convention-agnostic MC POT normalisation described above. The third block (<sup>†</sup>) is the two angular observables proposed as secondary scope (§11.3).

| Observable               | $\sigma_{\text{int}}$ | unf./truth | $\chi^2/\text{ndf}$ |
|--------------------------|-----------------------|------------|---------------------|
| $W_{\text{had}}$         | 0.314                 | 0.96       | 0.69/6              |
| $\delta p_T$             | 0.418                 | 1.10       | 0.52/2              |
| $\delta \alpha_T$        | 0.438                 | 1.08       | 0.19/2              |
| $\delta \phi_T$          | 0.389                 | 1.02       | 0.10/2              |
| $p_n$                    | 0.391                 | 1.00       | 1.06/2              |
| $p_\mu$                  | 0.380                 | 1.09       | 3.80/7              |
| $p_\pi$                  | 0.429                 | 1.09       | 1.21/2              |
| $\cos \theta_\mu$        | 0.392                 | 1.11       | 0.84/5              |
| $\cos \theta_\pi$        | 0.365                 | 1.10       | 6.01/4              |
| $\theta_{\mu\pi}$        | 0.367                 | 0.96       | 0.14/5              |
| $\theta_p^\dagger$       | 0.391                 | 0.99       | 0.04/4              |
| $\theta_{\pi p}^\dagger$ | 0.378                 | 1.02       | 0.12/3              |

Table 24: Combined FHC+RHC fake-data closure for the twelve proton-tagged observables at binning and columns as in Table 22. All ten close positive and every  $\chi^2$  is comfortably below its number of bins. This completes the proton-tagged measurement in all three beam configurations. Note that the combined result is an independent extraction, not a fluence-weighted average of the FHC and RHC ones, so it is not required to lie between them; see the discussion below. The third block ( $^\dagger$ ) is the two angular observables proposed as secondary scope (§11.3).

| Observable               | $\sigma_{\text{int}}$ | unf./truth | $\chi^2/\text{ndf}$ |
|--------------------------|-----------------------|------------|---------------------|
| $W_{\text{had}}$         | 0.345                 | 0.99       | 0.85/6              |
| $\delta p_T$             | 0.540                 | 1.09       | 0.17/2              |
| $\delta\alpha_T$         | 0.459                 | 1.01       | 0.02/2              |
| $\delta\phi_T$           | 0.503                 | 1.09       | 0.24/2              |
| $p_n$                    | 0.449                 | 1.02       | 0.85/2              |
| $p_\mu$                  | 0.437                 | 1.03       | 1.53/7              |
| $p_\pi$                  | 0.462                 | 1.03       | 0.40/2              |
| $\cos\theta_\mu$         | 0.410                 | 1.06       | 1.52/5              |
| $\cos\theta_\pi$         | 0.382                 | 1.02       | 3.40/4              |
| $\theta_{\mu\pi}$        | 0.445                 | 1.04       | 0.26/5              |
| $\theta_p^\dagger$       | 0.485                 | 1.01       | 0.67/4              |
| $\theta_{\pi p}^\dagger$ | 0.413                 | 1.01       | 0.06/3              |

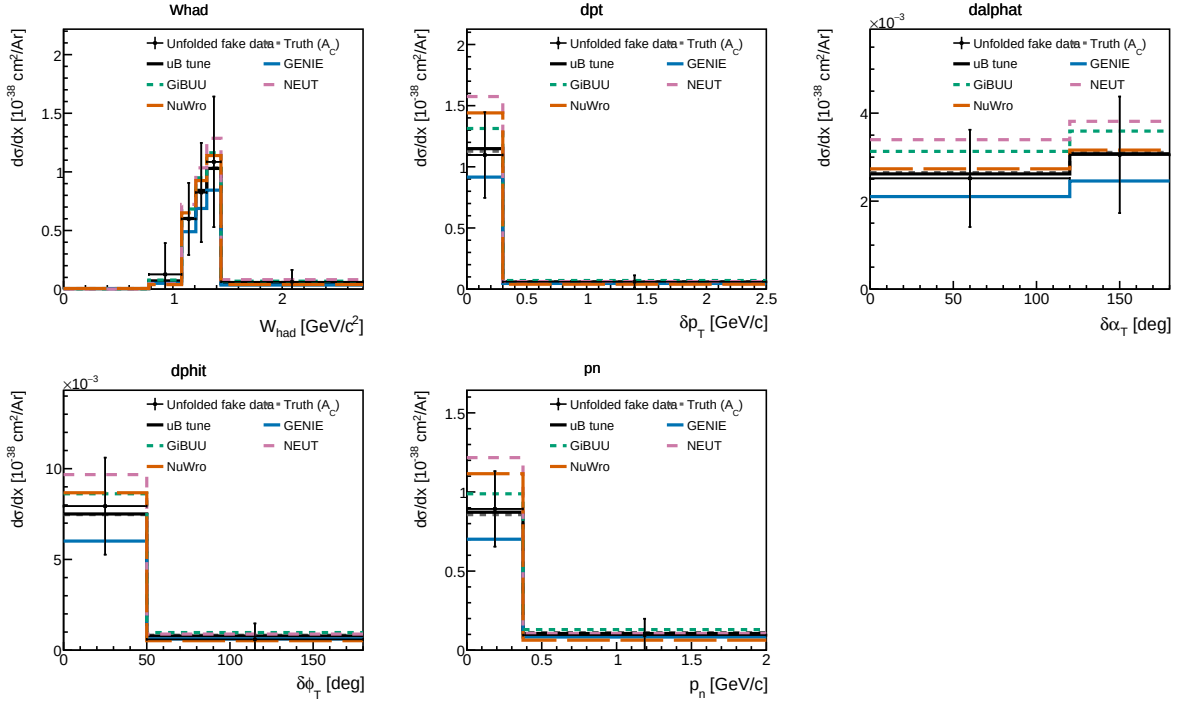


Figure 25: Proton-tagged released differential cross sections for FHC: the calorimetric hadronic mass  $W_{\text{had}}$  and the four transverse-kinematic-imbalance observables  $\delta p_T$ ,  $\delta\alpha_T$ ,  $\delta\phi_T$  and  $p_n$  (the withdrawn  $W_{\pi p}$  shape is not shown; see the technical supplement). Unfolded fake data (black points) are compared to the  $A_C$ -smeared fake-data truth (grey dashed), the MicroBooNE tune (black), and the GENIE (blue), GiBUU (green), NEUT (purple) and NuWro (orange) predictions built for the same proton-tagged phase space.  $\delta\alpha_T$  rises toward  $180^\circ$ , the signature of final-state interactions.

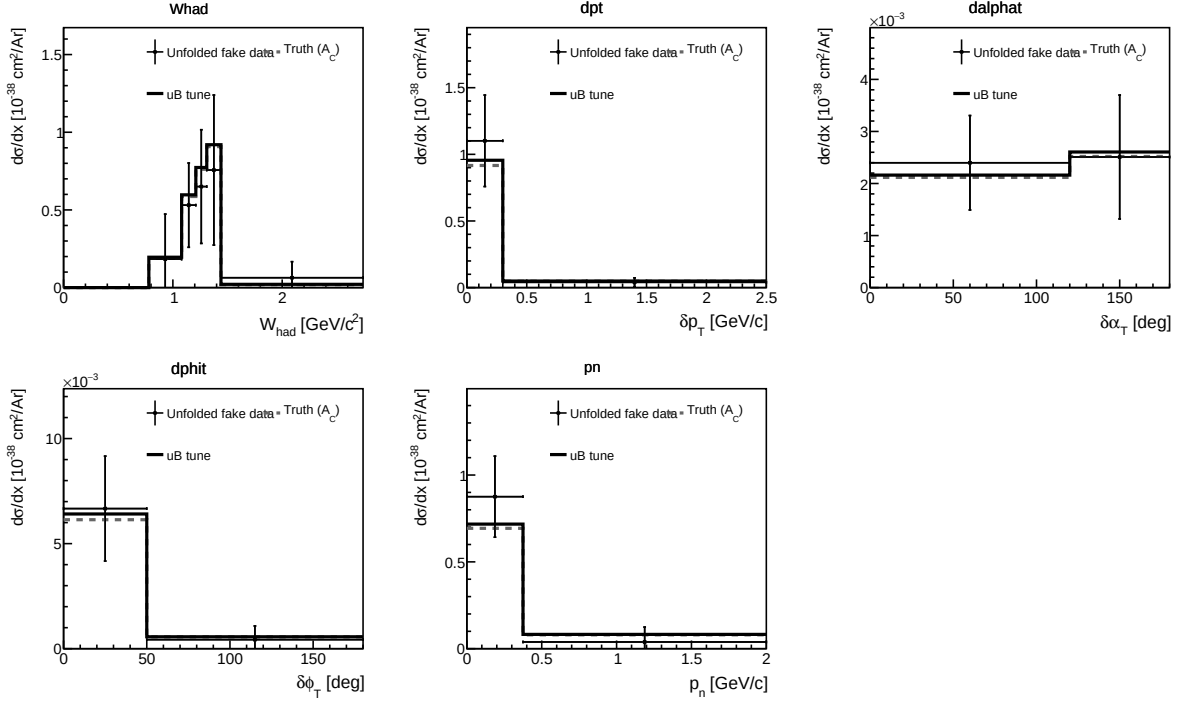


Figure 26: The same five proton-tagged observables for RHC. Curves and points as in Fig. 25.

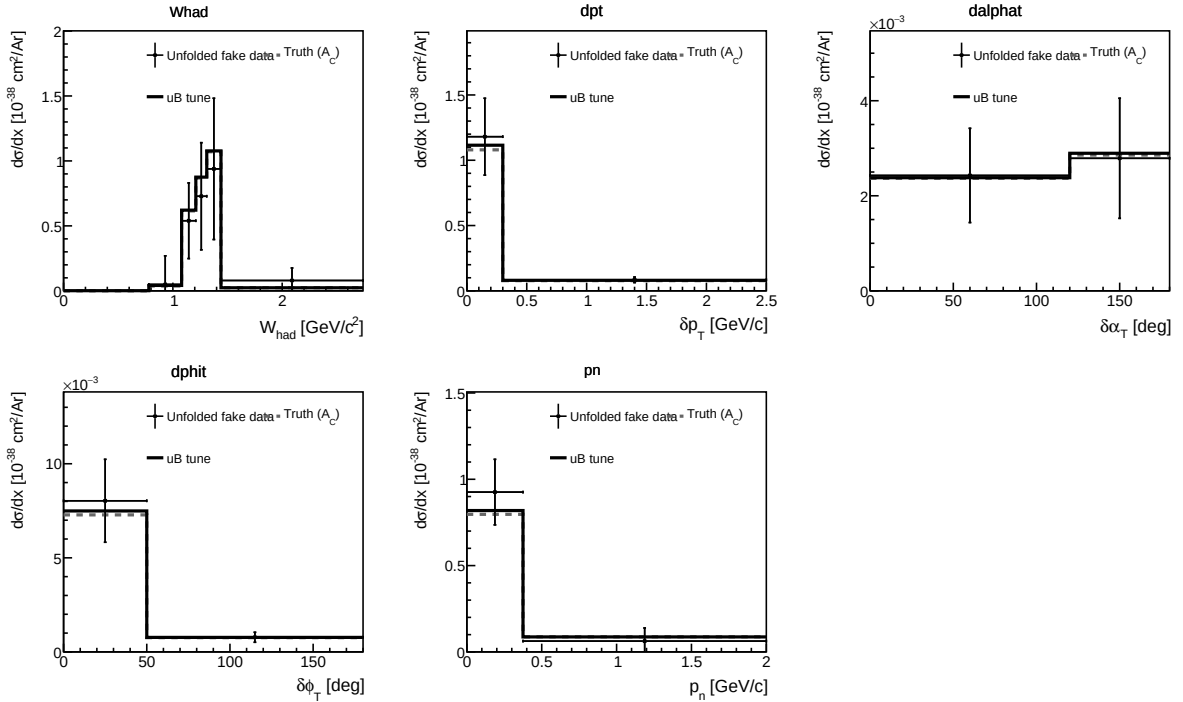


Figure 27: The same five proton-tagged observables for the combined configuration. Curves and points as in Fig. 25.

With all three configurations in hand, the combined  $\sigma_{\text{int}}$  lies between the FHC and RHC values for eight of the ten released observables;  $\delta p_T$  (0.540 against 0.461 and 0.418) and  $\delta\phi_T$  (0.503 against 0.474 and 0.389) lie above both, by amounts well inside their  $\sim 30\%$  uncertainties.

That is not a defect: the combined measurement is *not* required to lie between the two single-mode ones. It is an independent extraction with its own response matrix, its own regularisation

and its own  $A_C$ , not a fluence-weighted average, so excursions at the few-per-cent level would have been unremarkable against a  $\sim 30\%$  systematic uncertainty.

The spread across observables within a configuration, 0.35 to 0.54 for the combined set, is likewise not a physics spread but the residual of the additional smearing discussed in the next section: each observable's  $A_C$  rescales the integral by a different factor, so the ten numbers are not ten independent measurements of the same quantity.

## 11.2 What the proton candidate is

Every proton-tagged observable is built on the reconstructed proton candidate, while the truth-level definitions use the leading true proton. On selected signal events the candidate is the leading true proton 81.7% of the time, some true proton 87.4%, and not a proton at all 12.6% (10 105 events; `bkgfit/protonmatch.py`).

What that costs was measured for the two angular observables, and the answer depends on which question is asked. A correct pick reconstructs the angle tightly (interquartile range 0.073 rad in  $\theta_p$ , 0.124 rad in  $\theta_{\pi p}$ ); a wrong pick does not smear the angle, it randomises it (1.0–1.4 rad, three quarters of those events beyond 0.3 rad). Averaged over everything the candidate widens the *core* resolution by only a quarter to a third, which understates it badly, because a binned measurement is not limited by its core. Residuals beyond 0.3 rad occur in 18.9% of  $\theta_p$  events and 74% of those are *wrong picks* ( $\theta_{\pi p}$ : 25.9% and 51%). Candidate purity is therefore the dominant source of bin-to-bin migration in  $\theta_p$ , which is why its migration diagonal sits at the criterion and its RHC response is rank-deficient.

The same 18% wrong-pick rate applies to  $W_{\pi p}$ ,  $W_{\text{had}}$  and the TKI observables, which use the same candidate; the migration attribution above was measured for the angles only. The control region that would constrain this is the multi- $\pi$  sideband, which is already open: `pion_number` is defined by the same PID cut that picks the proton, so an otherwise-signal event whose proton fails it has two pion candidates and lands there. It constrains the proton  $\rightarrow$  pion direction rather than the pion  $\rightarrow$  proton one that produces the 12.6%, and it is the most signal-contaminated of the four regions, so a region inverting the proton PID at exactly one pion candidate is what would separate them.

## 11.3 Proton angular observables: $\theta_p$ and $\theta_{\pi p}$

Two angular observables of the proton-tagged sample are proposed as secondary scope. Like every other number in this note they are fake data. Both use the leading proton candidate:

$$\theta_p = \arccos(\hat{p}_p \cdot \hat{\nu}), \quad \theta_{\pi p} = \arccos(\hat{p}_\pi \cdot \hat{p}_p),$$

with  $\hat{\nu}$  the beam-frame neutrino direction of §3.2 — the same convention as every other angle here — while  $\theta_{\pi p}$  is an angle between two reconstructed tracks and so needs no beam direction at all, which makes it the one angular observable in this analysis that is insensitive to the beam-frame convention.  $\theta_p$  is available from the stored proton direction;  $\theta_{\pi p}$  required the proton-tagged tree to be reprocessed to write `pi_pr_opening_angle_{reco,true}`, gated on a bit-for-bit comparison of every existing branch (56/56 files) before the old trees were rotated aside.

They measure something the released set does not. The five W/TKI observables are built from the muon against the *summed* hadronic system, so they are insensitive to how that system shares its momentum between pion and proton;  $\theta_{\pi p}$  is exactly that sharing.  $\theta_p$  is the proton direction alone, where the released set has only the derived quantities.

Both unfold well (Tables 22–24).  $\theta_{\pi p}$  is the best-conditioned observable of the proton-tagged programme,  $s_{\min}/s_1 = 0.30\text{--}0.48$  against 0.27/0.24 for the released  $\cos\theta_\pi$ , at the price of a last bin whose total uncertainty reaches 93% (FHC) and 187% (RHC).  $\theta_p$  conditions at 0.383 (FHC) and 0.408 (combined) but is rank-deficient in RHC (0.001), so it is proposed for FHC and the combined configuration only; coarsening the RHC binning was tried and does not rescue

it. Its edges are derived on the RHC chain rather than the FHC one: at the same four bins they condition three times better in both usable configurations. Both distributions are strongly forward peaked, and the generators differ mainly in normalisation rather than shape, so these observables discriminate through the integral. The study behind all of this — binning, generator predictions, the 0.50-criterion screen and the two-dimensional programme — is a separate note.

Proton-tagged  $\text{CC}1\mu1\pi^\pm$ : new observables, FAKE DATA (signal region blind); curves and integrated  $\sigma [10^{-38} \text{ cm}^2/\text{Ar}]$  are  $A_C$ -smeared

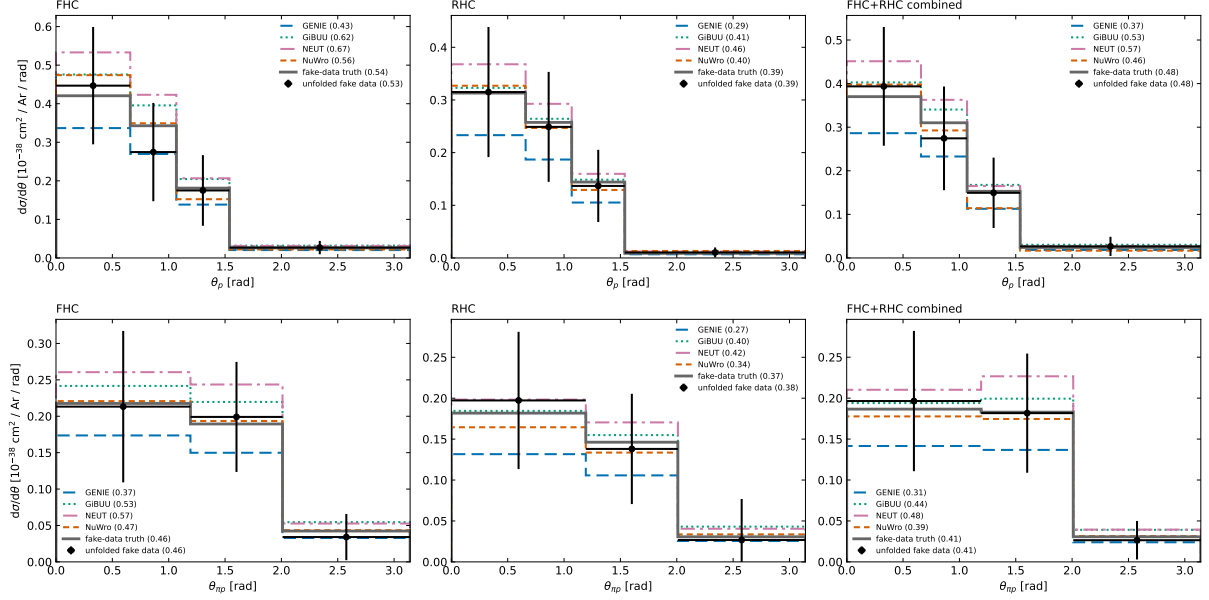


Figure 28: The two proton angular observables in all three configurations:  $\theta_p$  (top row) and  $\theta_{\pi p}$  (bottom row), for FHC, RHC and the combined exposure. Points are the unfolded fake data with the total uncertainty, the grey histogram is the  $A_C$ -smeared fake-data truth, and the four coloured histograms are the generator predictions; integrated cross sections are given in the legends. The RHC  $\theta_p$  panel is shown for completeness only: its response is rank-deficient and its shape is not a measurement.

## 11.4 Two-dimensional cross sections (proposed)

Eight two-dimensional pairs were screened and extracted in all three configurations. The framework needed no new unfolding machinery — a two-dimensional binning is a flat list of analysis bins ordered slice by slice — and the integrated normalisation of the two-dimensional chain agrees with the released one-bin total to a median of 0.7% once each is compared with its own  $A_C$ -smeared truth. Two pairs are worth proposing, both in the combined configuration only:  $\theta_p \times \delta p_T$  ( $s_{\min}/s_1 = 0.617$ ) and  $\cos\theta_\pi \times \cos\theta_\mu$  (0.563). Combining the beam modes roughly doubles their conditioning ( $0.46 \rightarrow 0.62$ ,  $0.25 \rightarrow 0.56$ ), which is the clearest benefit the joint exposure delivers anywhere in this analysis, and it comes from statistics and the shared flux rather than from any  $\nu/\bar{\nu}$  separation. Everything built on  $W_{\pi p}$  is unusable, as its 39% leakage between outer bins predicted.

One pair is worth recording as a caution.  $\theta_{\pi p} \times W_{\pi p}$  conditions at 0.560, as well as the two usable pairs, and its extraction is still worthless: a 58% data-statistical term, per-bin uncertainties to 545% and a closure ratio of  $-0.40$ . Conditioning bounds how much of the response can be inverted and says nothing about how much data survives in the inverted directions; the two have to be read together, here and everywhere else in this note.

## 12 Validation

Nothing in this section uses beam-on data. Every study here asks whether the machinery does what it claims on simulation, where the right answer is known: whether the extraction returns the truth it was given, whether the quoted uncertainties are the right size, and whether the background fit that the control regions feed is unbiased. The tests that do use beam-on data — the control-region comparison itself — are in the control-sample note.

### 12.1 Self-closure of the extraction

The first test is the weakest and the most necessary: unfold a spectrum thrown from the same simulation that built the response, and require the truth back. This catches an inverted matrix, a misaligned binning or a normalisation applied twice, and it is reported with every extraction in this note as the closure  $\chi^2$  against the  $A_C$ -smeared truth. It cannot detect a wrong model, because the model is the same on both sides, and it cannot detect a mis-sized uncertainty, because one throw says nothing about a distribution of throws. Those are the next two subsections.

### 12.2 Ensemble coverage: are the uncertainties the right size?

A single closure throw cannot say whether the quoted uncertainty is correctly sized, so thirty-two statistically independent Poisson realisations of the FHC data were each put through the whole chain, universes rebuilt from that throw, and the per-bin pull  $p_i = (\hat{x}_i - (A_C x_{CV})_i) / \sigma_i$  formed against the fixed central-value truth smeared by that member’s own  $A_C$ , for three extractions chosen to bracket the conditioning:  $p_\mu$ ,  $\cos \theta_\mu$  (whose  $A_C$  row sums fall to 0.17 in FHC) and the one-bin total (no regularisation).

| Reference: statistical covariance | pull mean | pull width | 68% / 95% coverage |
|-----------------------------------|-----------|------------|--------------------|
| $d\sigma/dp_\mu$                  | −0.05     | 1.08       | 62.1% / 92.0%      |
| $d\sigma/d \cos \theta_\mu$       | −0.18     | 0.97       | 71.2% / 93.1%      |
| one-bin total                     | −0.25     | 1.02       | 62.5% / 93.8%      |
| expected                          | 0         | 1          | 68% / 95%          |

The widths are consistent with unity, each determined to  $\pm 13\%$  with this many members, and the coverage is correct within the same statistics: **the statistical uncertainties are correctly sized, and the strongly suppressed observable behaves no differently from the well-conditioned one.** Against the full covariance the pull widths are 0.33–0.35 (total: 0.11) with essentially 100% coverage, because the ensemble varies only the data statistics while the quoted uncertainty carries the flux, detector and cross-section terms that are identical in every member; that is the measured form of the closure  $p$ -values clustered near unity. The ensemble means sit below the reference by 0.5% ( $p_\mu$ ), 1.0% (total) and 2.7% ( $\cos \theta_\mu$ ), at 0.4, 1.3 and 1.9 standard deviations of the mean, all from the same throws: one measurement of a small negative offset, not three, not significant, and not corrected. The throw itself is unbiased (mean selected count 0.999 of the expectation). The study covers three extractions in one configuration; RHC and the proton-tagged family have not been tested. The reference construction and the full-covariance pulls are discussed in Sec. 8.6 of the technical supplement.

### 12.3 The background fit, validated on pseudo-data

- **Toys:** see §12.4 for ensembles of up to  $10^5$  toys per model; the 300-toy sets of Fig. 29 give pull means 0.02–0.07, widths 0.89–1.03 and coverage 67–75%.
- **Injections** (Asimov, two-parameter totals fit): class shifts are recovered exactly; a 30% signal excess moves the normalisations by  $0.13\sigma$ ; an unfitted class at +30% by  $-0.07\sigma$ ; a

detector knob by at most  $0.14\sigma$ ; flux and GENIE universes by  $-0.97$  to  $+0.84\sigma$ , i.e. within the uncertainty they are part of.

- **FHC versus RHC**: a likelihood-ratio test of shared against beam-specific normalisations inside the joint covariance (a parameter goodness-of-fit test is not valid here, the modes share universes). A  $+15\%$  shift of the RHC neutrino Monte Carlo alone gives  $\Delta\chi^2 = 12.9/2$  ( $p = 0.002$ , totals) and  $20.3/2$  ( $p = 4 \times 10^{-5}$ , 2D).
- **MCMC** (adaptive Metropolis, four chains, flat priors): posterior means within  $0.02$  and widths within  $2\%$  of MIGRAD/MINOS on the Asimov dataset and on a  $CC0\pi \times 1.25$  injection,  $\hat{R} \leq 1.004$ . With a few parameters and a near-quadratic likelihood the profile likelihood is sufficient; the sampler is kept as a cross-check.

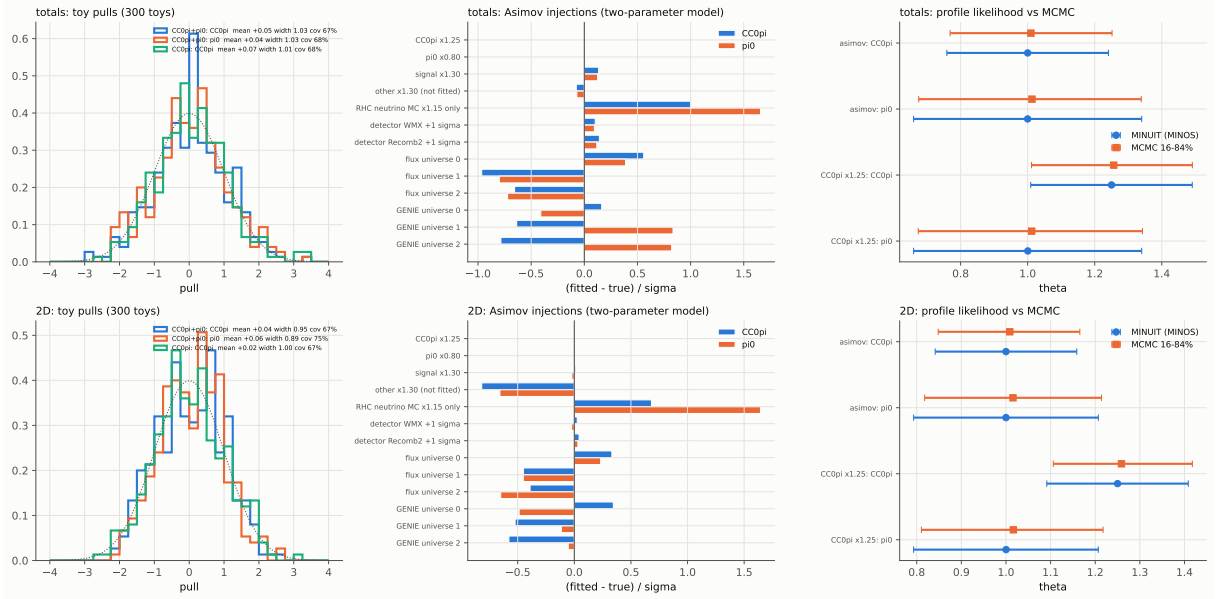


Figure 29: Validation, totals (top) and 2D (bottom): toy pulls, Asimov injections, profile likelihood against MCMC.

## 12.4 Pull-mean convergence of the background fit

With 300 toys the error on a pull mean is  $0.058$ , so the  $+0.07$  of the first validation could not be separated from zero. Ensembles of  $10^5$  toys per model (error  $0.003$ ) with three diagnostic variants, each of  $5 \times 10^4$ , settle it (Table 25, Fig. 30). With the combined Neyman–Pearson data variance the mean does *not* converge to zero in the 2D fits: a residual  $+0.019$  ( $CC0\pi$ ) and  $+0.034$ ,  $+0.031$  ( $CC0\pi + \pi^0$ ) remains at six to ten standard deviations of the ensemble. It persists when the Poisson fluctuation is replaced by a Gaussian one ( $+0.031$ ), so it is not Poisson counting in low bins; it disappears when the data variance is the prediction at the nominal parameters ( $+0.000$ ,  $-0.004$ ,  $-0.001$ ). The CNP variance depends on the fluctuating data, which correlates the weight of a bin with its own fluctuation. The bias is small —  $0.3$ – $0.7\%$  on  $\theta$  against a  $16$ – $21\%$  uncertainty — but it is avoidable, so the Pearson form is the default of the fit and every data result below uses it. Pull widths are  $1.00$  and coverage  $68\%$  in every variant.

Table 25: Pull means  $\pm$  their ensemble error. “standard”: systematic throw from the covariance, Poisson, CNP data variance ( $10^5$  toys); “Gaussian”: Gaussian instead of Poisson fluctuation; “Pearson”: standard toys fitted with the prediction as data variance ( $5 \times 10^4$  each).

| Bins   | Model (parameter)              | standard, CNP      | Gaussian, CNP      | standard, Pearson  |
|--------|--------------------------------|--------------------|--------------------|--------------------|
| totals | CC0 $\pi$                      | $+0.003 \pm 0.003$ | $+0.005 \pm 0.005$ | $-0.011 \pm 0.005$ |
| totals | CC0 $\pi + \pi^0$ (CC0 $\pi$ ) | $+0.012 \pm 0.003$ | $+0.014 \pm 0.005$ | $+0.003 \pm 0.005$ |
| totals | CC0 $\pi + \pi^0$ ( $\pi^0$ )  | $+0.016 \pm 0.003$ | $+0.014 \pm 0.005$ | $+0.008 \pm 0.005$ |
| 2D     | CC0 $\pi$                      | $+0.019 \pm 0.003$ | $+0.031 \pm 0.005$ | $+0.000 \pm 0.005$ |
| 2D     | CC0 $\pi + \pi^0$ (CC0 $\pi$ ) | $+0.034 \pm 0.003$ | $+0.031 \pm 0.005$ | $-0.004 \pm 0.005$ |
| 2D     | CC0 $\pi + \pi^0$ ( $\pi^0$ )  | $+0.031 \pm 0.003$ | $+0.029 \pm 0.005$ | $-0.001 \pm 0.005$ |

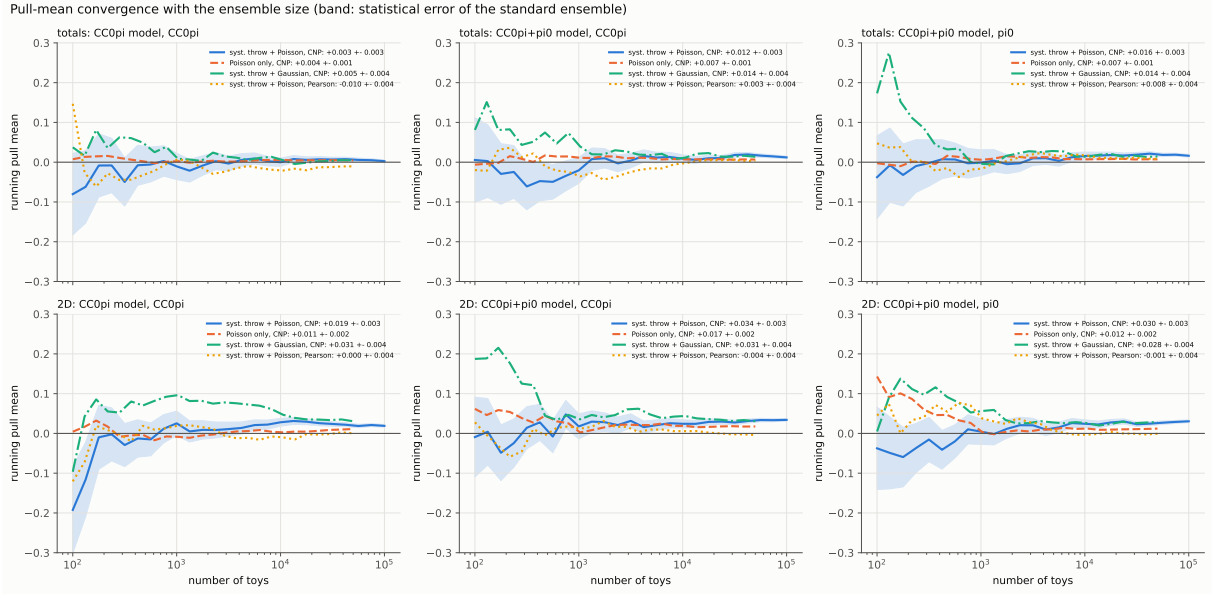


Figure 30: Running pull mean against the ensemble size for each model and parameter; band: statistical error of the standard ensemble. The Poisson-only variant has no systematic throw, so its pulls are narrower.

## 13 Systematic Uncertainties

The configured systematic covariance is propagated for all fifty-one extractions, complete with respect to the configured model (the identified unpropagated terms, the MCS momentum scale, the beamline geometry, the beam-off gate ratios and the Run-2 beam-off stand-in, are listed in §14). Every result carries flux, cross-section-model, hadron-reinteraction, detector-variation, POT, target-count and statistical terms, propagated through the unfolding by the error propagation matrix and released per component (Sec. 10.2). Table 26 lists the sources.

Table 26: Systematic sources and their bin-averaged fractional contribution, computed from the released per-component covariance rather than estimated. Ranges span observables and configurations; the two families are given separately because the proton-tagged sample is both statistically sparser and more detector-sensitive.

| Source                  | Method                            | Inclusive (%)    | Proton-tagged (%) |
|-------------------------|-----------------------------------|------------------|-------------------|
| Flux (PPFX multisims)   | PPFX multisim weights             | 24.2–32.5        | 18.3–74.6         |
| Detector response       | Dedicated samples                 | 15.6–35.0        | 10.5–108.5        |
| Cross-section model     | GENIE multisim and unisim weights | 8.5–16.9         | 7.3–53.8          |
| Hadron re-interaction   | Geant4 reweight                   | 3.5–11.9         | 3.6–24.9          |
| POT counting            | Beam toroids                      | 2.7–3.8          | 2.2–8.7           |
| Target count            | FV geometry                       | 1.4–1.9          | 1.1–4.4           |
| MC statistics           | universe spread                   | 1.9–6.3          | 2.3–20.0          |
| EXT statistics          | beam-off sample                   | 0.6–3.1          | 0.6–4.4           |
| Data statistics         | thrown fake data                  | 6.3–18.3         | 7.4–62.1          |
| <b>Prediction total</b> | quadrature sum                    | <b>31.0–50.2</b> | <b>23.2–133.6</b> |
| <b>Total</b>            | incl. data stats                  | <b>32.4–53.9</b> | <b>25.1–147.7</b> |

Flux and detector response are the two largest terms in both families; which of the two leads depends on the observable and configuration (Table 27; Table 16 of the technical supplement–Table 17 of the technical supplement). Every entry is propagated into the released covariance; no source declared in the release systematics configuration is left unpropagated. That is a statement about the configured model, not a completeness claim: the beamline-geometry flux weights exist in the ntuples but are not in the release configuration, and the other identified omissions (MCS momentum scale, beam-off gate ratio, the Run-2 beam-off stand-in) are listed in §14.

### 13.1 Systematic breakdown by source

The framework builds a covariance matrix for every systematic. The bin-averaged fractional uncertainty from each source group for the released FHC extractions is given below; the RHC and combined tables, generated from the same release dumps, are Table 16 of the technical supplement and Table 17 of the technical supplement; the numbers are those of Table 15.

Table 27: FHC (Runs 1, 2, 4, 5): bin-averaged fractional uncertainty (%) by source group for the six released extractions, read from the release systematics dumps “Prediction total” excludes the data statistics; the flux row is the PPFX multisim term. Every row is a bin average of the per-bin fractional uncertainties, so the rows do not combine in quadrature to the totals (the quadrature is taken per bin before averaging).

| Source                         | $p_\mu$     | $p_\pi$     | $\cos \theta_\mu$ | $\cos \theta_\pi$ | $\theta_{\mu\pi}$ | $\theta_\mu$ |
|--------------------------------|-------------|-------------|-------------------|-------------------|-------------------|--------------|
| <b>Prediction total</b>        | <b>40.7</b> | <b>50.2</b> | <b>42.0</b>       | <b>38.8</b>       | <b>36.5</b>       | <b>31.0</b>  |
| Cross section (GENIE)          | 10.7        | 16.9        | 10.3              | 11.3              | 10.5              | 8.5          |
| Flux (PPFX)                    | 28.0        | 32.5        | 28.0              | 27.0              | 28.3              | 24.7         |
| Detector                       | 24.7        | 30.3        | 28.3              | 22.9              | 19.6              | 15.6         |
| Reinteraction                  | 4.4         | 11.9        | 3.6               | 5.2               | 4.0               | 3.5          |
| MC stat                        | 4.9         | 6.3         | 5.3               | 4.3               | 2.7               | 2.9          |
| EXT stat                       | 0.9         | 3.1         | 2.3               | 1.5               | 1.2               | 1.4          |
| Data stat                      | 14.3        | 18.3        | 15.4              | 12.7              | 8.2               | 8.6          |
| POT + targets                  | 3.4         | 4.2         | 3.6               | 3.4               | 3.5               | 3.0          |
| <b>Total (incl. data stat)</b> | <b>43.5</b> | <b>53.9</b> | <b>44.9</b>       | <b>41.0</b>       | <b>37.5</b>       | <b>32.4</b>  |

The NuMI flux uncertainty has two parts: the PPFX hadron-production term (the dominant systematic) and a beamline-geometry term. The latter is built from the 12 geometry-variation weights (`Horn_2kA`, `Horn1/2_x/y_3mm`, `Beam_spot_*`, `Beam_shift_*`, `Horns_*_water`, `Target_z_7mm`) injected into the numuMC overlay, summed in quadrature as two-sided unisim variations. Evaluated in a dedicated pass it is 1.6–3.3% across the observables and sub-dominant; it is *not* part of the released covariance, whose configuration omits the beamline block. Added in quadrature to the  $\sim 28\%$  PPFX term it would raise the flux row by  $\lesssim 0.2\%$  absolute, so the omission is immaterial to the budget, but the term is listed as absent rather than propagated. The per-bin breakdowns are shown in Fig. 31.

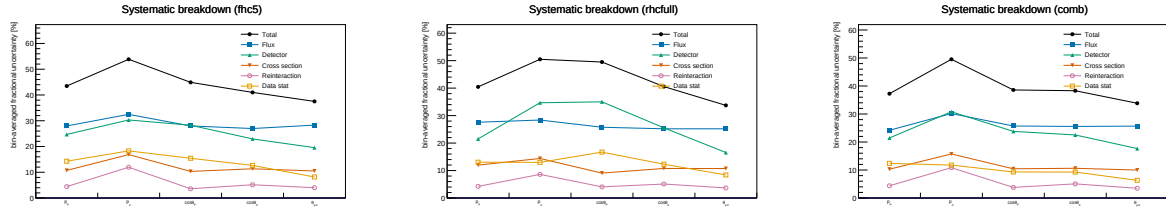


Figure 31: Bin-averaged fractional systematic uncertainty by source vs. observable, for FHC (left), RHC (centre) and combined (right); the figure form of Table 15 of the technical supplement–Table 17 of the technical supplement. Colours use the Okabe-Ito colourblind-safe palette with distinct markers per source: total (black), flux (blue), detector (green), cross section (vermillion), reinteraction (purple), data stat (orange).

### 13.2 Anatomy of the flux term: why 17% becomes 30%

**Why a 17% flux uncertainty becomes 28–30% on the total.** A flux universe scales signal and neutrino background together, and the extracted signal is  $(N_{\text{pred}} - B)/(\epsilon\Phi N_{\text{Ar}})$ , so its fractional flux variation is  $\delta\Phi/\Phi \times (1 + B/S)$ :  $17.7\% \times (1 + 637/943) = 29.7\%$  for FHC against the one-bin value 30.25%. The one-bin total has no regularisation, so this is background subtraction, not unfolding amplification. The lever is purity or a data-driven background constraint (§10.5).

The unfolded PPFX term is 24–33% (Tables 15–17) and the one-bin total carries 28–30% (Table 19), against the  $\sim 17\%$  normalisation uncertainty of the NuMI flux itself. The difference is not amplification by the unfolding operator: the one-bin total, which has no regularisation, carries it too. The factor is the background subtraction. A flux universe scales the whole reconstructed prediction, signal and neutrino background alike, and the framework’s covariance is that of the prediction of the measured quantity,  $(N_{\text{pred}} - B)/(\epsilon\Phi N_{\text{Ar}})$  with  $B$  and  $\Phi$  moving together in every universe; relative to the signal that is a fractional variation of  $\delta\Phi/\Phi \times (1 + B/S)$ , with  $B$  the neutrino background (the beam-off and dirt carry no flux weight). With  $B/S = 637/943$  (FHC, Table 10) and a reconstructed-level flux term of 17.7% this gives 29.7% against the one-bin

FHC value of 30.25%; RHC ( $15\% \times (1 + 720/1015) = 25.6\%$  against 27.7%) and the proton-tagged FHC total ( $17.7\% \times (1 + 338/376) = 33.6\%$  against 34.9%, a sample with a different purity) follow the same arithmetic. The differential flux terms of 24–33% are the same effect bin by bin, with the response conditioning adding a few per cent in the worst-conditioned bins ( $p_\pi$ ,  $\cos\theta_\mu$ ) and nothing in the best ( $\theta_{\mu\pi}$ ).

Two consequences follow. First, the flux term is fully correlated between the flux normalisation and the subtracted background, which is the correct treatment when both come from the same flux prediction; a reader who wants the “pure” normalisation term should read 17%, and the remainder is the flux dependence of the background model. Second, the lever for reducing it is the purity of the selection (or a data-driven background constraint from the control regions of §10.5), not the regularisation. The reconstructed-level flux term is 16.4–17.7% bin-averaged and flat across observables, as expected of a normalisation term (the 600 PPFX universes reweight the same events, so it carries no Monte Carlo statistics); that is the 17% referred to throughout.

## 14 Limitations

This section states what limits the measurement as it stands. The studies behind the scope decisions are in the technical supplement; the history of the analysis is in the change log.

**The analysis is blind, and that is the dominant limitation.** No beam-on signal region has been opened. Every “data” point is Poisson-thrown from the central-value Monte Carlo, so everything demonstrated here concerns the extraction machinery rather than nature. In particular, no statement in this note tests the background model against data. At  $\approx 59\%$  purity, with proton-to-pion misidentification the largest neutrino background, the measurement depends on that model substantially, and fake-data closure cannot probe it: the same simulation supplies both the pseudo-data and the background estimate. This is the single largest untested assumption in the analysis.

**Pion momentum is not recoverable, and that sets the scope.** Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near  $0.34\text{ GeV}/c$  and carries little information above roughly  $0.4\text{ GeV}/c$ . Every alternative estimator was tried — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, the proton-tagged selection — and a truth-level golden selection performs no better than the trained classifier, which places the limit in the physics rather than the estimator. This is why  $p_\pi$  is reported in two coarse bins and why the differential  $W_{\pi p}$  shape is withdrawn: at this exposure no binning satisfies both the migration and the statistics requirement.

**The extraction may carry a small negative offset.** The ensemble of thirty-two throws on the current release puts the mean below the reference by 0.5% ( $p_\mu$ ), 1.0% (one-bin total) and 2.7% ( $\cos\theta_\mu$ ), at 0.4, 1.3 and 1.9 standard deviations of the mean, all from the same throws. It is not significant individually, but it has the same sign everywhere and is largest in the most regularised observable. No correction or uncertainty is assigned; a larger ensemble, and one on RHC, would resolve it.

**The  $W_{\text{had}}$  binning has not been edge-scanned.** The migration-level edge scan that fixed the  $p_\pi$  boundary (Sec. 1.1 of the technical supplement) was not run for  $W_{\text{had}}$ , whose upper bins are narrower than its resolution half-width; the six-bin  $W_{\text{had}}$  result is released with its closure and generator overlays, and its shape should be read with that caveat until the scan is made (not gating).

**No alternative-generator fake-data test exists for the released chain.** Every closure in this note is a same-sample self-closure: the fake data are thrown from the central-value tune that also supplies the response. The model dependence of the efficiency is therefore untested where it matters most: the one-bin total (a single efficiency averaged over the whole phase space) and the open-ended  $p_\pi > 0.205$  GeV/ $c$  bin, inside which the efficiency varies strongly with the true momentum. A NuWro or GiBUU fake-data pass through the released configurations is required before the signal region is opened; it is not required for the control-region opening, which compares reconstruction-level yields and does not involve the unfolding.

**Coverage is demonstrated for three extractions, in one configuration.** The pull and coverage study covers FHC  $p_\mu$ ,  $\cos\theta_\mu$  and the one-bin total. All give widths consistent with unity, and  $\cos\theta_\mu$  was chosen because its  $A_C$  suppression is the most severe in the analysis, but RHC and the proton-tagged family — where the statistics are sparser — have not been tested.

**The proton-tagged results are secondary, and their model comparisons are the weakest part.** Their covariance is complete with respect to the configured model and their central values current, but the exposure does not support a differential  $W_{\pi p}$  shape, and the sample has no dedicated control region for the  $\approx 13\%$  proton mis-tag: every existing sideband applies the proton requirement rather than inverting it, so nothing directly constrains a wrong-track proton assignment, which is exactly what distorts the hadronic observables.

**The TKI prescription is borrowed from the single-proton case.**  $\delta p_T$ ,  $\delta\alpha_T$  and  $\delta\phi_T$  are exact for the  $\mu + (\pi p)$  system, but  $p_n$  and  $W_{\text{had}}$  use the single-nucleon residual-mass and removal-energy assumptions of that tool with the summed hadronic system in the proton slot (§11.1). Data and generators are treated identically, so the comparison is consistent, but no truth-level closure of this prescription against a pion-production-specific definition has been made and the bias from the residual-mass assumption is not quantified;  $p_n$  should be read as a proxy.

**No dedicated systematic covers the MCS muon momentum.** Sixty-three per cent of the selected muons exit the detector and take their momentum from multiple Coulomb scattering, whose scale and resolution are not varied by the eight detector-variation knobs; the estimator enters only through the response matrix. A data-driven MCS-versus-range comparison on contained muons is the standard way to bound it and has not been done here. Before the signal region is opened one of three things is required: that comparison, on contained muons in the beam-off and overlay samples; a propagated MCS scale and resolution variation through the response; or a quantitative argument that the existing detector variations, several of which change the MCS estimate through the space-charge and recombination models, already cover it. The control-region comparison does not depend on it, since it is made at reconstruction level with the same estimator on both sides.

**Two beam-off terms are not in the covariance.** The released covariance carries the beam-off statistical term but no uncertainty on the per-run gate ratios or on the use of the Run-1 beam-off sample for Run 2, which has no sample of its own (14% of the FHC and 23% of the RHC exposure). With the beam-off at 1.9% of the selected sample a 20% uncertainty on either would add 0.4% to the total; they are omitted for that reason and stated here. The beamline-geometry flux term (1.6–3.3%) is likewise absent (§13).

**$A_C$  is not norm preserving, so two different totals exist.** The Wiener-SVD differential result estimates  $A_C x_{\text{true}}$ , and the row sums fall as low as 0.050. Summing it over bins therefore does not give a physical total cross section; the physical total is the one-bin extraction, quoted separately. Two matrices have negative row sums in their least-determined bin.

**Actions required before unblinding.** The prerequisites, split by the decision they gate, are the checklist of Sec. C of the technical supplement. In brief: the four inclusive control regions are to be opened first, under the frozen procedure of Sec. D.5 of the technical supplement once the conveners have ratified it; the signal region only after any control-region discrepancy has been treated under that procedure, the flux constants have been confirmed as PPFX-corrected central values (§3.4), the real beam-on gate counts have replaced the fake-data ones (the Run-1 beam-off sample stands in for Run 2 permanently, §4), and a final clean-cache reproduction has been made; and any proton-tagged result only after the inverted-proton-PID control region exists (Sec. D.6 of the technical supplement). Nothing has to be switched for the beam-off treatment: the framework detects fake data by the presence of MC truth and selects the treatment itself, and refuses a real beam-on file unless an explicit unblinding switch is set.

The five automated guards that protect against regression — the normalisation manifest, the systematics/file-list coupling assertion, the staleness check, the table-consistency check and the blinding guard — are described in Sec. F of the technical supplement.

## 15 Summary

We have developed a measurement of  $\nu_\mu + \bar{\nu}_\mu$  charged-current single charged-pion production on argon using the full MicroBooNE NuMI exposure: forward horn current ( $8.857 \times 10^{20}$  POT), reverse horn current ( $1.1082 \times 10^{21}$  POT), and their combination ( $1.9939 \times 10^{21}$  POT).

**The analysis is blind.** No beam-on signal region has been opened. Every “data” point in this note is Poisson-thrown from the central-value Monte Carlo, so the results demonstrate that the extraction works; they are not measurements, and no statement here about a generator is a statement about nature.

**Measured phase space and selection.**  $|\text{PDG}_\nu| = 14$ , charged current, true vertex in the fiducial volume, exactly one muon and one charged pion, no neutral pions or kaons, any number of protons, with  $p_\mu > 0.15 \text{ GeV}/c$ ,  $p_\pi > 0.175 \text{ GeV}/c$  and  $\theta_{\mu\pi} < 2.6 \text{ rad}$ . The selection reaches  $\approx 17.5\%$  efficiency at  $\approx 59\%$  purity. Angles and transverse observables are defined about the per-event neutrino direction, not the detector  $z$  axis.

**What is measured.** Fifty-one differential extractions, eighteen inclusive and thirty-three proton-tagged, across the three configurations, plus six one-bin totals. The primary scope is the total flux-averaged cross section,  $d\sigma/dp_\mu$ , the muon and pion angular cross sections,  $d\sigma/d\theta_{\mu\pi}$ , and a deliberately coarse two-bin  $d\sigma/dp_\pi$ . Inclusive differential total uncertainties run 32.4–53.9%, with the detector term 15.6–35.0% and flux the largest single contribution; the one-bin totals carry 37.6–38.7%.

The proton-tagged subsample is secondary scope, with differential total uncertainties 25.1–147.7% (the largest for the six-bin combined  $W_{\text{had}}$ ) and one-bin totals of 41–47%. Its covariance is complete with respect to the configured model, but the exposure does not support a differential  $W_{\pi p}$  shape at any binning, so that result is withdrawn and only the integrated proton-tagged cross section is offered.

**What limits the measurement.** Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near  $0.34 \text{ GeV}/c$ . A survey of alternatives — golden-pion tagging, multiple Coulomb scattering, calorimetry, multivariate regression, daughter-energy recovery, and the proton-tagged selection — recovers none of it, and a truth-level golden selection performs no better than the classifier, which places the limit in the physics rather than the estimator. This is what reduces  $p_\pi$  to two bins and  $W_{\pi p}$  to an integrated quantity, and it is the single most consequential limitation of the analysis.

**Validation.** The extraction reproduces its own input: the mean closure ratio over the eighteen inclusive extractions is 1.010 with second-derivative regularisation. The algebraic identity  $UR = A_C$  holds to  $4.4 \times 10^{-6}$  (App. A of the technical supplement), which tests the unfolding construction rather than only its behaviour on one sample. The released  $A_C$  row sums establish that the observable-to-observable spread of the Wiener-SVD integrals is the additional smearing rather than physics or acceptance; D’Agostini iterative unfolding, whose integral is flat across observables, is consistent with that (it returns its prior on central-value input, so it cannot test more).

The closure  $\chi^2$  values should not be read as agreement: their  $p$ -values cluster near unity because the measurement and the truth it is compared against share their systematics, which cancel in the residual while still inflating the covariance used to judge it.

**Reproducibility from raw inputs.** The release is built from the processed ntuples with every cached universe deleted, in a single pass, so no preserved intermediate exists. An independent rebuild of the same kind, into a separate tree so that a failure could not damage the release it was verifying, reproduced an earlier release *bin by bin*, with no bin differing by more than  $10^{-9}$  in any of the fifty-one extractions; repeating that comparison on the release to be unblinded is a prerequisite (Sec. F of the technical supplement). This is a different question from the closure and ensemble tests: those ask whether the extraction is right, this asks whether what is published is what the code produces. The external-reproduction check of the released  $A_C$  prescription (§10.2) belongs to the same class: a user must be able to regenerate every published generator curve from the released matrix, and can.

**History.** The analysis reached this state through a series of corrections, each recorded in the change log with the values before and after; the automated guards of Sec. F of the technical supplement exist because of them. This note describes the analysis as it stands and does not repeat that history.

**Before unblinding.** Control regions must be opened in data first, with the permitted response to any discrepancy fixed beforehand, and a clean rebuild from empty caches must reproduce the release. The configuration-to-systematics coupling is asserted by the framework (Sec. F of the technical supplement). The ensemble of throws covers three FHC extractions; extending it to RHC and to the proton-tagged family would turn the closure caveat above into a coverage statement for every result.

## A Observable Binning

Generated directly from the released `curves_*.tsv` files, so these are by construction the bin edges of the published results rather than a transcription. The binnings are identical across the three beam configurations; the inclusive  $\cos \theta_\pi$  uses the forward-split five bins below, the proton-tagged  $\cos \theta_\pi$  four (Sec. 6 of the technical supplement).

Table 28: Bin edges of every released observable, read from the data release.

| Observable                             | Bins | Edges                                       |
|----------------------------------------|------|---------------------------------------------|
| <i>Inclusive</i>                       |      |                                             |
| $p_\mu$ (GeV/c)                        | 7    | 0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3 |
| $p_\pi$ (GeV/c)                        | 2    | 0.175, 0.205, 1*                            |
| $\cos \theta_\mu$                      | 5    | -1, 0.45, 0.65, 0.8, 0.9, 1                 |
| $\cos \theta_\pi$                      | 5    | -1, -0.1, 0.35, 0.6, 0.8, 1                 |
| $\theta_{\mu\pi}$ (rad)                | 5    | 0, 0.6, 0.85, 1.3, 1.85, 2.6                |
| $\theta_\mu$ (rad)                     | 5    | 0, 0.43, 0.62, 0.84, 1.17, 3.15             |
| <i>Proton-tagged</i>                   |      |                                             |
| $p_\mu$ (GeV/c)                        | 7    | 0.15, 0.35, 0.55, 0.75, 0.95, 1.25, 1.75, 3 |
| $p_\pi$ (GeV/c)                        | 2    | 0.175, 0.205, 1*                            |
| $\cos \theta_\mu$                      | 5    | -1, 0.45, 0.65, 0.8, 0.9, 1                 |
| $\cos \theta_\pi$                      | 4    | -1, -0.1, 0.35, 0.75, 1                     |
| $\theta_{\mu\pi}$ (rad)                | 5    | 0, 0.6, 0.85, 1.3, 1.85, 2.6                |
| $W_{\pi p}$ (GeV/c <sup>2</sup> )      | 6    | 1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.9     |
| $W_{\text{had}}$ (GeV/c <sup>2</sup> ) | 6    | 0, 0.78, 1.08, 1.21, 1.31, 1.44, 2.74       |
| $\delta p_T$ (GeV/c)                   | 2    | 0, 0.3, 2.5                                 |
| $\delta \phi_T$ (deg)                  | 2    | 0, 50, 180                                  |
| $\delta \alpha_T$ (deg)                | 2    | 0, 120, 180                                 |
| $p_n$ (GeV/c)                          | 2    | 0, 0.375, 2                                 |
| $\theta_p$ (rad) <sup>†</sup>          | 4    | 0, 0.659, 1.068, 1.539, $\pi$               |
| $\theta_{\pi p}$ (rad) <sup>†</sup>    | 3    | 0, 1.193, 2.010, $\pi$                      |

<sup>†</sup> Proposed secondary scope: the proton polar angle and the pion–proton opening angle (§11.3).  $\theta_p$  uses edges derived on the RHC chain, which condition three times better than the FHC-derived ones at the same bin count.

\* Numerical plotting boundary only; the upper truth bin is open-ended ( $p_\pi > 0.205$  GeV/c), and the released density over the conventional width 0.795 GeV/c represents the bin integral (Table 28 of the technical supplement).

The migration-diagonal criterion itself was also relaxed, from 0.68 to 0.50 admits twenty-six new candidate binnings, and *every one of them* fails the secondary gate on the conditioning of  $A_C$ , which is not the diagonal. The only healthy candidate is a two-bin  $W_{\text{had}}$ , and the six-bin scheme is kept, since two bins buy conditioning by discarding the shape the observable is measured for. No binning in this note changes as a result.

Every released binning was tested for whether finer or more bins are admissible (Sec. 6.3 of the technical supplement):  $p_\mu$ ,  $\cos \theta_\mu$ ,  $\theta_\mu$  and  $p_\pi$  cannot go finer,  $\theta_{\mu\pi}$  gains bins but no resolved information, and  $\cos \theta_\pi$  admits a fifth bin in the forward hemisphere (edges  $-1, -0.1, 0.35, 0.6, 0.8, 1$ ) with better-conditioned  $A_C$  and unchanged per-bin uncertainties, which is the released binning; the generator predictions are computed at these edges from the event-level files.

The top  $p_\pi$  bin is defined at truth level as  $p_\pi > 0.205$  GeV/c with no upper edge; the 1 GeV/c listed above is the finite boundary used in the numerical release to assign that bin a width, not a kinematic limit of the sample. The bin is an inclusive high-momentum region, not a differential point, for the reasons set out in the technical supplement; the bin-integrated partial cross sections are given in Table 28 of the technical supplement.

## B Alternative binnings considered

Every released binning is the coarsest that satisfies the migration-diagonal criterion, and finer schemes were tested rather than assumed unusable. The criterion itself was also relaxed, from the 0.68 column-normalised diagonal to 0.50, to see what that would admit: twenty-six further candidates pass the relaxed diagonal, and all but one fail the secondary requirement, which is the conditioning of the additional-smearing matrix  $A_C$ . A binning that migrates acceptably can

still give a response that cannot be inverted, and that — not the diagonal — is what limits this measurement. Table 29 lists the alternatives and their fate.

Table 29: Alternative binnings examined.  $s_{\min}/s_1$  is the conditioning of  $A_C$  (FHC/RHC unless stated); a value near zero means the regularised response cannot be inverted, whatever the migration diagonal says. Only the two-bin  $W_{\text{had}}$  is healthy, and it was not adopted.

| Observable        | Alternative        | $s_{\min}/s_1$       | Status                                               |
|-------------------|--------------------|----------------------|------------------------------------------------------|
| $\cos \theta_\mu$ | 6 bins             | 0.11/0.09            | not adopted: conditioning                            |
|                   | 7 bins             | 0.02/0.05            | not adopted: conditioning                            |
| $\theta_\mu$      | 7, 8 bins          | 0.00/0.00            | not adopted: rank-deficient                          |
| $p_\pi$           | 3 bins             | 0.05/0.01            | not adopted: conditioning                            |
| $\delta p_T$      | 3 bins             | 0.00/0.00            | not adopted: rank-deficient                          |
| $\delta \alpha_T$ | 3 bins             | 0.03/0.01            | not adopted: conditioning                            |
| $W_{\pi p}$       | 2 bins             | 0.26/0.15            | sound response, no shape (App. C)                    |
| $W_{\text{had}}$  | 2 bins             | <b>0.58/0.54</b>     | healthy; <i>not</i> adopted, six bins keep the shape |
| $\cos \theta_p$   | 4 bins             | 0.45/0.02            | superseded by $\theta_p$                             |
|                   | 6 bins             | 0.00/0.00            | not adopted: rank-deficient                          |
| $\theta_p$        | 5, 7 bins          | $\leq 0.03$          | not adopted: conditioning                            |
|                   | 3 bins (RHC edges) | 0.16/0.05; COMB 0.51 | alternative to the released four bins                |
| $\cos \theta_\pi$ | 4 bins             | 0.27/0.24            | superseded by the five-bin scheme                    |

Two entries deserve a sentence. The two-bin  $W_{\text{had}}$  is the only candidate whose response is healthier than the released one, and it was still not adopted: two bins buy that conditioning by discarding almost all of the shape, and the shape is what  $W_{\text{had}}$  is measured for. The three-bin  $\theta_p$  reaches the best combined conditioning of anything tried (0.51), but the released four-bin scheme is better balanced across the two configurations in which  $\theta_p$  is quoted, and keeps a bin.

## C Withdrawn results

Two results were extracted, examined and withdrawn. They are recorded here rather than in the body because the reasons are properties of the response, not of the measurement being reported, and because their files remain in the data release, where they could otherwise be mistaken for results. **Neither is to be quoted.**

### C.1 The differential $W_{\pi p}$ shape

The directly measured pion–proton invariant mass is the observable that would show the  $\Delta(1232)$  directly, and it is the reason the proton-tagged subsample was built. At this exposure its shape is not measurable at any binning tried. The six-bin scheme that resolves the resonance region has a top bin only 2% diagonal: essentially every event reconstructed there comes from somewhere else. Coarsening to two bins restores the response ( $s_{\min}/s_1 = 0.26$  FHC, 0.15 RHC) but leaves an observable with no shape to speak of, and the two-bin scheme was not adopted for that reason. Closure against the  $A_C$ -smeared truth does not rescue the six-bin result: closure tests whether the extraction inverted the response it was given, and cannot restore information the response does not contain.

What the proton-tagged sample does support is the *integrated* cross section, which is released, and the calorimetric hadronic mass  $W_{\text{had}}$ , whose response is sound at six bins.

### C.2 The five-bin $d\sigma/dp_\pi$

Charged pions interact hadronically before ranging out, so the reconstructed momentum saturates near 0.34 GeV/ $c$  and the migration matrix loses its diagonal above it: four of the five bins

fall below the 68% criterion, the top bin reaching 6%. The released  $d\sigma/dp_\pi$  is therefore a two-bin result — a near-threshold interval and one broad inclusive interval — and is a two-region partition rather than a spectrum. The limitation is the estimator, not the binning: range-based momentum is only valid for a pion that stops, and no rebinning of a saturated variable recovers the information.

## D Selection Cut Parameter Summary

Table 30: All selection threshold constants.

| Parameter                                  | Value                   | Ntuple variable                         |
|--------------------------------------------|-------------------------|-----------------------------------------|
| Topological score                          | $> 0.67$                | Pandora score                           |
| Muon track score                           | $> 0.80$                | Pandora track score                     |
| Muon vertex distance                       | $\leq 4.0$ cm           |                                         |
| Muon track length                          | $> 10.0$ cm             |                                         |
| Muon LLR PID                               | $> 0.20$                | LLR PID                                 |
| Muon MIP BDT                               | $\geq -0.10$            | TMVA BDT                                |
| Pion track length                          | $> 20.0$ cm             |                                         |
| Pion vertex distance                       | $< 4.0$ cm              |                                         |
| Pion LLR PID                               | $> 0.10$                | LLR PID                                 |
| Pion MIP BDT                               | $> -0.10$               | TMVA BDT                                |
| Pion pion-BDT                              | $> -0.10$               | TMVA BDT                                |
| Pion Bragg score                           | $\geq 0.08$             | Bragg likelihood                        |
| Max. opening angle                         | $< 2.6$ rad             |                                         |
| Shower Y-plane hits                        | $< 50$ (if 1 shower)    |                                         |
| Shower vertex distance                     | $> 10$ cm (if 1 shower) |                                         |
| Non-proton multiplicity                    | $< 4$                   | LLR PID $> 0.1$                         |
| Primary track multiplicity                 | $< 5$                   |                                         |
| Cosmic BDT                                 | $> 0.0$ (disabled)      | TMVA BDT                                |
| Software trigger                           | $= 1$                   |                                         |
| <i>Proton-tagged selection only (§7.4)</i> |                         |                                         |
| Leading-proton LLR PID                     | $< 0.05$                | most proton-like non- $\mu/\pi$ primary |
| Proton end point                           | in FV                   |                                         |
| Proton momentum                            | $\geq 0.3$ GeV/c        | proton-hypothesis calorimetric KE       |

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